Pulmonary Function Test and CT features during follow-up after SARS, MERS and COVID-19: A Systematic Review and Meta-Analysis

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Supplementary Material

Table S1: Study Selection Inclusion and Exclusion Criteria

	Inclusion Criteria	Exclusion Criteria
Participants	Adults aged ≥ 18 years	Non-human studies Children (i.e. participants aged less than 18 years of age).
Exposures	SARS, MERS or COVID-19 infection since the year 2000.	Non-coronavirus infection or coronavirus infection before the year 2000.
Outcome	Thoracic CT (any technique) or pulmonary function test results reported after infection (defined as >= 28 days after coronavirus symptom onset and the participant is discharged from hospital (if admitted))	Participants who have deceased whilst in-hospital during their acute infection phase of the coronavirus. The pulmonary disease, complication or sequelae arose < 28 days after the coronavirus infection or the participant is still an inpatient from the acute infection. Other radiological modality reported.
Study types	Randomised control trial, Observational (Cohort, case-control/ case-referent, cross-sectional) or national guidelines, reports and statements to be included	Data not extractable. Opinion piece not based on data. Case reports and series.

***Table S2a*: Embase Database search**

Coronavirus related terms	AND	Respiratory related complication terms	AND	Follow-up terms
Covid-19.mp. Covid*.mp Coronavirinae/ Coronavirus.mp Betacoronavirus/ Coronavirus infection/ Coronavirus*.mp SARS coronavirus/ SARS-related coronavirus/ Sars*.mp Sars-cov.mp Sars-cov-2.mp Severe acute respiratory syndrome.mp Severe acute respiratory syndrome/ Mers*.mp Middle East Respiratory Syndrome Coronavirus/ Middle East respiratory syndrome/ Middle East respiratory syndrome.mp		Fibrosing alveolitis/ Ipf.mp Idiopathic pulmonary fibrosis.mp Pulmonary fibrosis.mp Lung fibrosis.mp Fibrosis/ or fibrosis.mp Fibrotic lung.mp Ild.mp Interstitial Lung Disease/ Interstitial lung disease.mp Pe.mp Pulmonary emboli*.mp Lung embolism/ Lung emboli*.mp Vte.mp Venous thromboembolism/ Thromboembolism/ Venous thromboembolism*.mp Nodul*.mp Nodular lung disease*.mp Opaci*.mp Ggo.mp Groundglass.mp Ground-glass.mp Lung disease*.mp or Lung Disease/ Pulmonary disease*.mp Pleural effusion/ or Pleural effusion*.mp Honeycomb*.mp Reticula*.mp Lung function/ or pulmonary function*.mp or lung function test/ Lung function*.mp Respiratory function/ or respiratory function*.mp Radiography/ or radiograph*.mp or thorax radiography/ or X ray film/ Radiol*.mp or radiology x-ray*.mp or X ray/ x ray*.mp ct.mp or computer assisted tomography/ hrct.mp or high resolution computer tomography/ mortal*.mp death*.mp surviv*.mp or survival/		Sequel*.mp Follow-Up studies/ Follow-up*.mp Long-term*.mp Impact*.mp Consequenc*.mp Persist*.mp Chronic*.mp Permanent*.mp Progress*.mp Advanc*.mp Regress*.mp Resolv*.mp Effect*.mp Outcome*.mp Result*.mp Complicat*.mp

Limited to year 2000 to current.

***Table S2b*: Medline Database search**

Coronavirus related terms	AND	Respiratory related complication terms	AND	Follow-up terms
Covid-19.mp. Coronavirus infections/ Covid.mp. Coronavirus.mp Coronavirus/ Betacoronaviruses.mp SARS virus/ Severe Acute Respiratory Syndrome/ Sars.mp Severe acute respiratory syndrome.mp Sars-cov.mp Sars-cov-2.mp Mers.mp Middle East Respiratory Syndrome Coronavirus/ Middle East respiratory syndrome.mp		Idiopathic pulmonary fibrosis/ Pulmonary fibrosis/ Ipf.mp Idiopathic pulmonary fibrosis.mp Pulmonary fibrosis.mp Lung fibrosis.mp Fibrotic.mp Fibrotic lung.mp Fibrosis/ Fibrosis.mp Lung Diseases, Interstitial/ Ild.mp Interstitial lung disease*.mp Pulmonary Embolism/ Pe.mp Pulmonary emboli*.mp Venous thrombosis/ Lung emboli*.mp Venous thromboembolism/ Thromboembolism/ Vte.mp Nodul*.mp Nodular lung disease*.mp Opaci*.mp Ggo.mp Groundglass.mp Ground-glass.mp Lung disease*.mp Lung Diseases/ Pulmonary disease*.mp Pleural effusion/ Pleural effusion*.mp Breathless*.mp Respiratory Function tests/ Lung function*.mp Respiratory function*.mp Pulmonary function*.mp Tomography, X-ray Computed/ Radiograph*.mp Radiology/ Radiolo*mp X-rays/ x-ray*.mp ct*.mp hrct.mp mortal*.mp Death/ Death*.mp Surviv*.mp		Sequel*.mp Complication/ or complicat*.mp Follow up/ or follow-up*.mp Long-term*.mp Impact*.mp Consequen*.mp Persist*.mp Chronic*.mp Permanent.mp Progress*.mp Advance*.mp Regress*.mp Develop*.mp Resolv*.mp Effect*.mp Outcome*.mp Result*.mp

Limited to year 2000 to current.

***Table S3*: Follow-up time period coding**

Final Code	Range of days included	Range of months included	Range of years included
0-2 months	0-60	0-2	
3-4 months	61-120	3-4	
5-6 months	121-180	5-6	
7-9 months	181-270	7-9	
10-12 months	271-365	10-12	
18 months		13-18	
2 years		19-24	
3 years		25-36	$2 < y \leq 3$
4 years		37-48	$3 < y \leq 4$
5 years		49-60	$4 < y \leq 5$
6 years		61-72	$5 < y \leq 6$
7 years		73-84	$6 < y \leq 7$
8 years		85-96	$7 < y \leq 8$
9 years		97-108	$8 < y \leq 9$
10 years		109-120	$9 < y \leq 10$
11 years		121-132	$10 < y \leq 11$
12 years		133-144	$11 < y \leq 12$
13 years		145-156	$12 < y \leq 13$
14 years		157-168	$13 < y \leq 14$
15 years		169-180	$14 < y \leq 15$

Table S4: CT Radiological terminology group coding

Final descriptive term used in analysis	Terms from literature included in this description
Air trapping	Air trapping
Architectural Distortion	Bronchovascular bundle distortion Interlobular pleural traction Architectural distortion
Cavity	Cavity
Consolidation	Consolidation Patchy shadow
Crazy paving	Crazy paving
Emphysema	Emphysema
Ground Glass Opacity (GGO)	GGO
Halo	Halo sign
Honeycombing	Honeycombing
Linear Opacities	Linear opacity/ opacities Linear density Fibrous bands Fibrous stripes Interstitial thickening Septal thickening Irregular interfaces Irregular lines Interlobular septal thickening Parenchymal bands Subpleural lines
Non ground glass nodules	Nodules
Pleural effusion	Pleural effusion
Pleural thickening	Pleural thickening Pleural adhesion or hypertrophy
Pulmonary Embolism	Pulmonary embolism/ PE
Reticulations	Reticulation Meshwork
Reverse halo	Reverse Halo sign Atoli sign
Traction bronchiectasis and bronchiolectasis	Traction dilatation Traction bronchiectasis Bronchiectasis

Definitions of terms as per Hansell DM, Bankier AA, MacMahon H, McLoud TC, Muller NL, Remy J. Fleischner Society: Glossary of Terms for Thoracic Imaging. Radiology. 2008; 246(3): 697-722.

Table S5: WHO COVID-19 Clinical Management Guidelines – Acute Disease Severity Classification for Adults

COVID-19 Disease Severity	Describing Feature	Description
Mild		Symptomatic patient who meet the case definition for COVID-19 as defined by the WHO without evidence of viral pneumonia or hypoxia
Moderate	Pneumonia	Clinical signs of pneumonia (fever, cough, dyspnoea, fast breathing) but no signs of severe pneumonia, including oxygen saturation (SpO2) ≥ 90% on room air
Severe	Severe Pneumonia	Clinical signs of pneumonia (fever, cough, dyspnoea, fast breathing) plus one of the following: respiratory rate >30 breaths/minute; severe respiratory distress; or SpO2 < 90% on room air.
Critical	Acute Respiratory Distress Syndrome (ARDS)	**Onset:** within 1 week of a known clinical insult (i.e. pneumonia) or new or worsening respiratory symptoms. **Chest imaging** (radiograph, CT or lung ultrasound)**:** bilateral opacities, not fully explained by volume overload, lobar or lung collapse or nodules. **Origin of pulmonary infiltrates:** respiratory failure not fully explained by cardiac failure or fluid overload. Need objective assessment (e.g. echocardiography) to exclude hydrostatic cause of infiltrates/ oedema if no risk factor present **Oxygen impairment:** - Mild ARDS – 200mmHg < PaO_2/FiO2 ≤ 300mmHg (with PEEP or CPAP ≥ 5cmH_2O) - Moderate ARDS – 100mmHg < PaO_2/FiO2 ≤ 200mmHg (with PEEP or CPAP ≥ 5cmH_2O) - Severe ARDS – PaO_2/FiO2 ≤ 100mmHg (with PEEP ≥ 5cmH_2O)
	Sepsis	Acute life-threatening organ dysfunction caused by a dysregulated host response to suspected or proven infection. Signs of organ dysfunction include: altered mental status, difficult or fast breathing, low oxygen saturation, reduced urine output, fast heart rate, weak pulse, cold extremities or low blood pressure, skin mottling, laboratory evidence of coagulopathy, thrombocytopenia, acidosis, high lactate or hyperbilirubinaemia
	Septic Shock	Persistent hypotension despite volume resuscitation, requiring vasopressors to maintain MAP ≥ 65mmHg and serum lactate level > 2mmol/L.

Adapted from: World Health Organisation. COVID-19 Clinical Management – Living guidance. World Health Organisation. 25th January 2021. Available from: https://www.who.int/publications/i/item/WHO-2019-nCoV-clinical-2021-1. Accessed 25th October 2021.

Table S6a: Included Study Risk of Bias Assessment – Newcastle Ottawa Quality Assessment Form for Cross-Sectional and Longitudinal Studies

(References as per main paper)

Study	Representativeness of Sample	Sample size	Non-respondents	Exposure confirmation	Comparability	Assessment of outcome	Statistical test
Antonio GE, et al. 2003 [11]	Non-random sampling	Not justified	Unsatisfactory response rate	Specific test	Controls additional factors	Independent blind assessment	Appropriate
Chang YC, et al. 2005 [18]	No description	Not justified	Unsatisfactory response rate	Specific test	Controls additional factors	Record linkage	Appropriate
Chen JH, et al. 2006 [19]	No description	Not justified	Unsatisfactory response rate	No description	No control of factors	No description	Inappropriate/ Incomplete/ not described
Chiang CH, et al. 2004 [20]	Non-random sampling	Justified/ satisfactory	No description	Specific test	No control of factors	Record linkage	Appropriate
Han Y, et al. 2003 [21]	No description of strategy	Not justified	No description	Specific test	No control of factors	Record linkage	Inappropriate/ Incomplete/ not described
Hsu HH, et al. 2004 [22]	Selected group	Not justified	Unsatisfactory response rate	Specific test	Controls most important factors	Independent blind assessment	Inappropriate/ Incomplete/ not described
Hui DS, et al. 2005 [23]	Non-random sampling	Not justified	Unsatisfactory response rate	Specific test	No control of factors	Independent blind assessment	Appropriate
Jin ZY, et al. 2003 [24]	No description of strategy	Not justified	No description	No description	No control of factors	No description	Inappropriate/ Incomplete/ not described
Li L, et al. 2015 [25]	No description	Not justified	No description	No description	No control of factors	Record linkage	Appropriate
Li TST, et al. 2006 [26]	All subjects/ random sampling	Justified/ satisfactory	No description	Specific test	Controls additional factors	Record linkage	Inappropriate/ Incomplete/ not described
Liu Y, et al. 2007 [27]	No description	Not justified	No description	Specific test	No control of factors	Independent blind assessment	Appropriate
Ngai JC, et al. 2010 [28]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	No control of factors	Record linkage	Appropriate
Ong KC, et al. 2005 [29]	All subjects/ random sampling	Justified/ satisfactory	Satisfactory	Specific test	Controls most important factors	Self report	Appropriate
Ong KC, et al. 2004 [30]	Selected group	Justified/ satisfactory	No description	Specific test	No control of factors	Record linkage	Appropriate
Su MC, et al. 2007 [31]	All subjects/ random sampling	Justified/ satisfactory	No description	No description	No control of factors	Record linkage	Appropriate
Tansey CM, et al. 2007 [32]	Non-random sampling	Justified/ satisfactory	Satisfactory	Specific test	No control of factors	Record linkage	Appropriate
Wang CH, et al. 2005 [33]	No description	Not justified	No description	Specific test	Controls most important factors	Self report	Inappropriate/ Incomplete/ not described
Wong KT, et al. 2004 [34]	Non-random sampling	Not justified	Unsatisfactory response rate	Specific test	No control of factors	Self report	Inappropriate/ Incomplete/ not described

Wu X, et al. 2016 [35]	Non-random sampling	Not justified	No description	No description	No control of factors	Independent blind assessment	Inappropriate/ Incomplete/ not described
Xie L, et al. 2005 [36]	Selected group	Not justified	No description	Specific test	No control of factors	Record linkage	Appropriate
Zhang P, et al. 2020 [37]	Selected group	Justified/ satisfactory	No description	No description	No control of factors	Record linkage	Appropriate
Park WB, et al. 2018 [38]	All subjects/ random sampling	Justified/ satisfactory	Satisfactory	Specific test	Controls most important factors	No description	Appropriate
Anastasio F, et al. 2021 [39]	All subjects/ random sampling	Justified/ satisfactory	No description	Specific test	Controls most important factors	Record linkage	Appropriate
Aparisi A, et al. 2021 [40]	Non-random sampling	Justified/ satisfactory	No description	Specific test	Controls additional factors	Record linkage	Appropriate
Armange L, et al. 2021 [41]	Non-random sampling	Justified/ satisfactory	Satisfactory	Specific test	No control of factors	Record linkage	Appropriate
Arnold DT, et al. 2020 [42]	Non-random sampling	Justified/ satisfactory	Satisfactory	Specific test	Controls most important factors	No description	Appropriate
Balbi M, et al. 2021 [43]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls additional factors	Record linkage	Appropriate
Barisione G, et al. 2021 [44]	All subjects/ random sampling	Justified/ satisfactory	No description	Specific test	Controls additional factors	Record linkage	Appropriate
Bellan M, et al. 2021 [45]	All subjects/ random sampling	Justified/ satisfactory	No description	Specific test	Controls most important factors	Record linkage	Appropriate
Boari GEM, et al. 2021 [46]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls most important factors	Record linkage	Appropriate
Cao J, et al. 2021 [47]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls most important factors	Record linkage	Appropriate
Cortes-Telles A, et al. 2021 [48]	No description	Not justified	No description	Specific test	Controls most important factors	Record linkage	Appropriate
Crisafulli E, et al. 2021 [49]	Non-random sampling	Justified/ satisfactory	No description	Specific test	Controls additional factors	Record linkage	Appropriate
D'Cruz RF, et al. 2021 [50]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls additional factors	Record linkage	Appropriate
Daher A, et al. 2021 [51]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls additional factors	Record linkage	Appropriate
Darley DR, et al. 2020 [52]	Non-random sampling	Not justified	No description	Specific test	Controls most important factors	Record linkage	Appropriate
De Graaf MA, et al. 2021 [53]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls most important factors	Record linkage	Appropriate
Debeaumont D, et al. 2021 [54]	Selected group	Justified/ satisfactory	No description	No description	Controls most important factors	Record linkage	Appropriate
Dorelli G, et al. 2021 [55]	Selected group	Not justified	No description	No description	Controls additional factors	Record linkage	Appropriate
Ego A, et al. 2021 [56]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	No description	No control of factors	Record linkage	Appropriate

Frija-Masson J, et al. 2021a [57]	No description	Not justified	No description	No description	No control of factors	Record linkage	Appropriate
Frija-Masson J, et al. 2021 [58]	Non-random sampling	Not justified	Unsatisfactory response rate	Specific test	Controls additional factors	Independent blind assessment	Appropriate
Froidure A, et al. 2021 [59]	Non-random sampling	Justified/ satisfactory	No description	Specific test	No control of factors	Independent blind assessment	Appropriate
Gianella P, et al. 2021 [60]	Non-random sampling	Justified/ satisfactory	No description	Specific test	Controls additional factors	Independent blind assessment	Appropriate
Gonzalez J, et al. 2021 [61]	Non-random sampling	Justified/ satisfactory	No description	Specific test	No control of factors	Record linkage	Appropriate
Han X, et al. 2021 [64]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls additional factors	Independent blind assessment	Appropriate
Jiang A, et al. 2021 [67]	No description	Not justified	No description	No description	Controls most important factors	No description	Inappropriate/ Incomplete/ not described
Joris M, et al. 2021 [68]	Non-random sampling	Not justified	Unsatisfactory response rate	No description	No control of factors	Record linkage	Appropriate
Komici K, et al. 2021 [69]	All subjects/ random sampling	Justified/ satisfactory	Satisfactory	Specific test	No control of factors	Record linkage	Inappropriate/ Incomplete/ not described
Labaraca G, et al. 2021 [70]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls most important factors	Independent blind assessment	Appropriate
Lerum TV, et al. 2021 [71]	No description	Justified/ satisfactory	No description	Specific test	Controls most important factors	Independent blind assessment	Appropriate
Li X, et al. 2021 [72]	Non-random sampling	Justified/ satisfactory	No description	Specific test	Controls additional factors	Independent blind assessment	Appropriate
Liang L, et al. 2020 [74]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls most important factors	Independent blind assessment	Appropriate
Liu D, et al. 2020 [75]	No description	Not justified	No description	Specific test	Controls additional factors	Independent blind assessment	Appropriate
Liu C, et al. 2020 [76]	Non-random sampling	Not justified	No description	Specific test	No control of factors	Independent blind assessment	Inappropriate/ Incomplete/ not described
Liu M, et al. 2021a [77]	No description	Not justified	No description	No description	Controls additional factors	Independent blind assessment	Appropriate
Liu M, et al. 2021b [78]	No description	Not justified	No description	No description	Controls most important factors	Independent blind assessment	Appropriate
Lombardi F, et al. 2021 [79]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls additional factors	Record linkage	Appropriate
Lopez-Romero S, et al. 2021 [80]	No description	Not justified	No description	No description	No control of factors	Record linkage	Appropriate
Marvisi M, et al. 2020 [81]	Non-random sampling	Not justified	Satisfactory	Specific test	Controls most important factors	Independent blind assessment	Appropriate
McGroder CF, et al. 2021 [82]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls additional factors	Independent blind assessment	Appropriate
Miwa M, et al. 2021 [83]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls additional factors	Independent blind assessment	Appropriate

Mohr A, et al. 2021 [84]	Non-random sampling	Justified/ satisfactory	No description	No description	No control of factors	Record linkage	Appropriate
Myall KJ, et al. 2021 [85]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	No control of factors	Record linkage	Appropriate
Noel-Savina E, et al. 2021 [86]	Non-random sampling	Not justified	Unsatisfactory response rate	Specific test	Controls most important factors	Record linkage	Appropriate
Nunez-Fernandez M, et al. 2021 [87]	Non-random sampling	Justified/ satisfactory	No description	Specific test	Controls additional factors	Record linkage	Appropriate
Pan M, et al. 2021 [88]	Non-random sampling	Not justified	No description	Specific test	Controls most important factors	Record linkage	Appropriate
Parker AJ, et al. 2021 [89]	Non-random sampling	Justified/ satisfactory	No description	Suspected	Controls additional factors	Record linkage	Appropriate
Parry AH, et al. 2021 [90]	Selected group	Not justified	No description	Specific test	Controls additional factors	Independent blind assessment	Appropriate
Pasau T, et al. 2021 [91]	No description	Not justified	No description	No description	No control of factors	Record linkage	Appropriate
Polese J, et al. 2021 [92]	No description	Not justified	No description	Specific test	No control of factors	Record linkage	Appropriate
Qin W, et al. 2021 [93]	Non-random sampling	Justified/ satisfactory	No description	Specific test	Controls most important factors	Independent blind assessment	Appropriate
Remy-Jardin M, et al. 2021 [95]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls additional factors	Independent blind assessment	Appropriate
Riou N, et al. 2021 [96]	Non-random sampling	Justified/ satisfactory	No description	No description	Controls most important factors	No description	Appropriate
Santus P, et al. 2021 [98]	Non-random sampling	Not justified	No description	Specific test	Controls most important factors	Independent blind assessment	Appropriate
Shah AS, et al. 2020 [99]	Non-random sampling	Not justified	Unsatisfactory response rate	Specific test	Controls most important factors	No description	Appropriate
Sibila O, et al. 2021 [100]	Non-random sampling	Justified/ satisfactory	Satisfactory	No description	Controls additional factors	Record linkage	Appropriate
Sonnweber T, et al. 2021 [101]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls additional factors	Record linkage	Inappropriate/ Incomplete/ not described
Strumiliene E, et al. 2021 [102]	Non-random sampling	Justified/ satisfactory	Satisfactory	Specific test	Controls most important factors	Independent blind assessment	Inappropriate/ Incomplete/ not described
Tabatabaei SMH, et al. 2020 [103]	Selected group	Not justified	Satisfactory	Specific test	Controls most important factors	Independent blind assessment	Appropriate
Trinkmann F, et al. 2021 [104]	Non-random sampling	Justified/ satisfactory	No description	Specific test	Controls additional factors	Record linkage	Inappropriate/ Incomplete/ not described
Truffaut L, et al. 2021 [105]	Non-random sampling	Not justified	Unsatisfactory response rate	Specific test	Controls most important factors	No description	Appropriate
Van den Borst B, et al. 2020 [106]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls most important factors	Independent blind assessment	Appropriate
Van der Sar-van der Brugge S, et al. 2021 [107]	Non-random sampling	Not justified	No description	Specific test	Controls most important factors	Record linkage	Appropriate

Van Gassel RJJ, et al. 2021a [108]	All subjects/ random sampling	Justified/ satisfactory	Unsatisfactory response rate	No description	Controls additional factors	Record linkage	Appropriate
Van Gassel RJJ, et al. 2021b [109]	All subjects/ random sampling	Justified/ satisfactory	Unsatisfactory response rate	No description	No control of factors	Record linkage	Appropriate
Venturelli S, et al. 2021 [111]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	No control of factors	Record linkage	Appropriate
Von Zeller C, et al. 2021 [112]	All subjects/ random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls additional factors	Self report	Appropriate
Wang Z, et al. 2021 [113]	Non-random sampling	Not justified	No description	Specific test	No control of factors	Independent blind assessment	Appropriate
Writing committee for the COMEBAC study group. 2021 [114]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls most important factors	Independent blind assessment	Appropriate
Wu Q, et al. 2021 [115]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	No description	Controls most important factors	Independent blind assessment	Appropriate
Wu X, et al. 2021 [116]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls most important factors	Independent blind assessment	Appropriate
Yan X, et al. 2021 [118]	No description	Not justified	No description	Specific test	Controls most important factors	Record linkage	Appropriate
Yang ZL, et al. 2021 [119]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	Specific test	Controls additional factors	Independent blind assessment	Appropriate
Zampogna E, et al. 2021 [120]	Non-random sampling	Not justified	Unsatisfactory response rate	No description	Controls additional factors	Record linkage	Appropriate
Zhang S, et al. 2021 [121]	No description	Not justified	No description	Specific test	Controls most important factors	Record linkage	Appropriate
Zhong L, et al. 2020 [122]	Non-random sampling	Justified/ satisfactory	Unsatisfactory response rate	No description	Controls most important factors	Independent blind assessment	Appropriate
Zou JN, et al. 2021 [124]	All subjects/ random sampling	Justified/ satisfactory	No description	Specific test	Controls most important factors	Independent blind assessment	Appropriate

Table S6b: Included Study Risk of Bias Assessment – Newcastle Ottawa Quality Assessment Form for Case-Control Studies

(References as per main paper)

Study	Adequate case definition	Representative of cases	Selection of controls	Definition of controls	Comparability of cases and controls	Ascertainment of exposure	Exposure assessment same for cases and control	Non-response rate
Grist JT, et al. 2021 [62]	Yes – record linkage	Potential for selection bias	Hospital controls	No history of disease	No control of factors	Secure record	Yes	Rate different/ no description
Salem AM, et al. 2021 [97]	Yes – record linkage	Potential for selection bias	Hospital controls	No history of disease	Controls other factors	Secure record	Yes	Rate different/ no description
Varughese RA, et al. 2021 [110]	No description	Potential for selection bias	No description	No history of disease	No control of factors	No description	Yes	Rate different/ no description

Table S6c: Included Study Risk of Bias Assessment – Newcastle Ottawa Quality Assessment Form for Cohort Studies

(References as per main paper)

Study	Representativeness of cohort	Selection of non-exposed	Comparability of cohorts based on confounders	Assessment of outcome	Sufficient follow-up period to allow outcomes to develop	Adequacy of follow-up of cohorts
Guler SA, et al. 2021 [63]	No description	Same community	Controls for sex	Independent blind assessment	Yes	Complete – all subjects accounted for
Huang C, et al. 2021 [65]	Somewhat representative	Same community	Controls for sex	Independent blind assessment	Yes	Follow-up rate < 80%
Li H, et al. 2020 [73]	Somewhat representative	No description	Controls for other factors	No description	Yes	Complete – all subjects accounted for
Raman B, et al. 2021 [94]	Somewhat representative	Different source	Controls for sex	Independent blind assessment	Yes	Complete – all subjects accounted for
Xu J, et al. 2021 [117]	Somewhat representative	Different source	Controls for other factors	Record linkage	Yes	Complete – all subjects accounted for
Zhou M, et al. 2021 [123]	Somewhat representative	Different source	Controls for other factors	Independent blind assessment	Yes	Complete – all subjects accounted for

Figure S1a: Forest plot of FEV1 % predicted by follow-up time after SARS infection

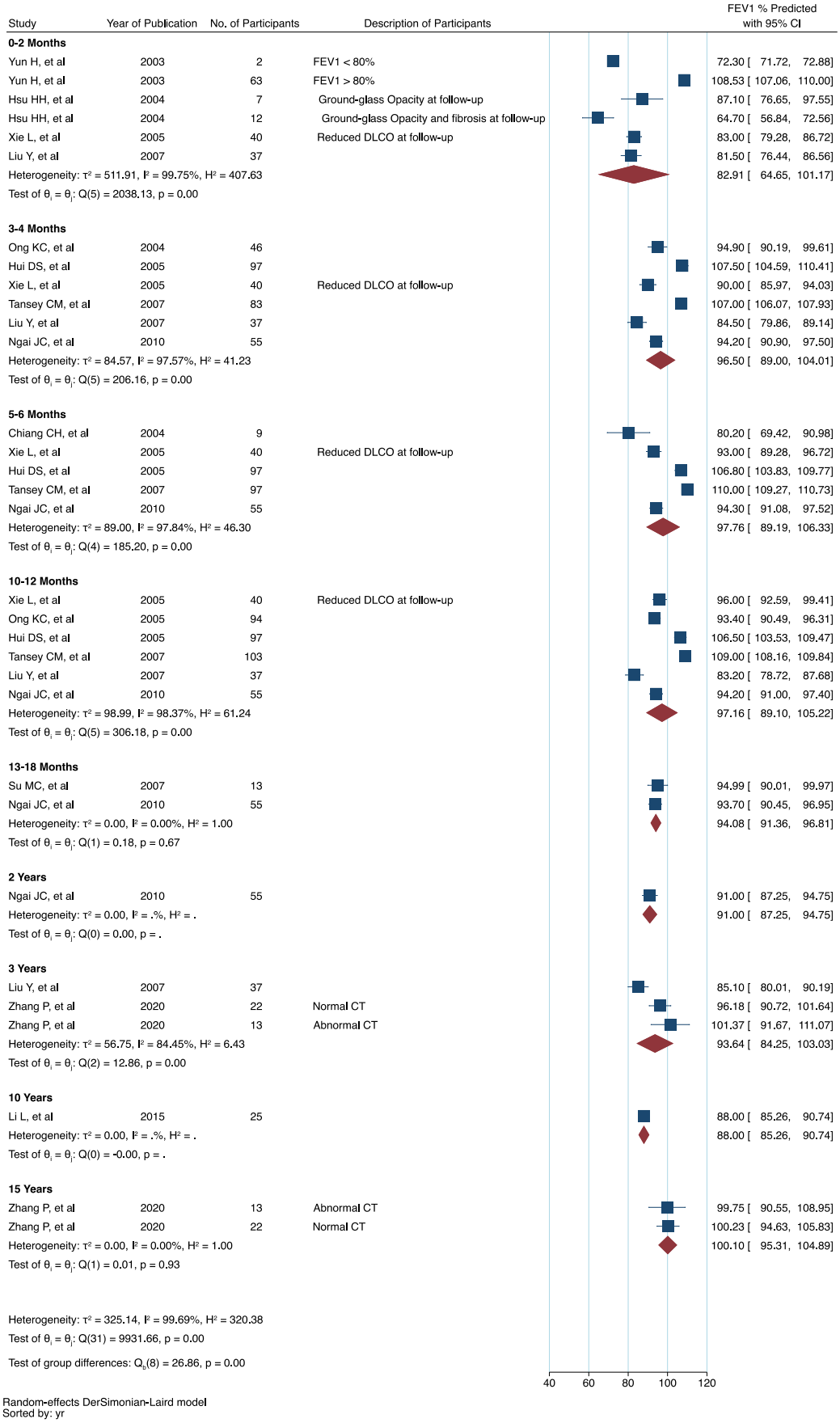


Figure S1b: Forest plot of FVC % predicted by follow-up time after SARS infection

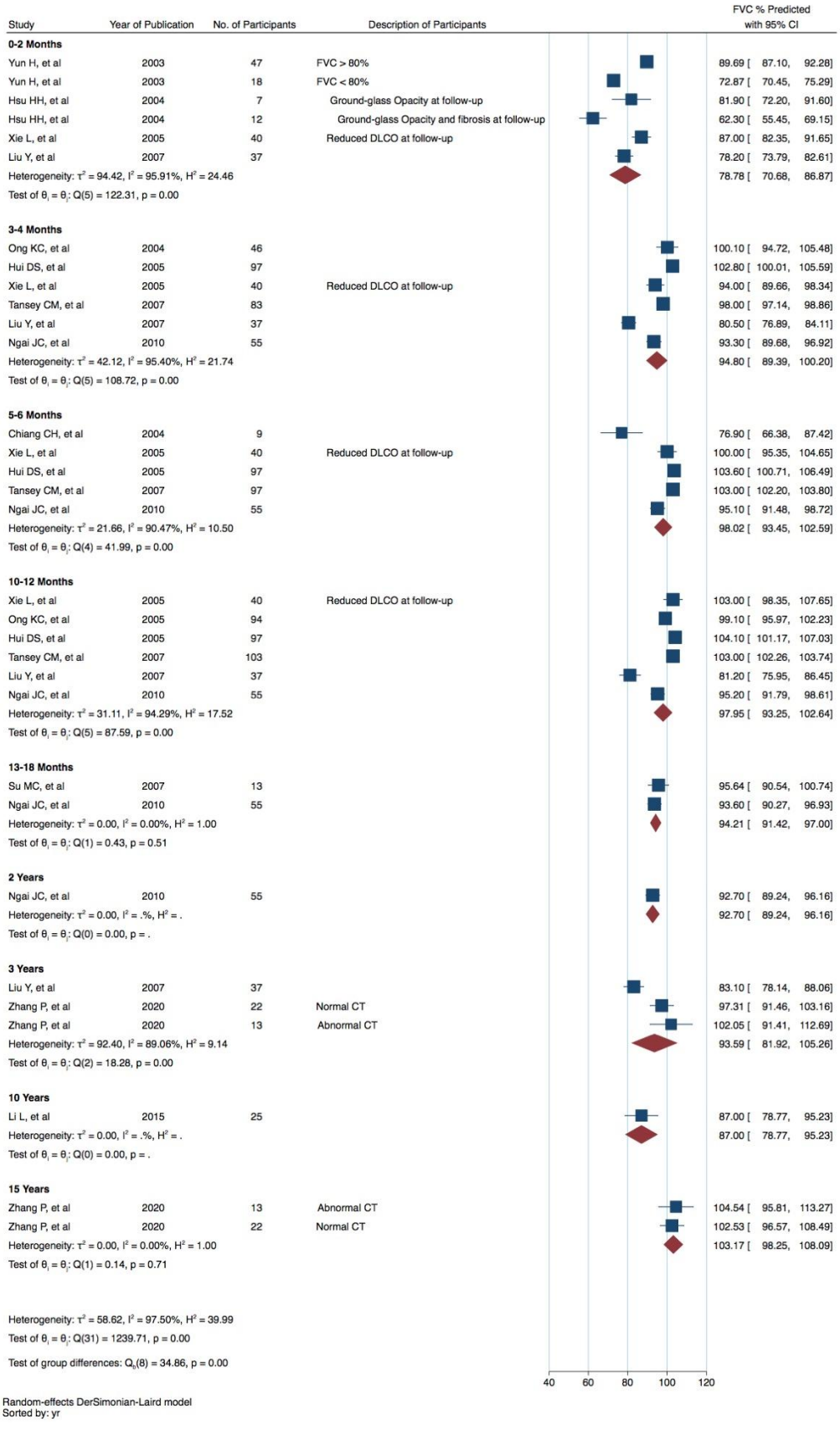


Figure S1c: Forest plot of DLco % predicted by follow-up time after SARS infection

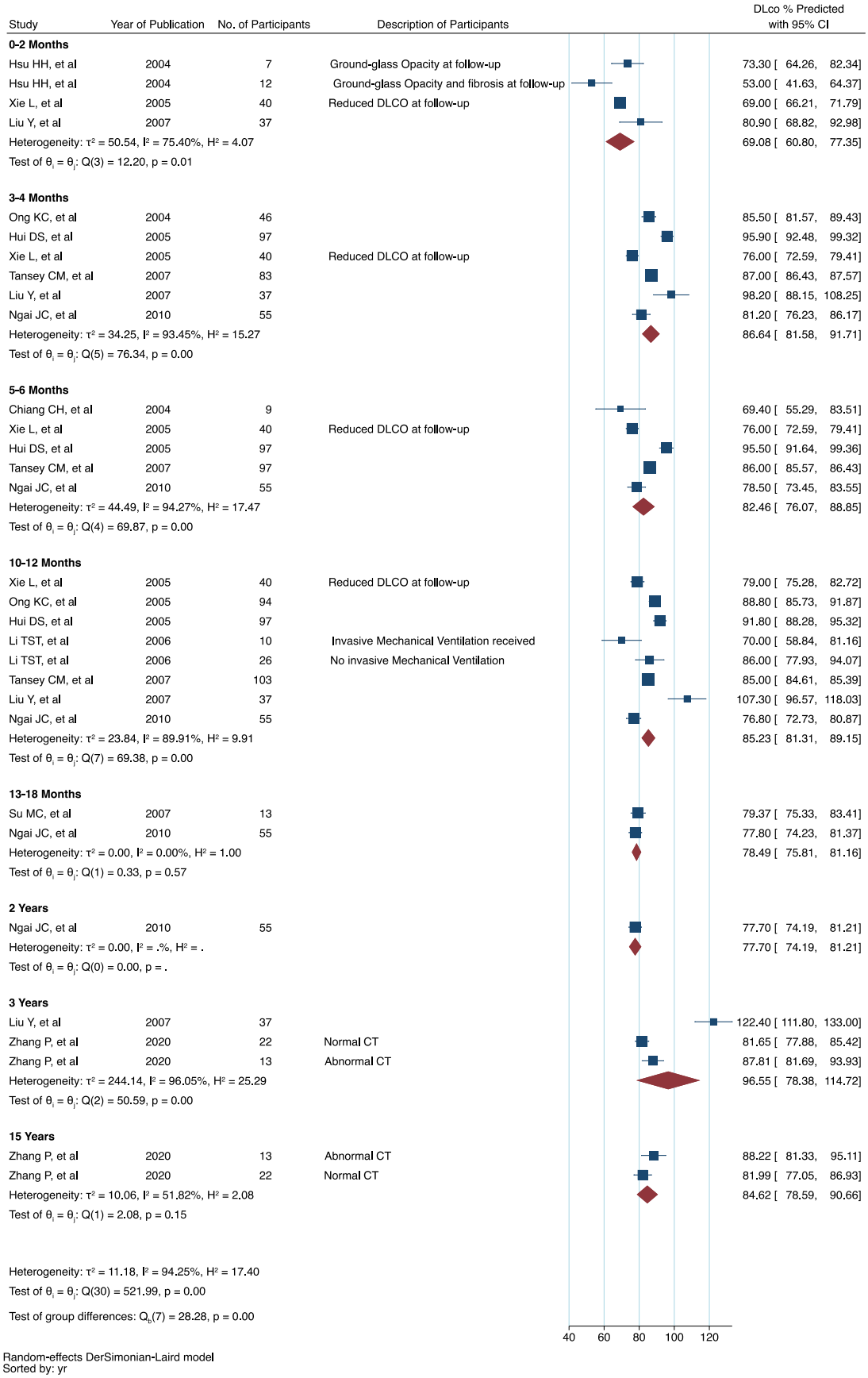


Figure S1d: Forest plot of Kco % predicted by follow-up time after SARS infection

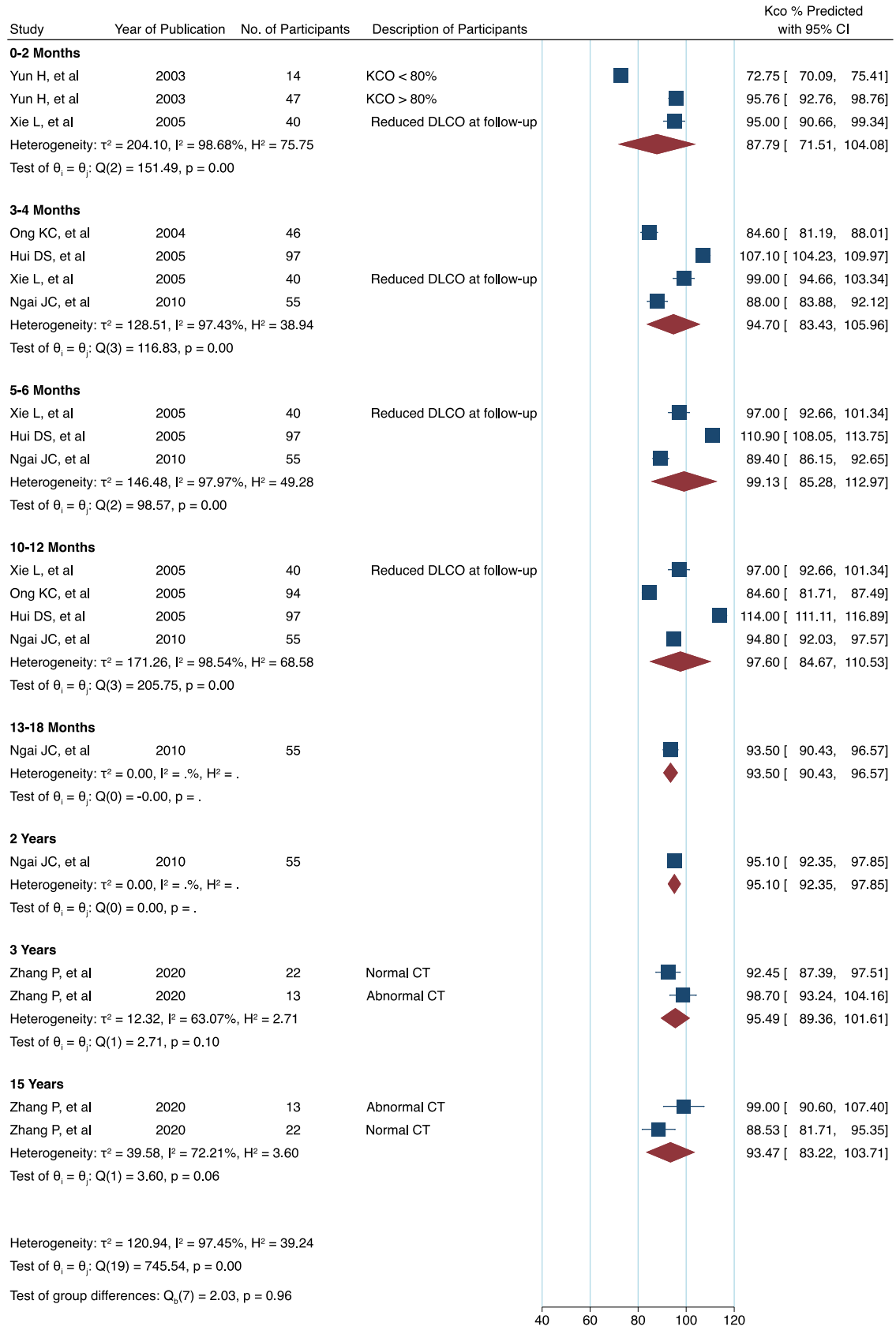


Figure S1e: Forest plot of TLC % predicted by follow-up time after SARS infection

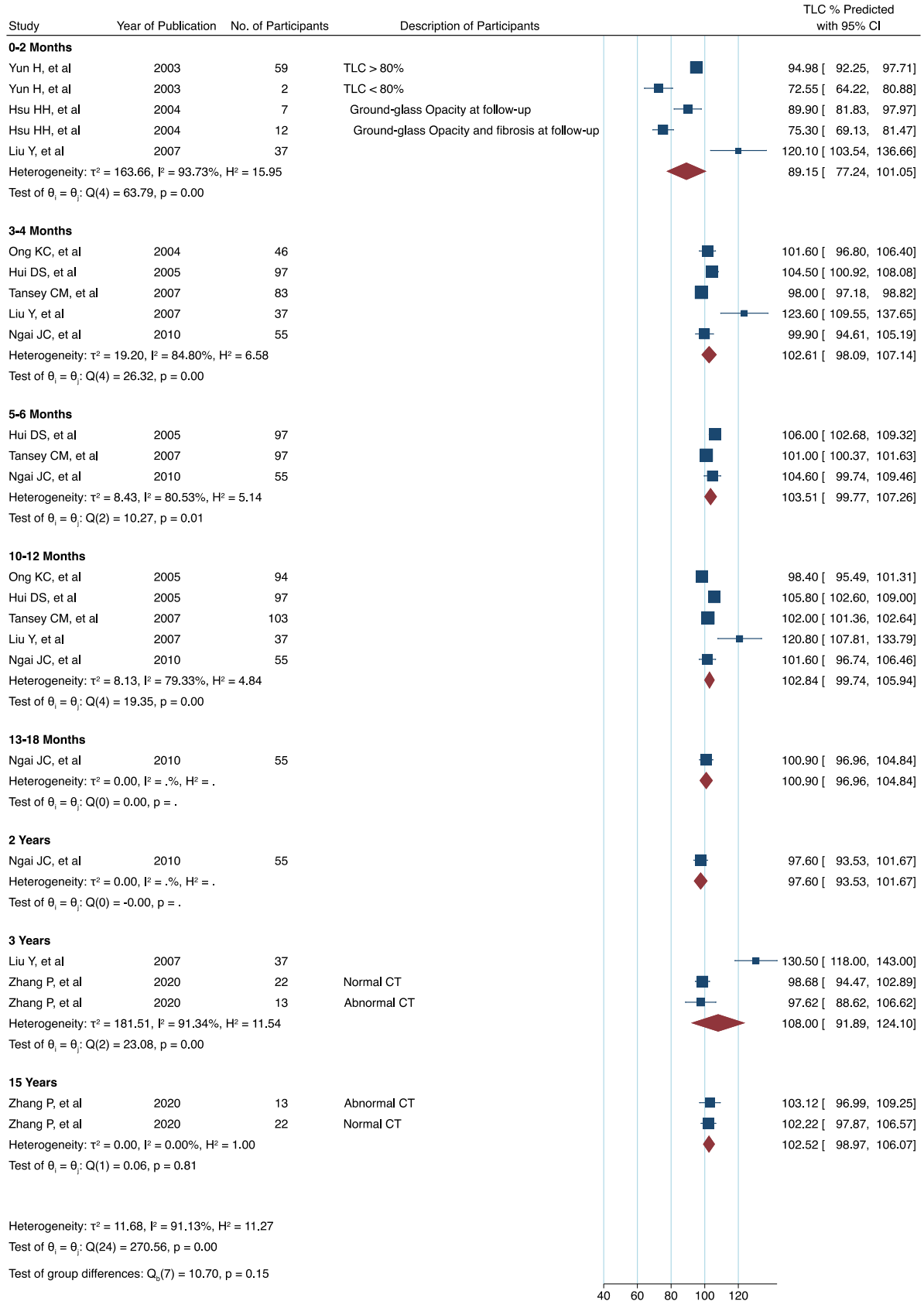


Figure S1f: Forest plot of RV % predicted by follow-up time after SARS infection

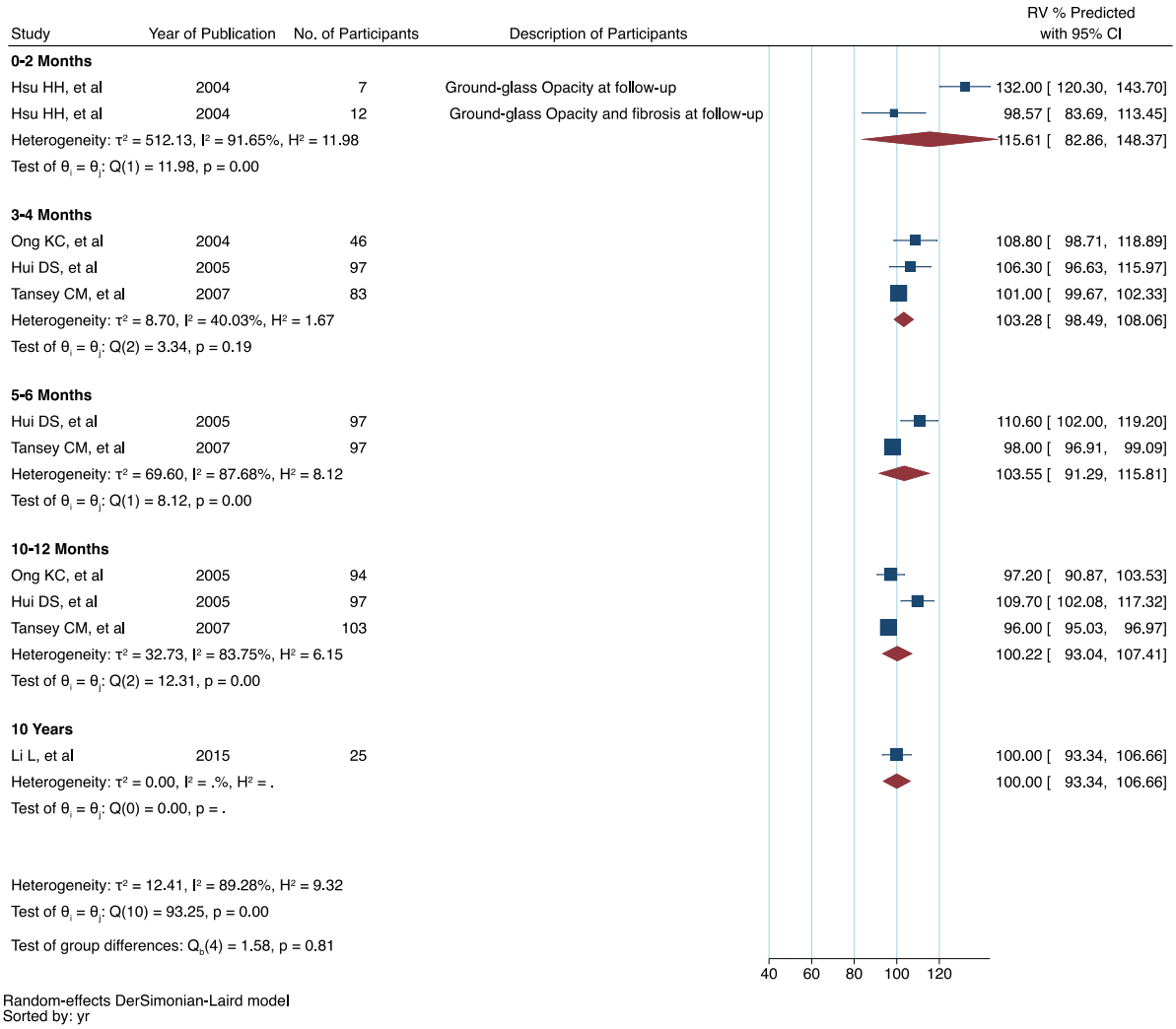


Figure S2a: Forest plot of FEV1 % predicted by follow-up time after MERS infection

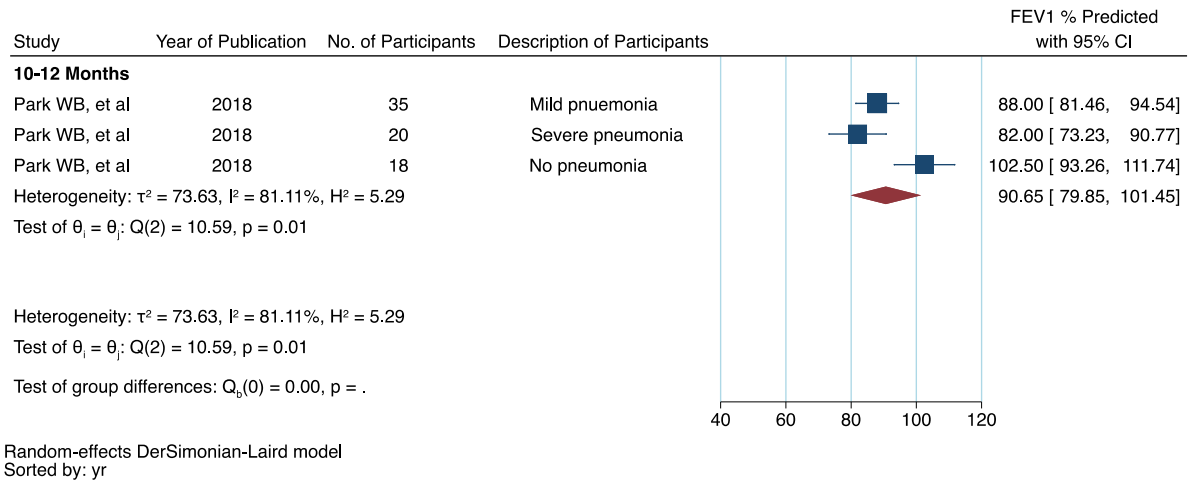


Figure S2b: Forest plot of FVC % predicted by follow-up time after MERS infection

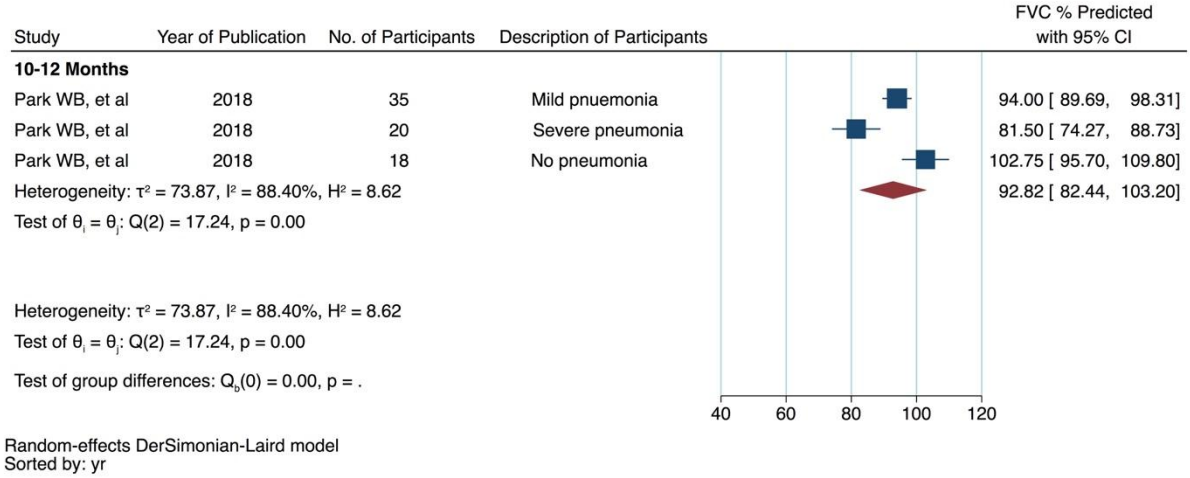


Figure S2c: Forest plot of DLco % predicted by follow-up time after MERS infection

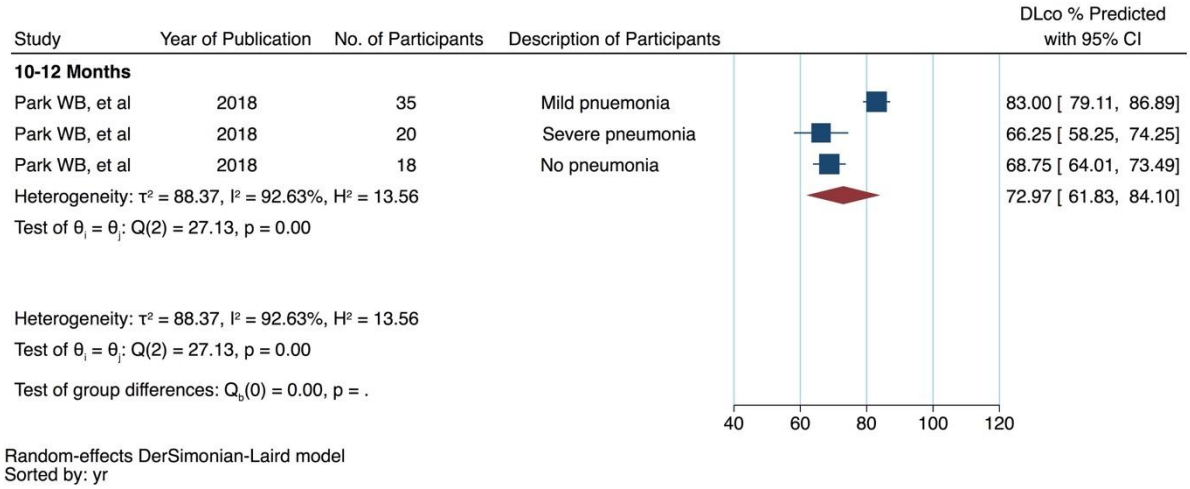


Figure S2d: Forest plot of TLC % predicted by follow-up time after MERS infection

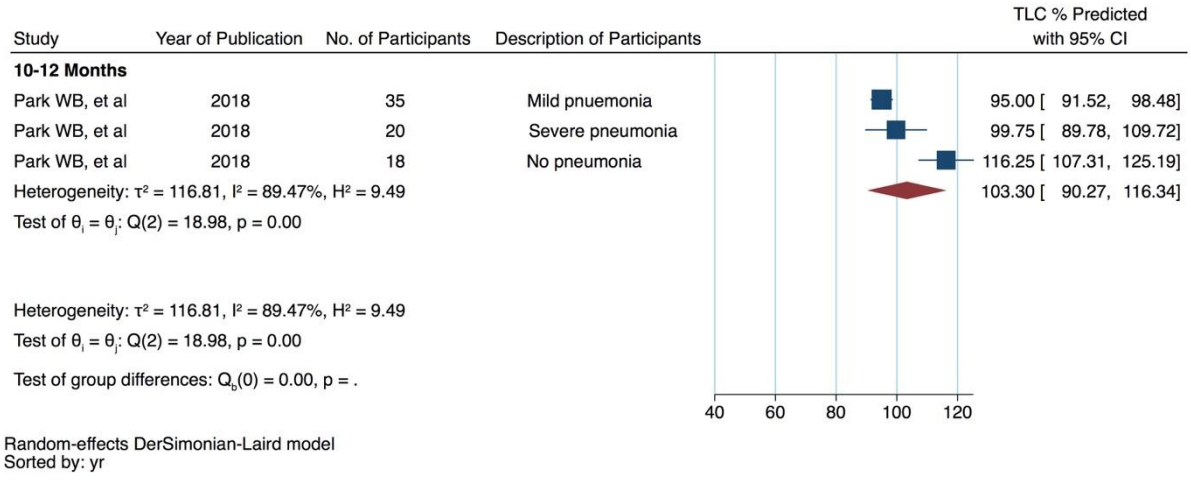


Figure S3a: Forest plot of FEV1 % predicted by follow-up time after COVID-19 infection

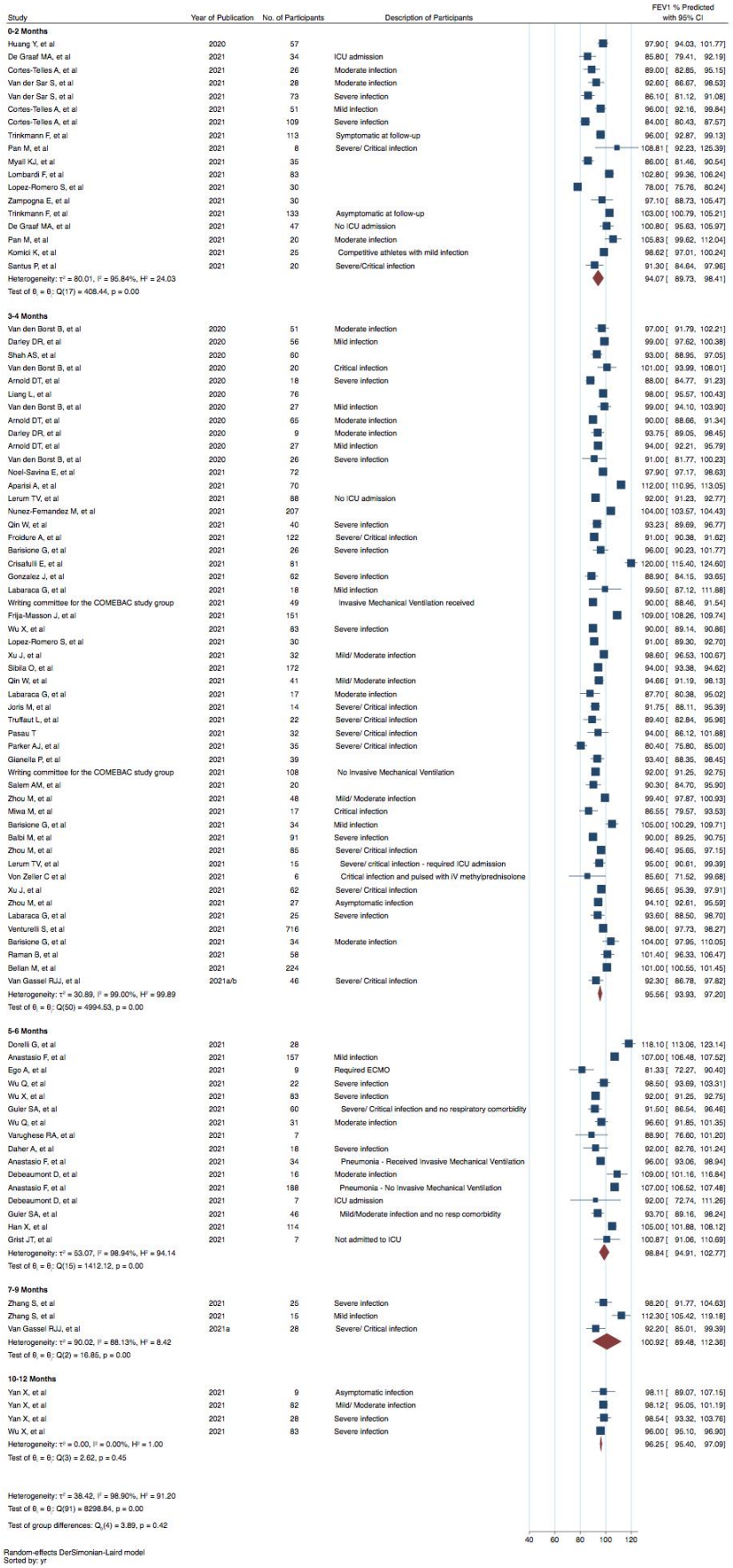


Figure S3b: Forest plot of FVC % predicted by follow-up time after COVID-19 infection

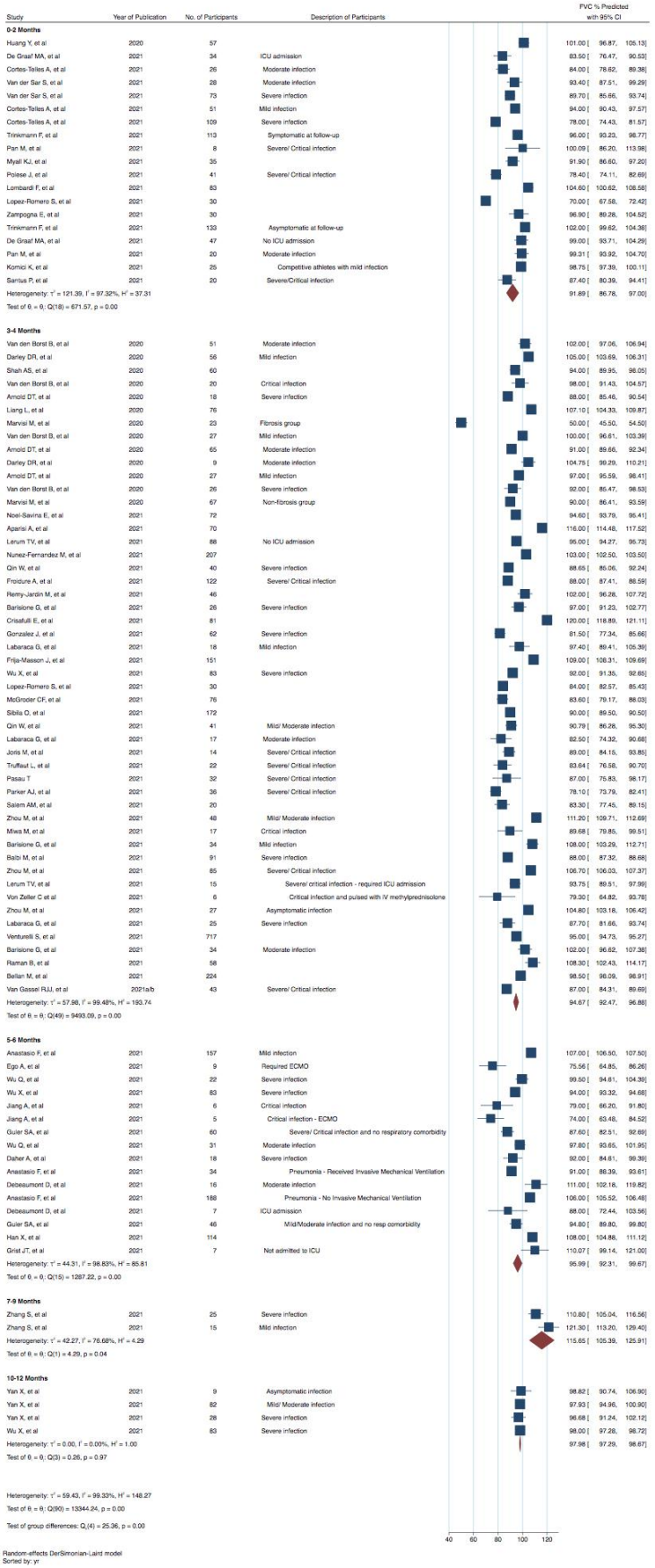


Figure S3c: Forest plot of DLco % predicted by follow-up time after COVID-19 infection

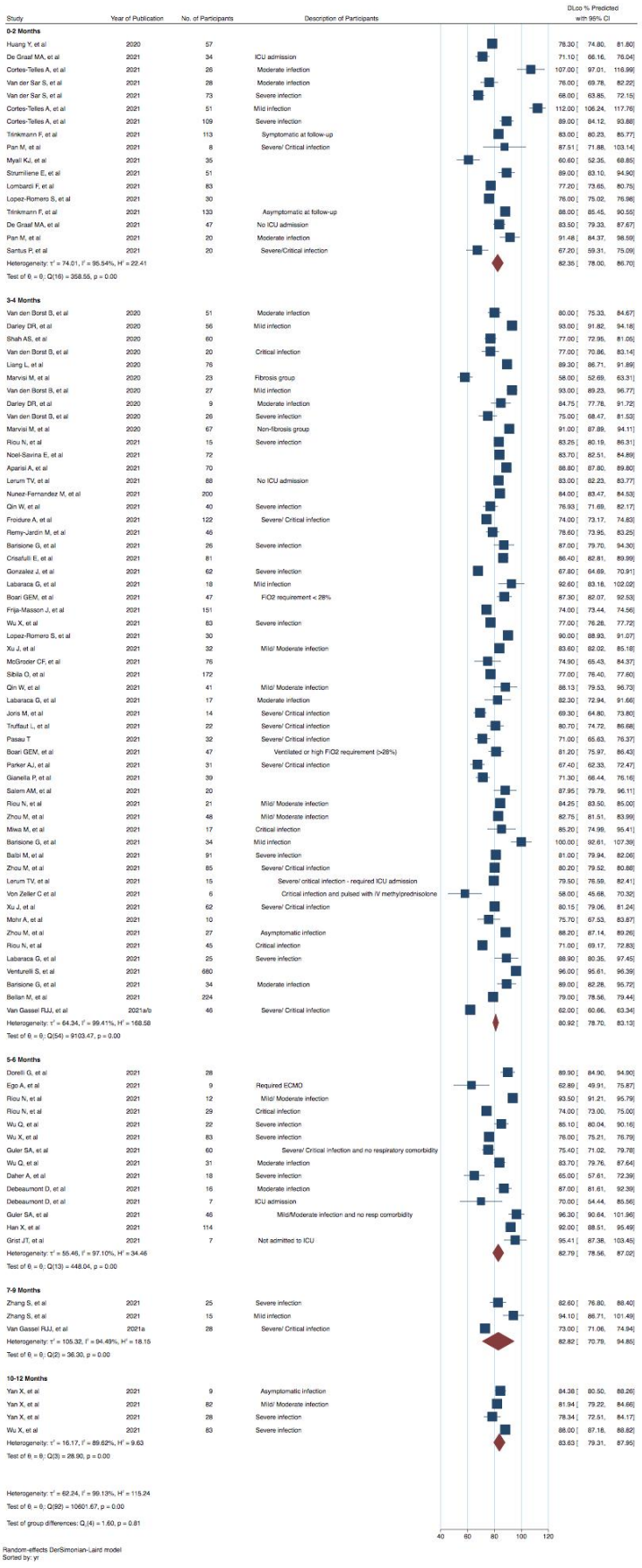


Figure S3d: Forest plot of Kco % predicted by follow-up time after COVID-19 infection

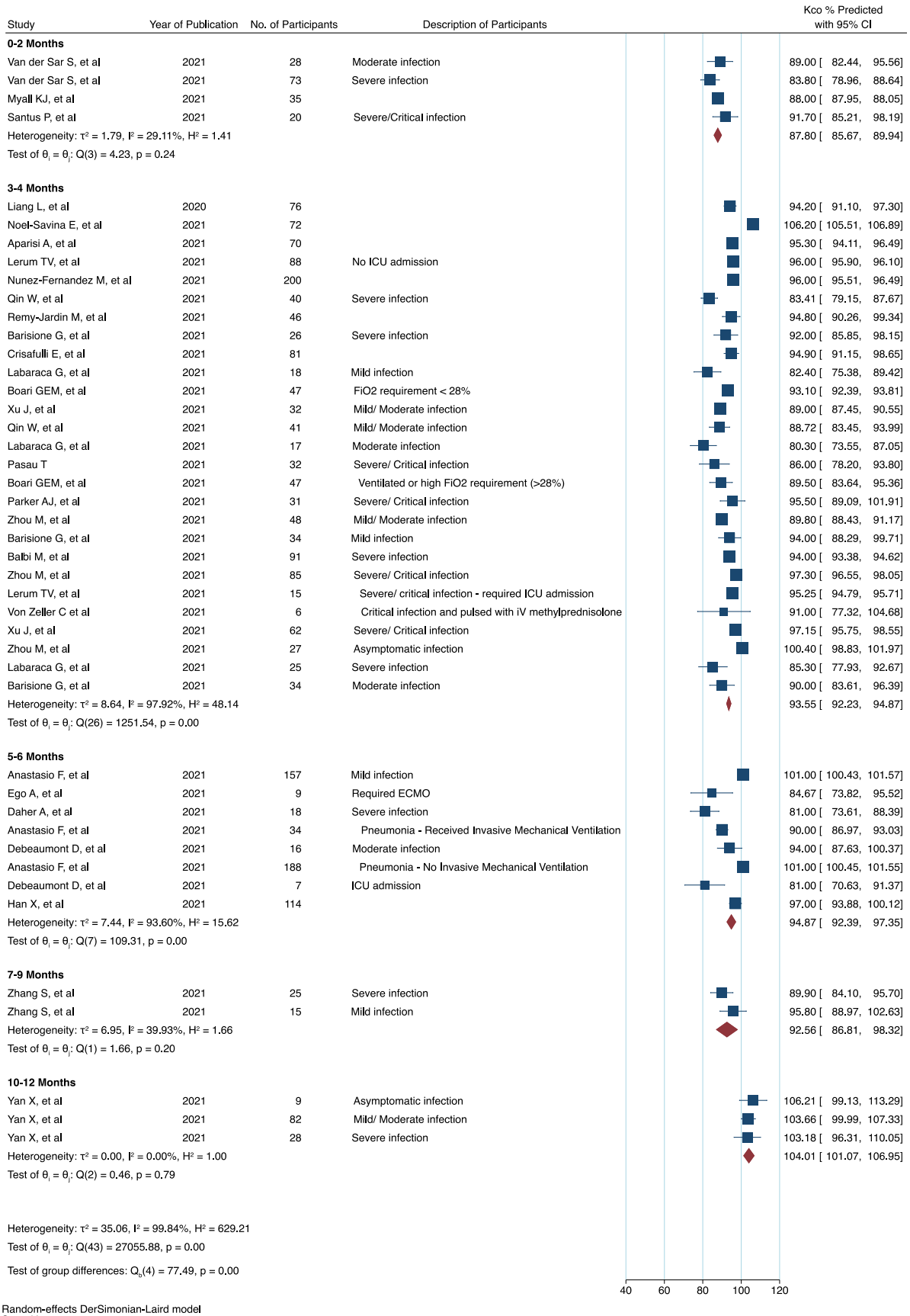


Figure S3e: Forest plot of TLC % predicted by follow-up time after COVID-19 infection

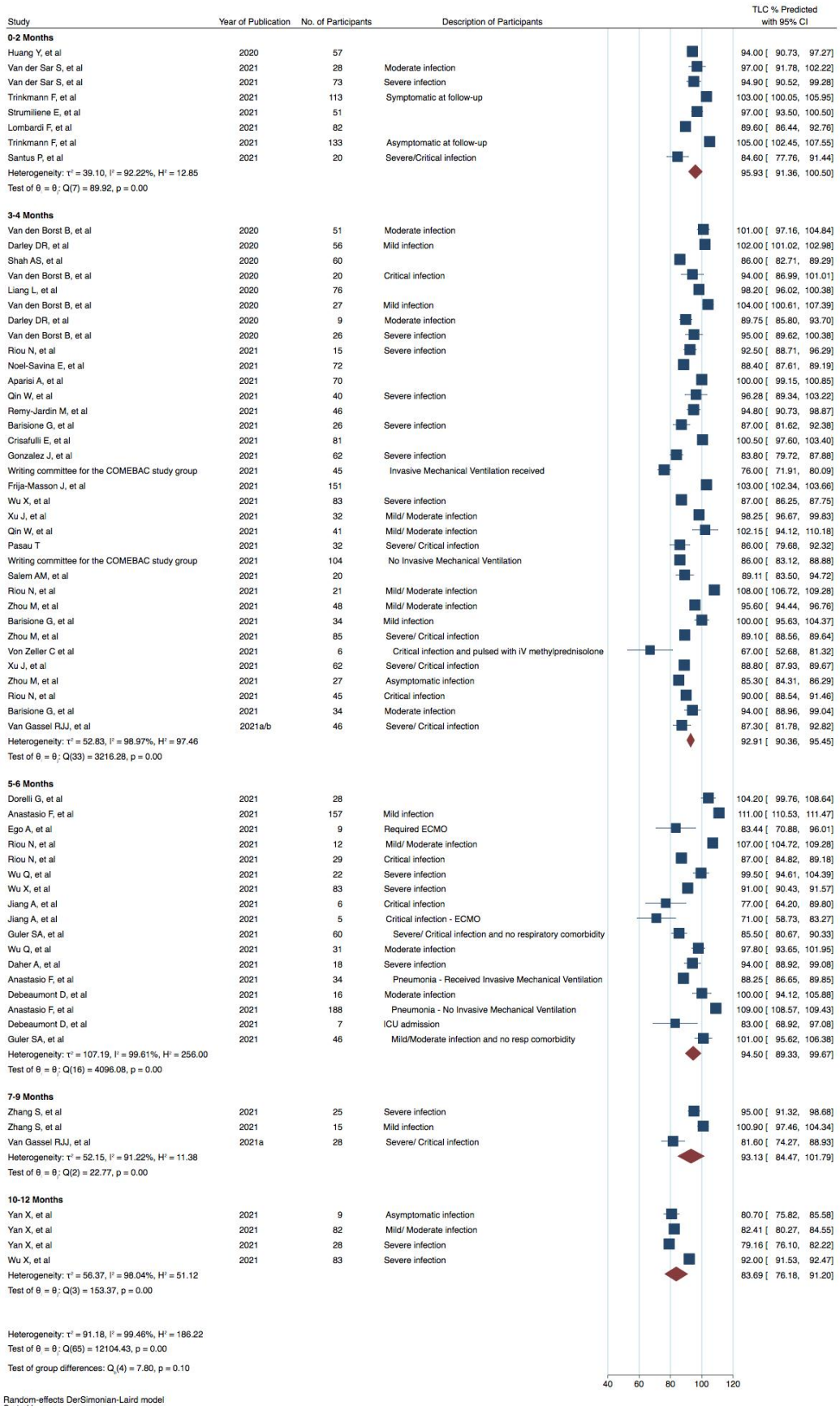


Figure S3f: Forest plot of RV % predicted by follow-up time after COVID-19 infection

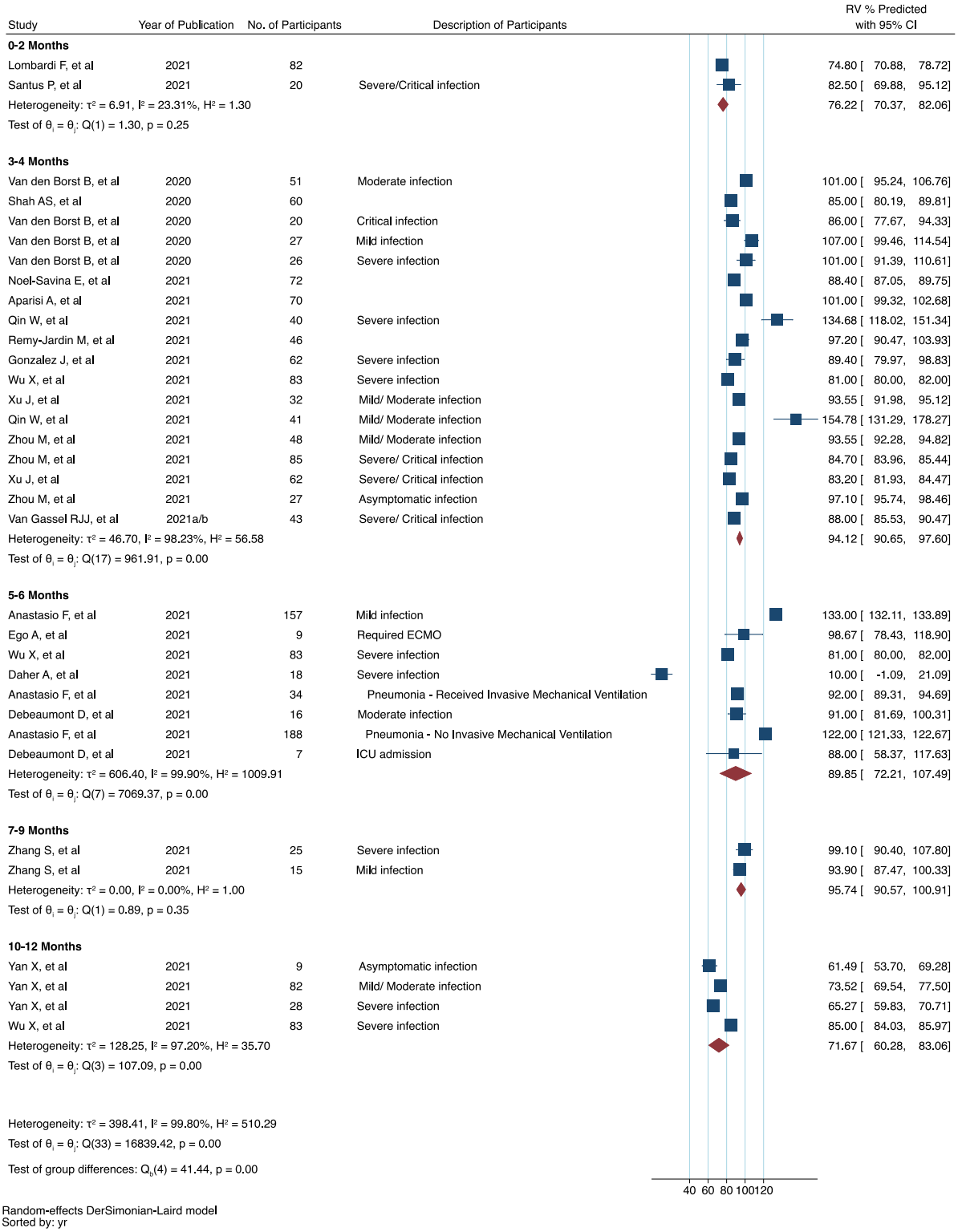


Figure S4a: Forest plot of FEV1 % predicted by follow-up time after mild/ moderate COVID-19 infection

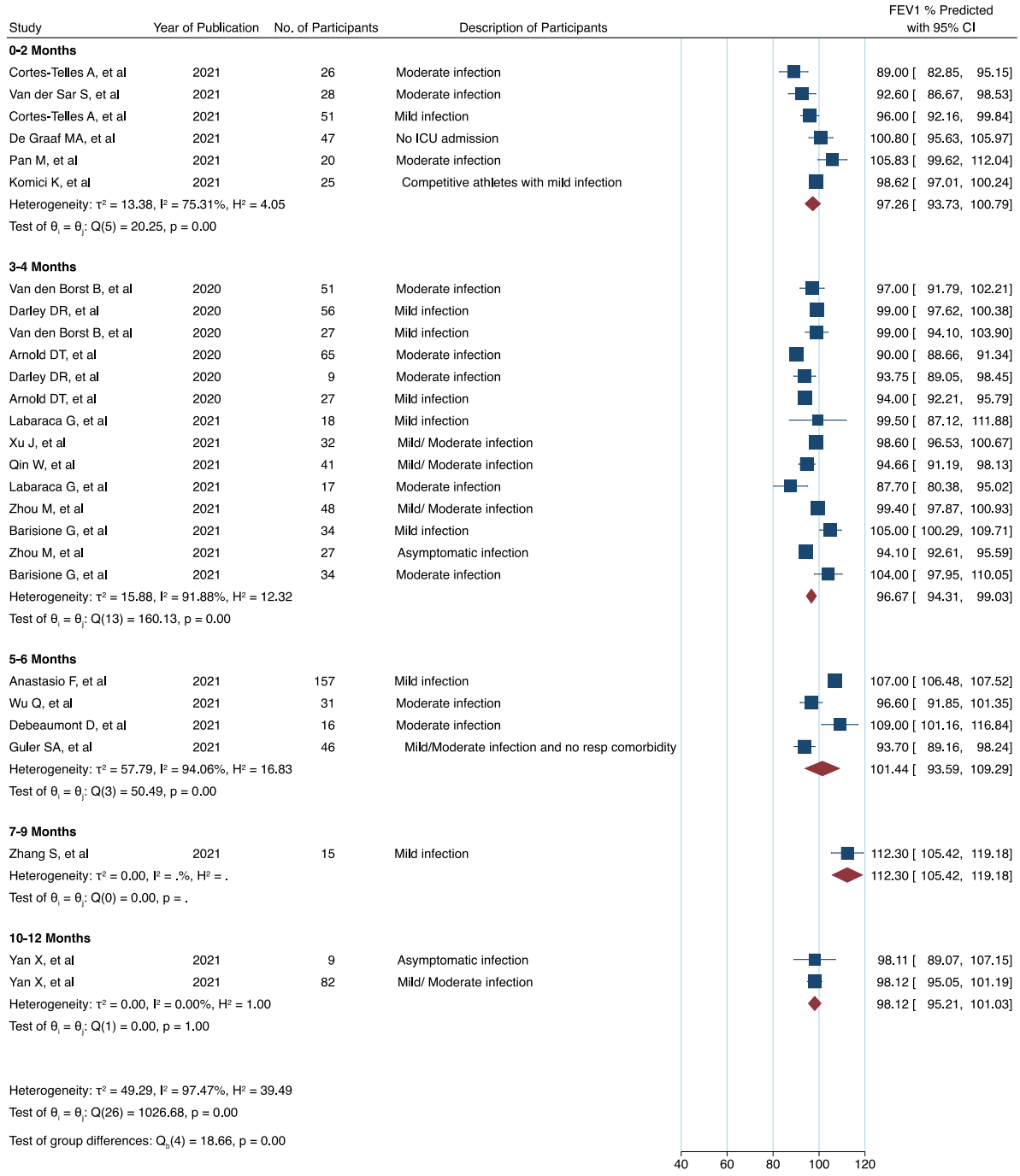
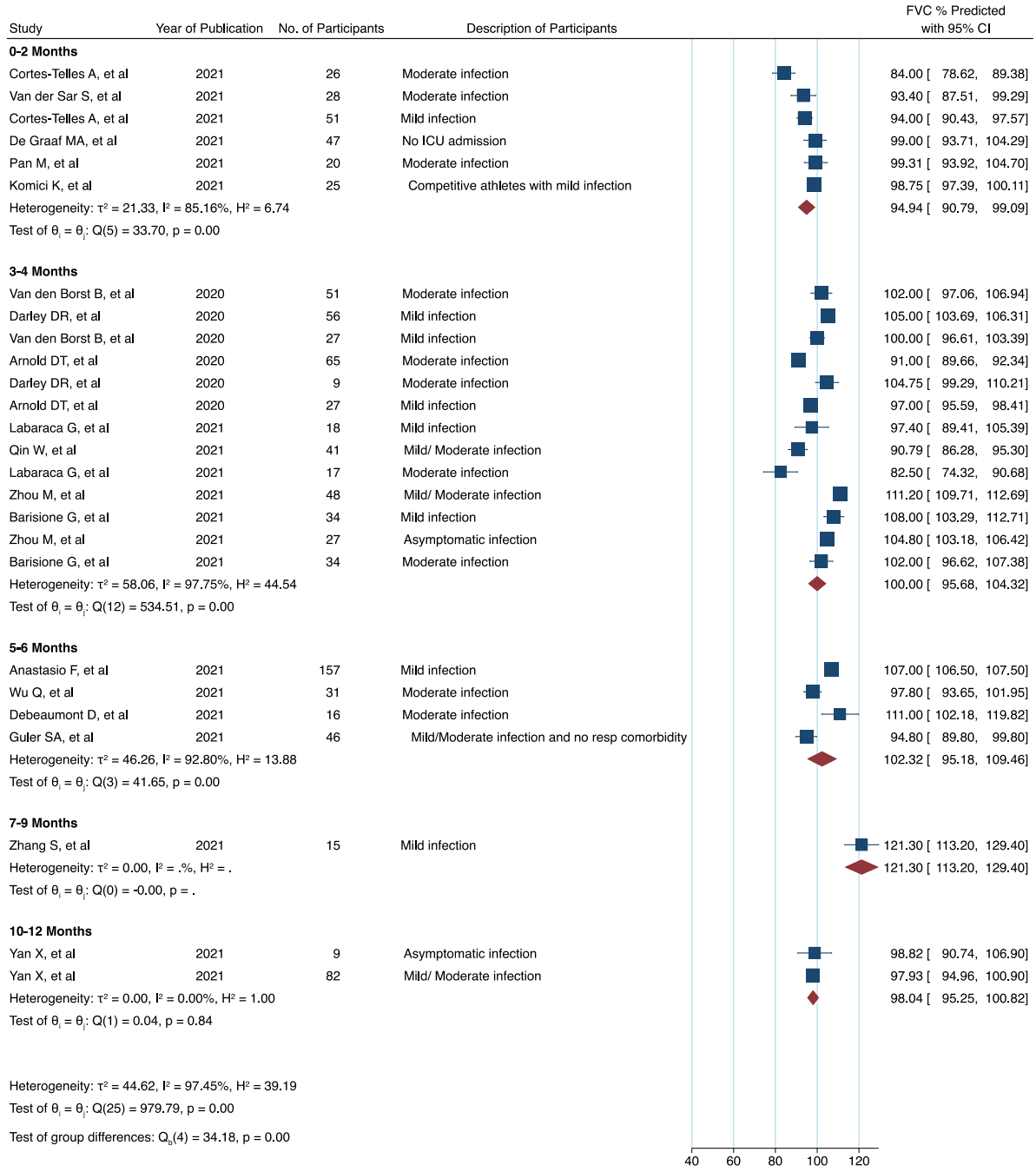


Figure S4b: Forest plot of FVC % predicted by follow-up time after mild/ moderate COVID-19 infection



Random-effects DerSimonian-Laird model
Sorted by: yr

Figure S4c: Forest plot of DLco % predicted by follow-up time after mild/ moderate COVID-19 infection

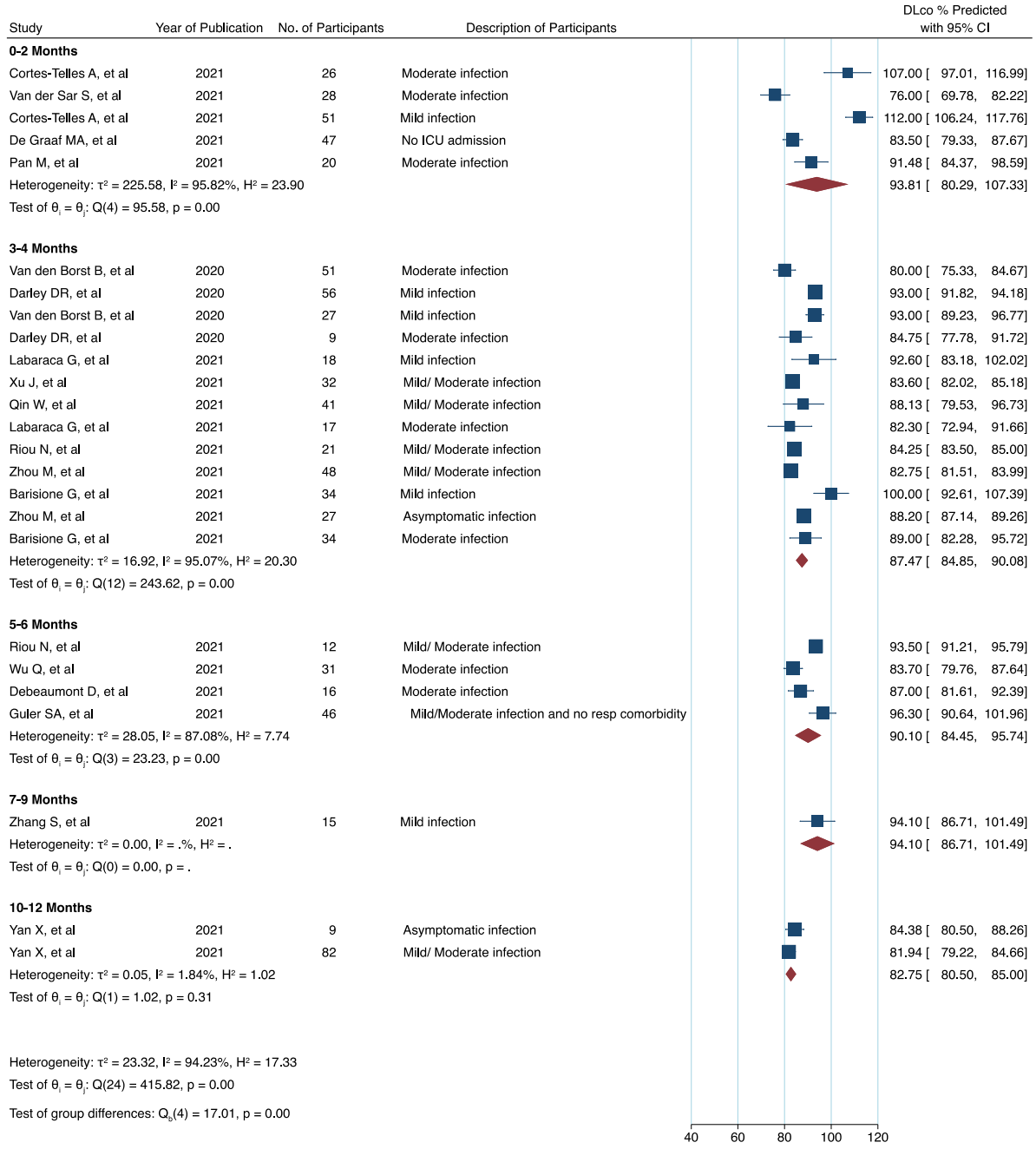
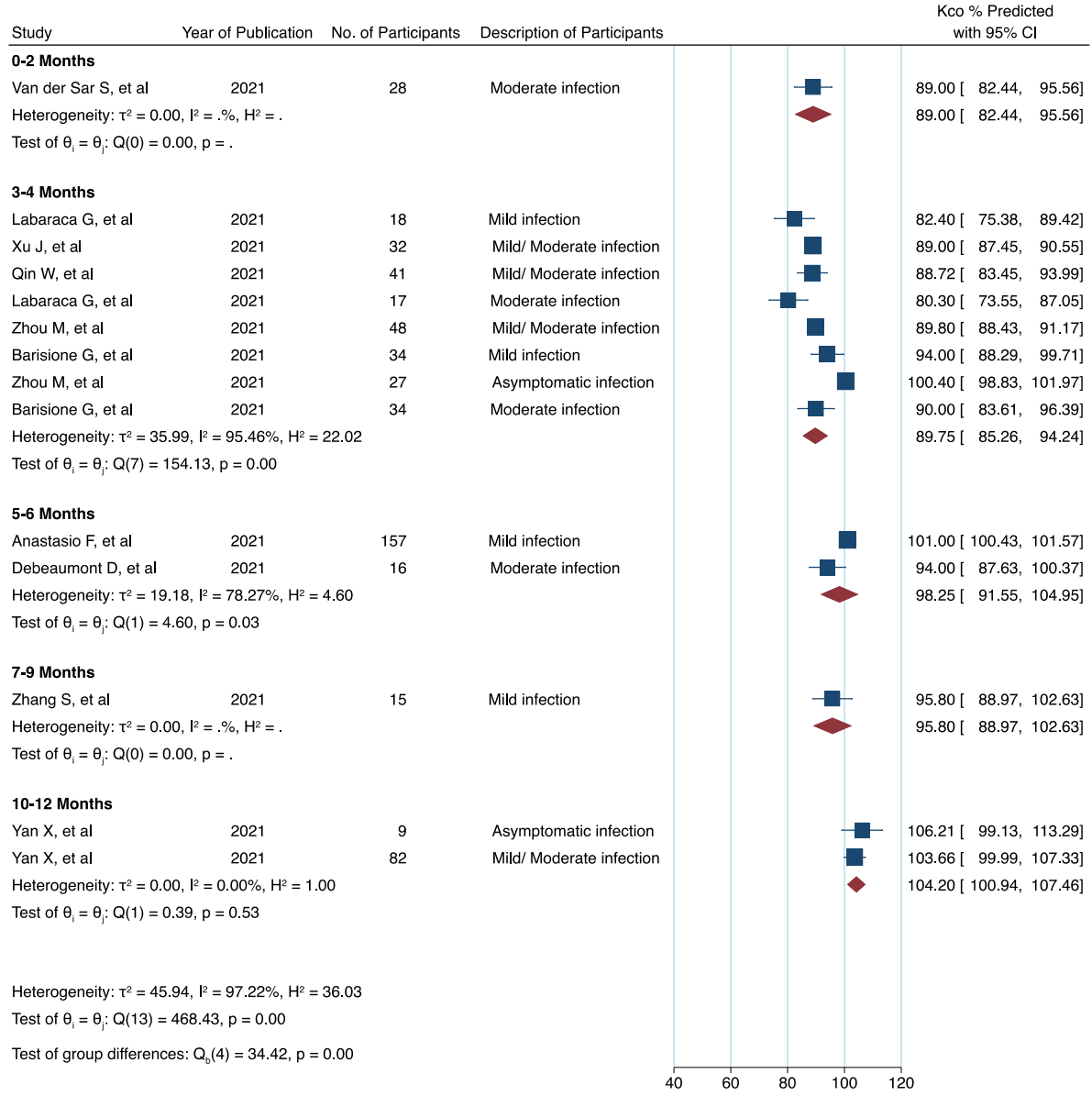


Figure S4d: Forest plot of Kco % predicted by follow-up time after mild/ moderate COVID-19 infection



Random-effects DerSimonian-Laird model
Sorted by: yr

Figure S4e: Forest plot of TLC % predicted by follow-up time after mild/ moderate COVID-19 infection

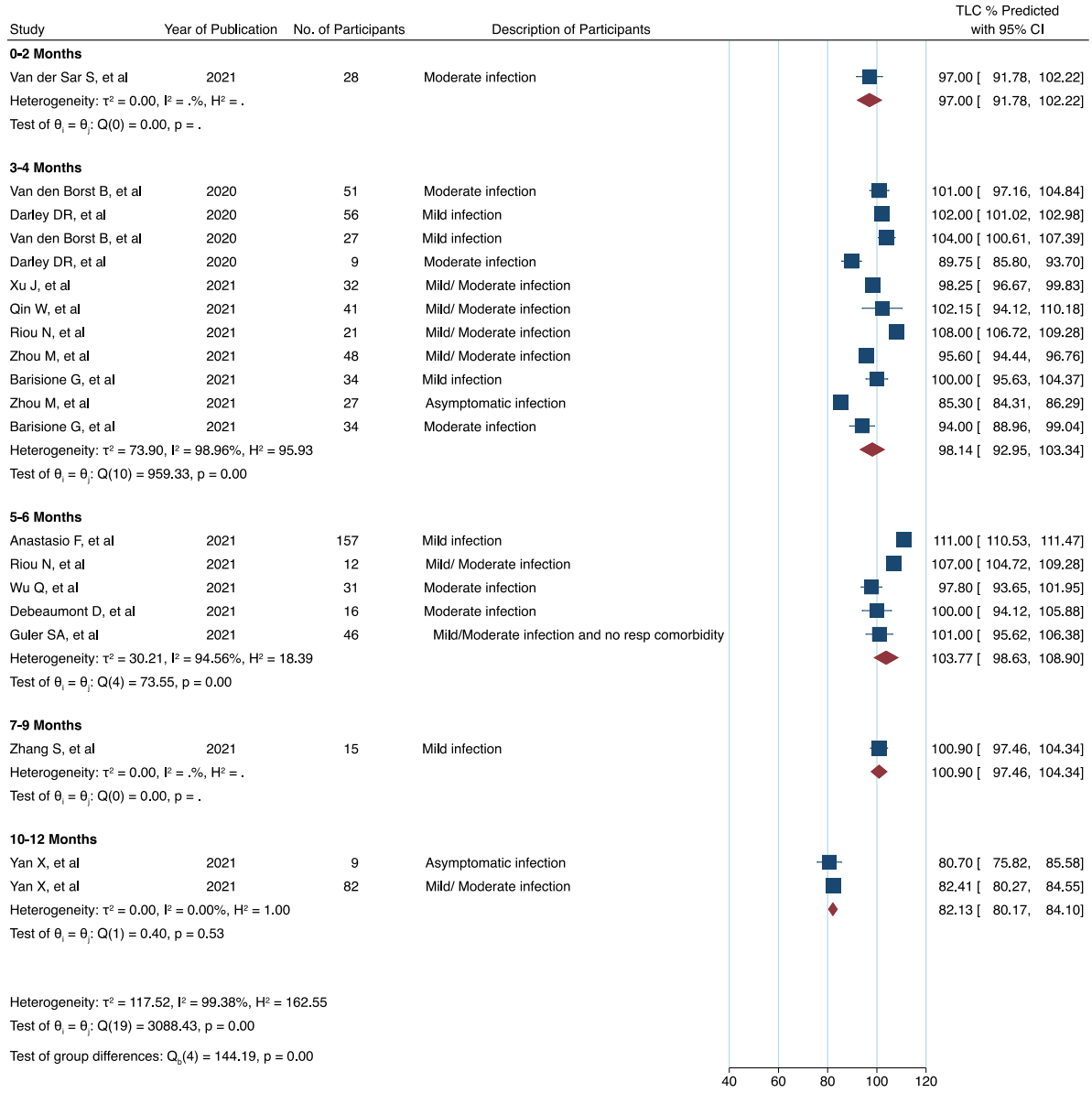
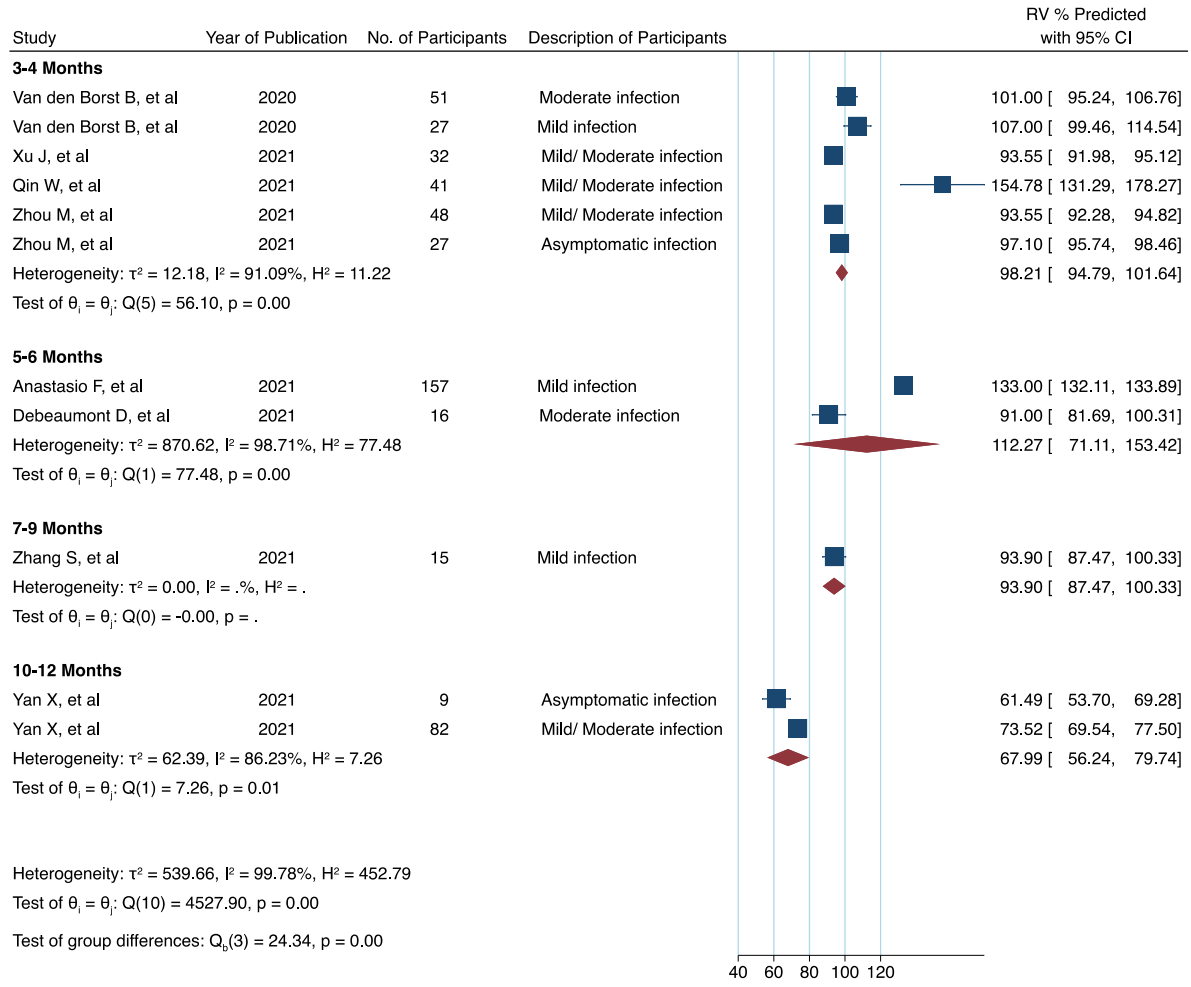


Figure S4f: Forest plot of RV % predicted by follow-up time after mild/ moderate COVID-19 infection



Random-effects DerSimonian-Laird model
Sorted by: yr

Figure S5a: Forest plot of FEV1 % predicted by follow-up time after severe/ critical COVID-19 infection

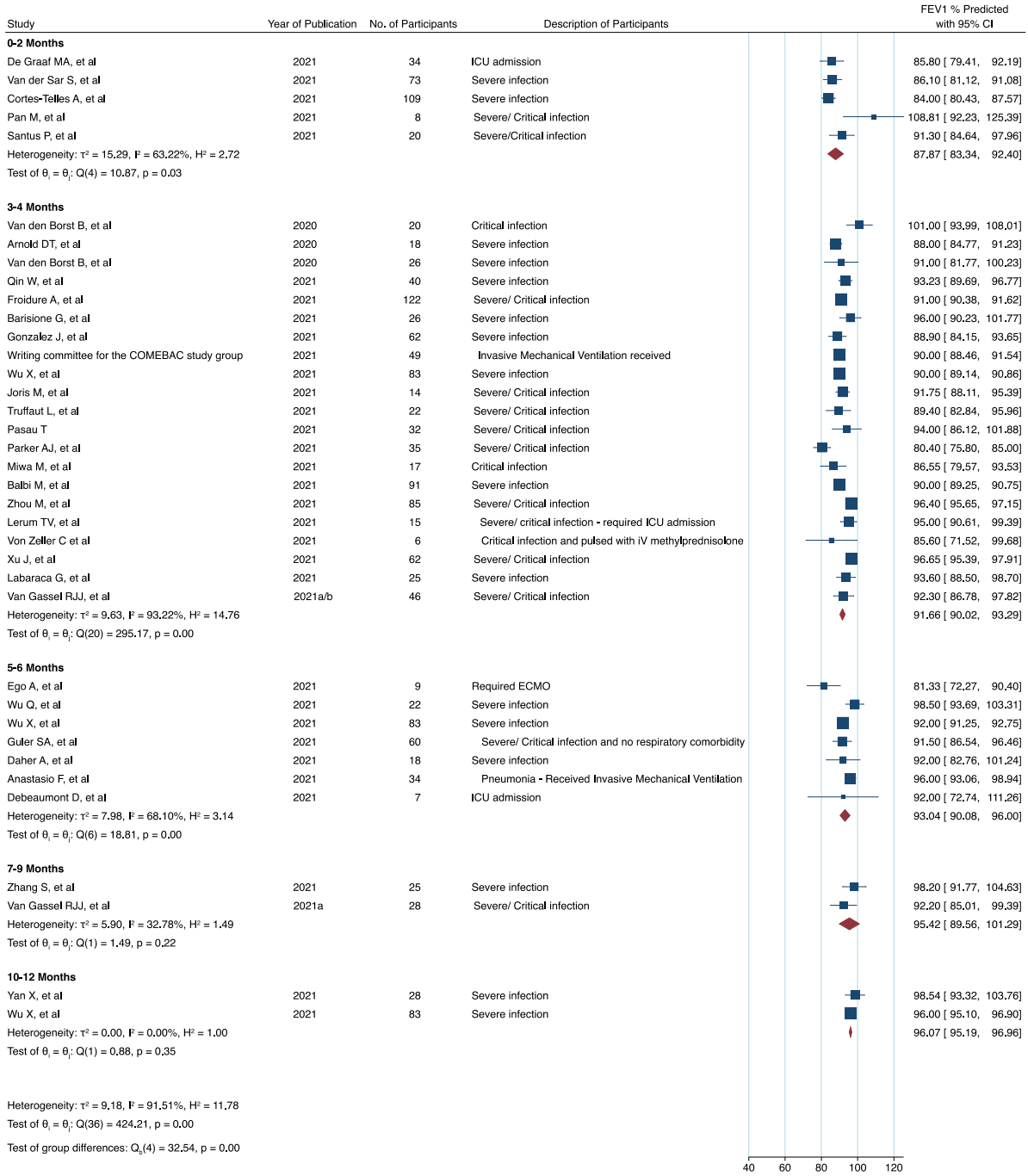


Figure S5b: Forest plot of FVC % predicted by follow-up time after severe/ critical COVID-19 infection

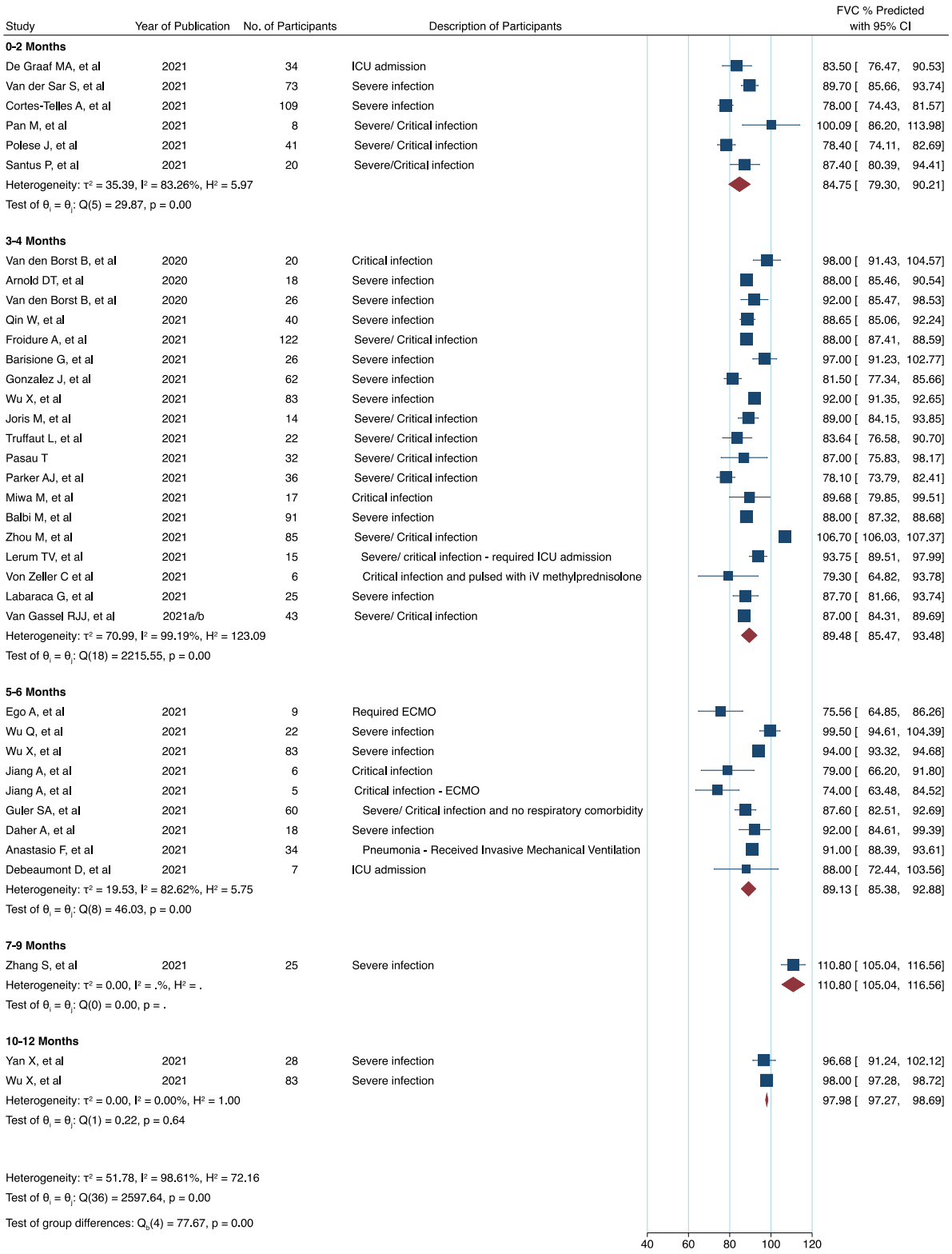


Figure S5c: Forest plot of DLco % predicted by follow-up time after severe/ critical COVID-19 infection

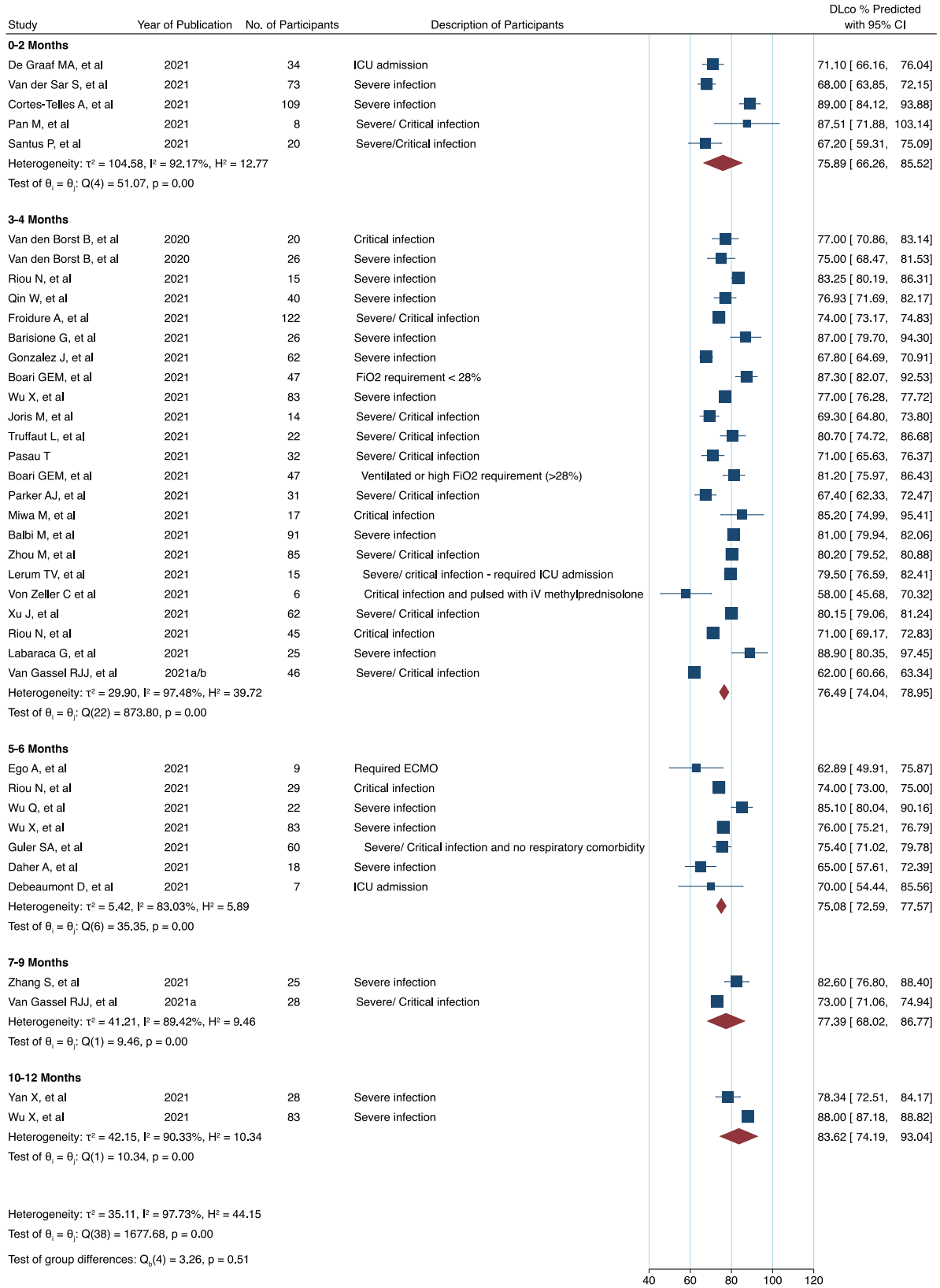


Figure S5d: Forest plot of Kco % predicted by follow-up time after severe/ critical COVID-19 infection

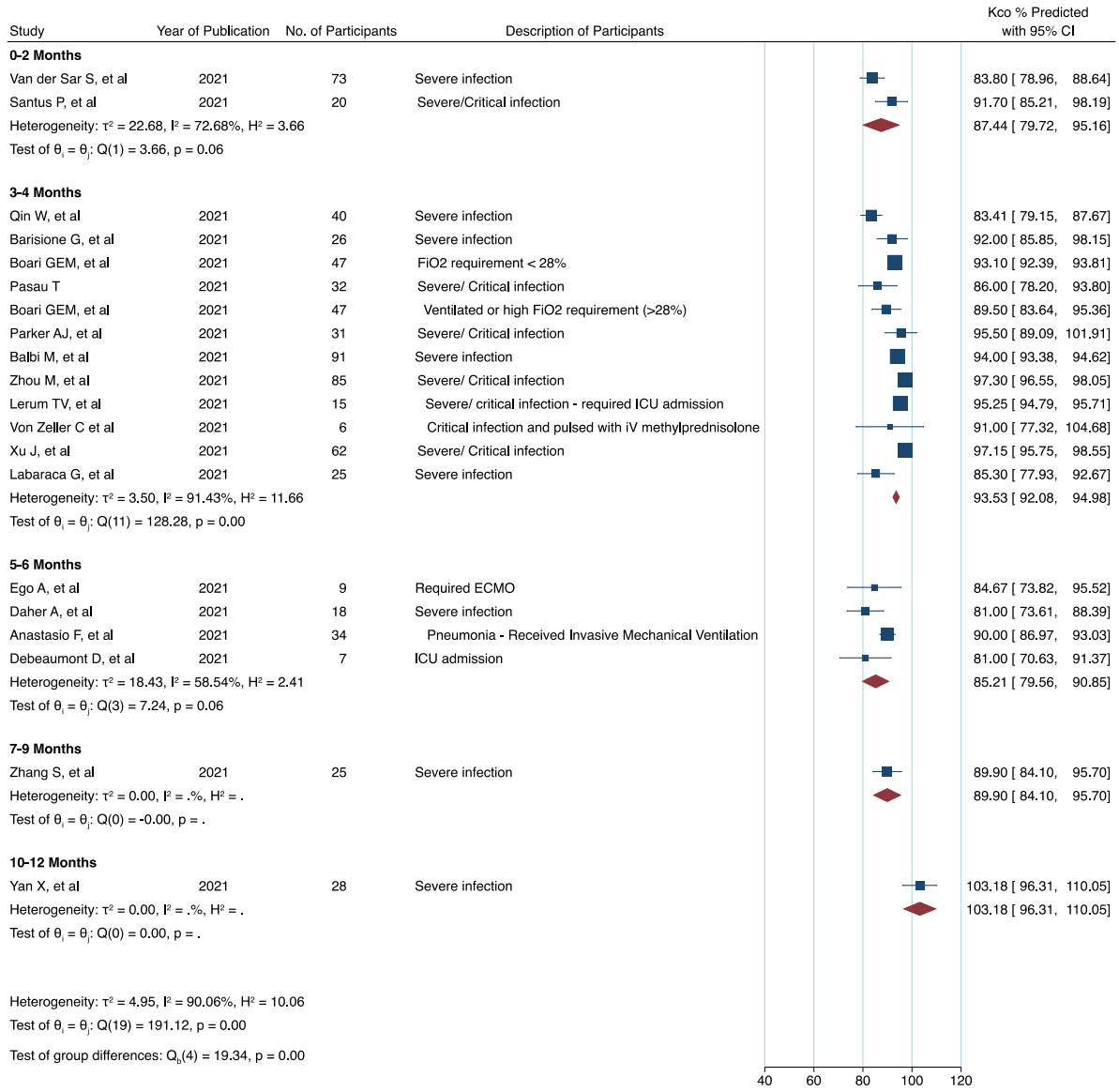


Figure S5e: Forest plot of TLC % predicted by follow-up time after severe/ critical COVID-19 infection

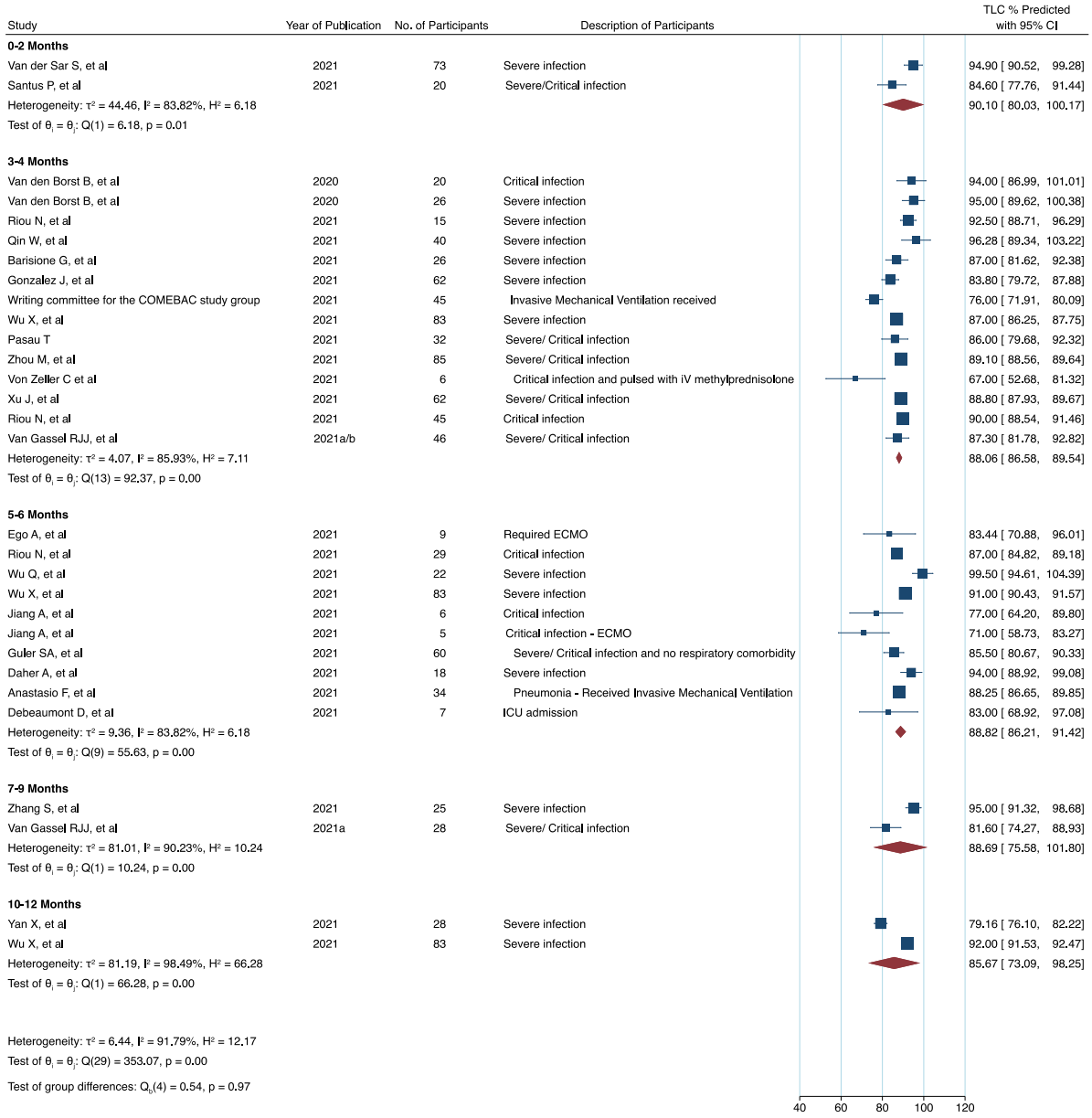
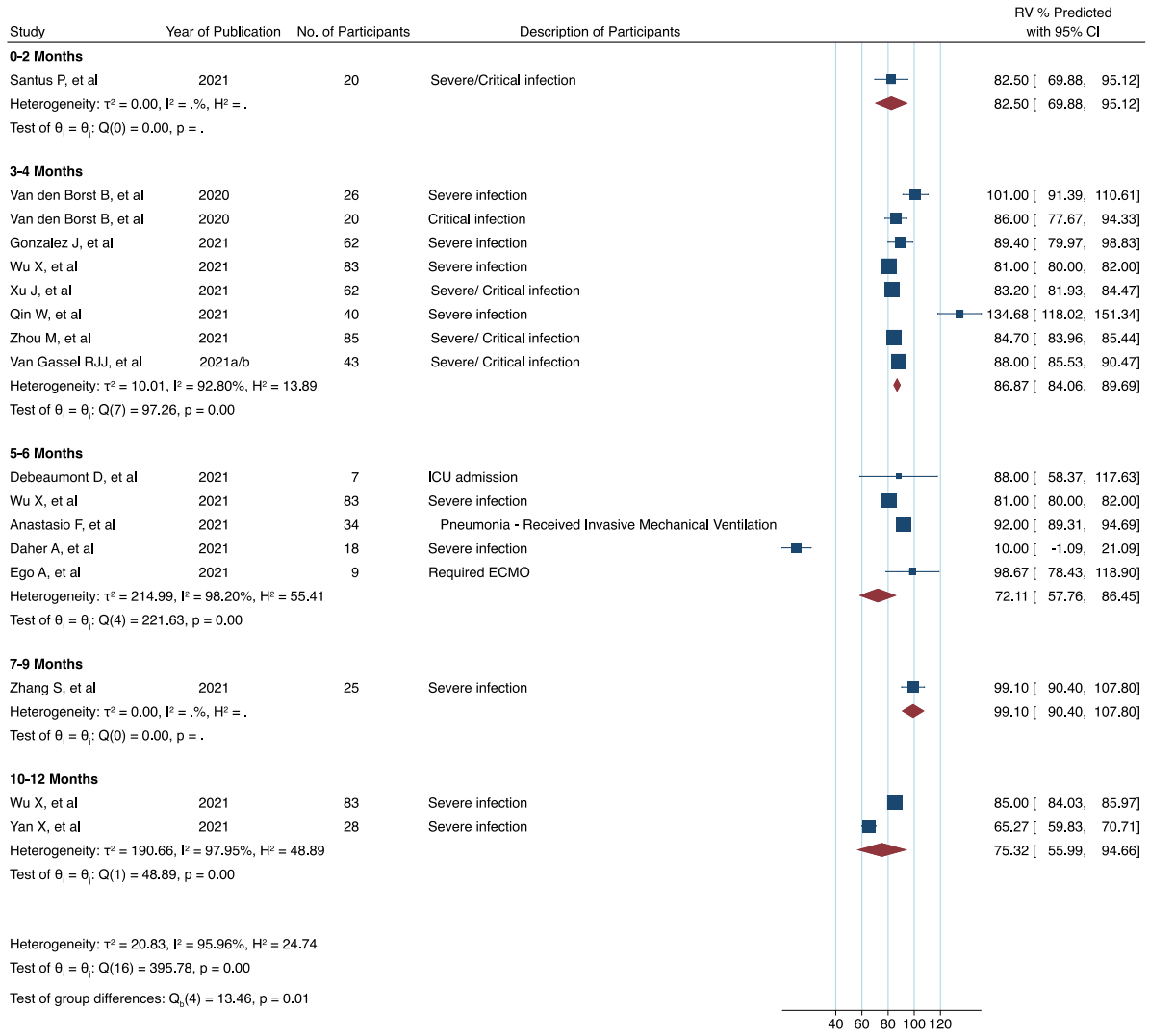


Figure S5f: Forest plot of RV % predicted by follow-up time after severe/ critical COVID-19 infection



Random-effects DerSimonian-Laird model
Sorted by: yr

Figure S6a: Forest plot of estimated proportion of follow-up CTs showing consolidation by time after SARS infection

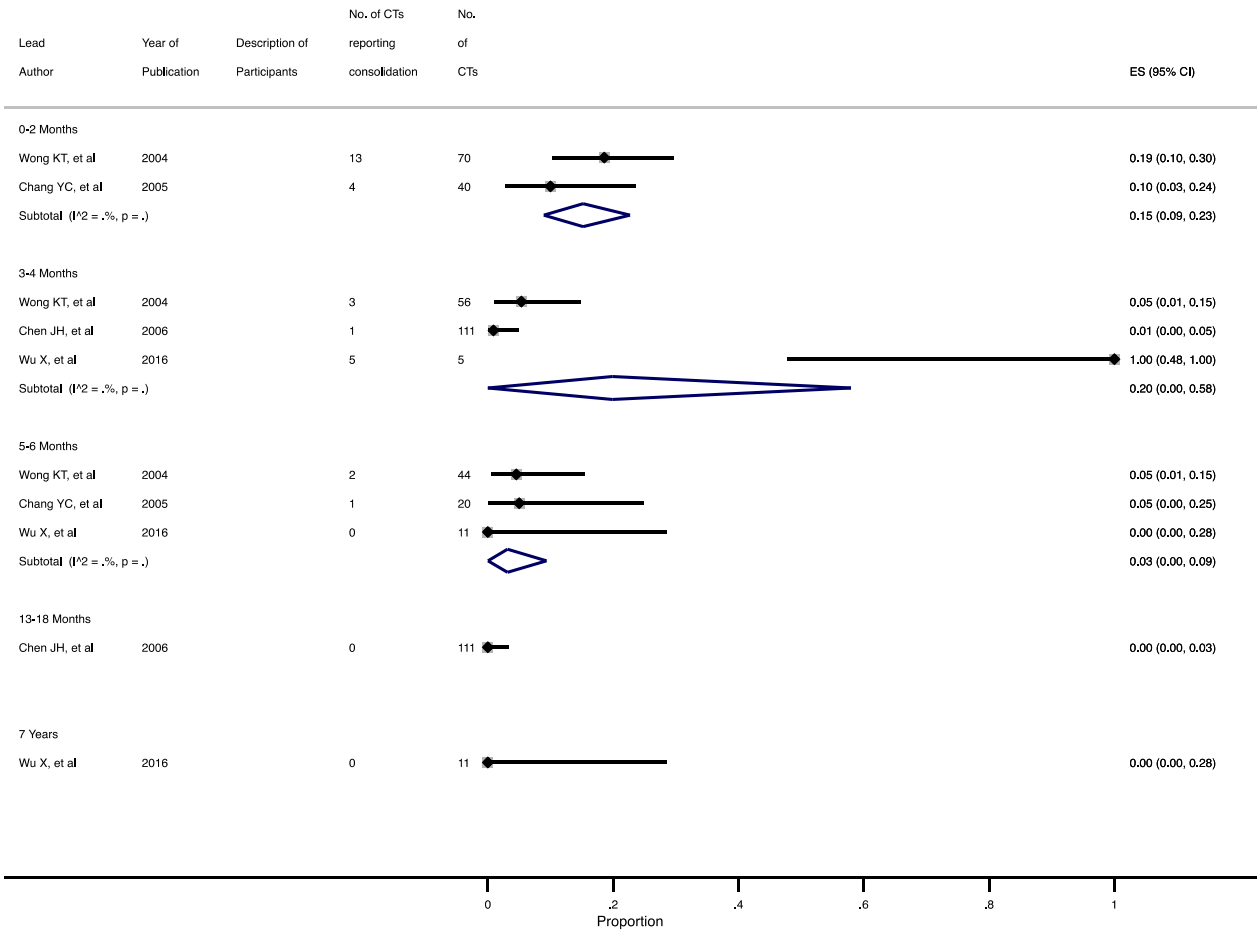


Figure S6b: Forest plot of estimated proportion of follow-up CTs showing ground-glass opacification by time after SARS infection

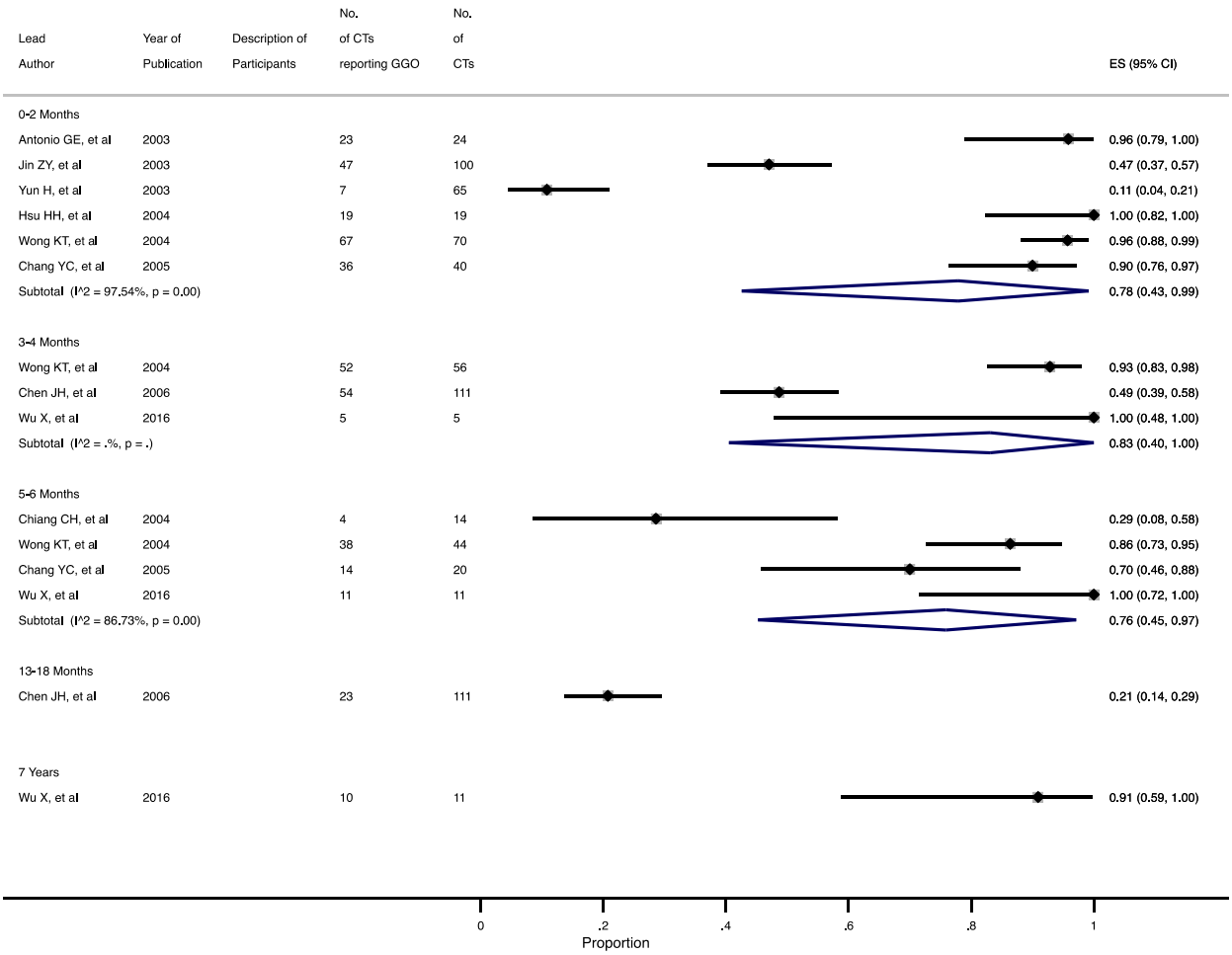


Figure S6c: Forest plot of estimated proportion of follow-up CTs showing honeycombing by time after SARS infection

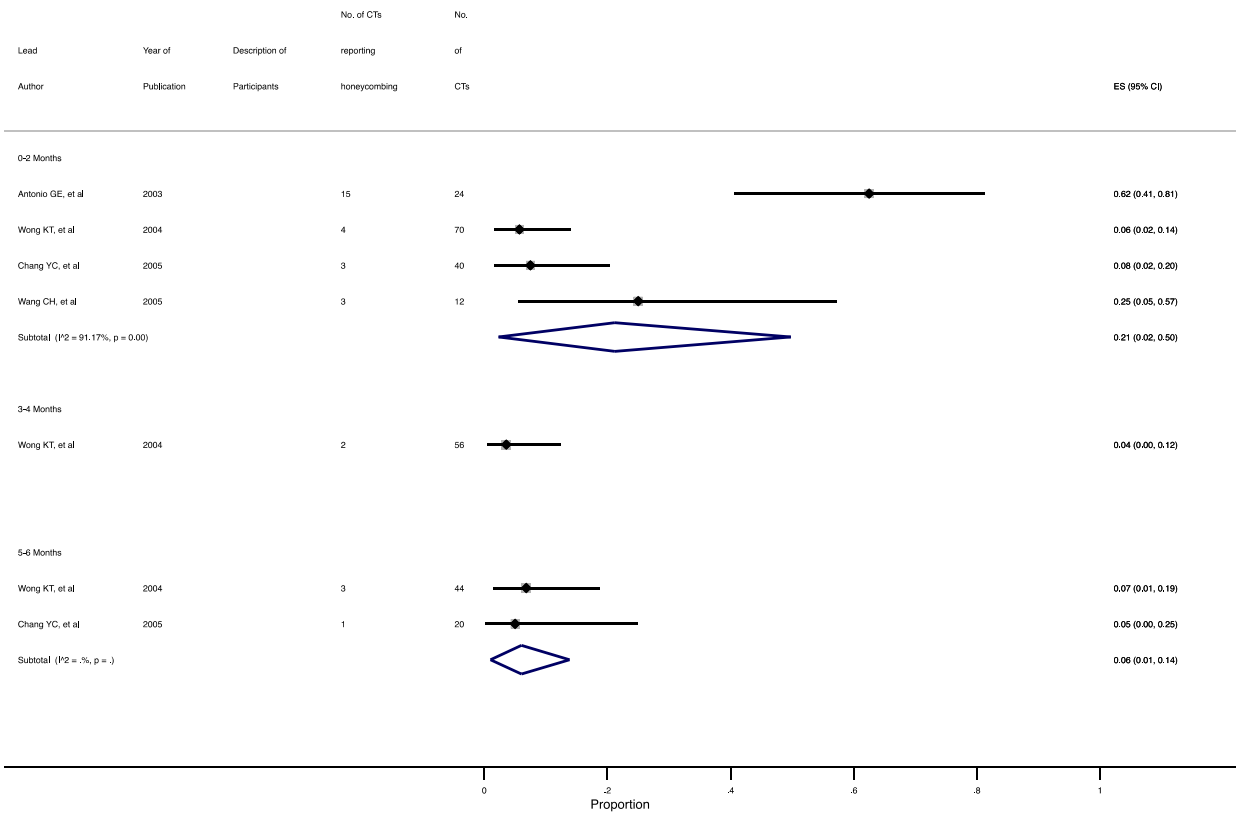


Figure S6d: Forest plot of estimated proportion of follow-up CTs showing linear opacities by time after SARS infection

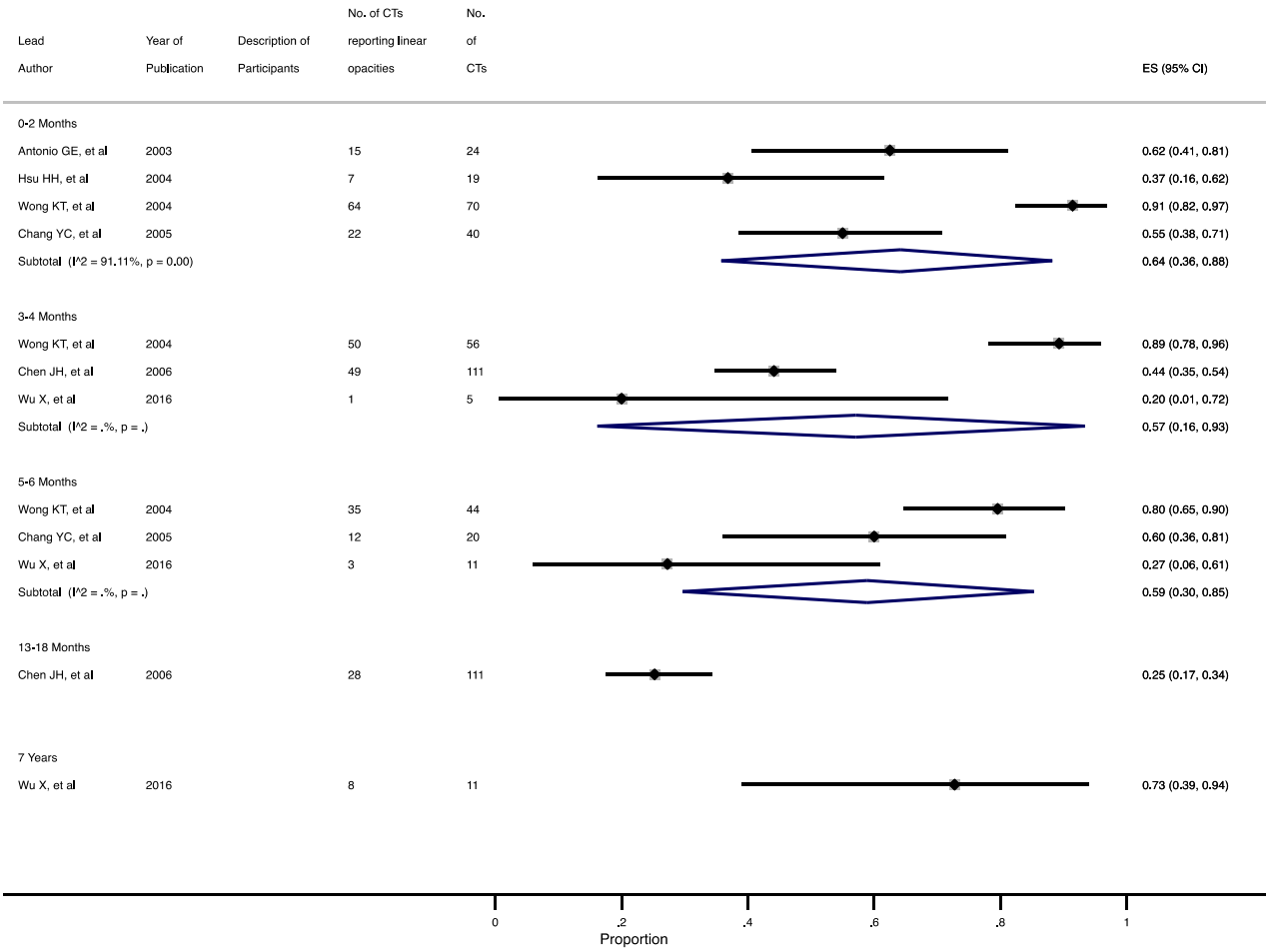


Figure S6e: Forest plot of estimated proportion of follow-up CTs showing reticulation by time after SARS infection

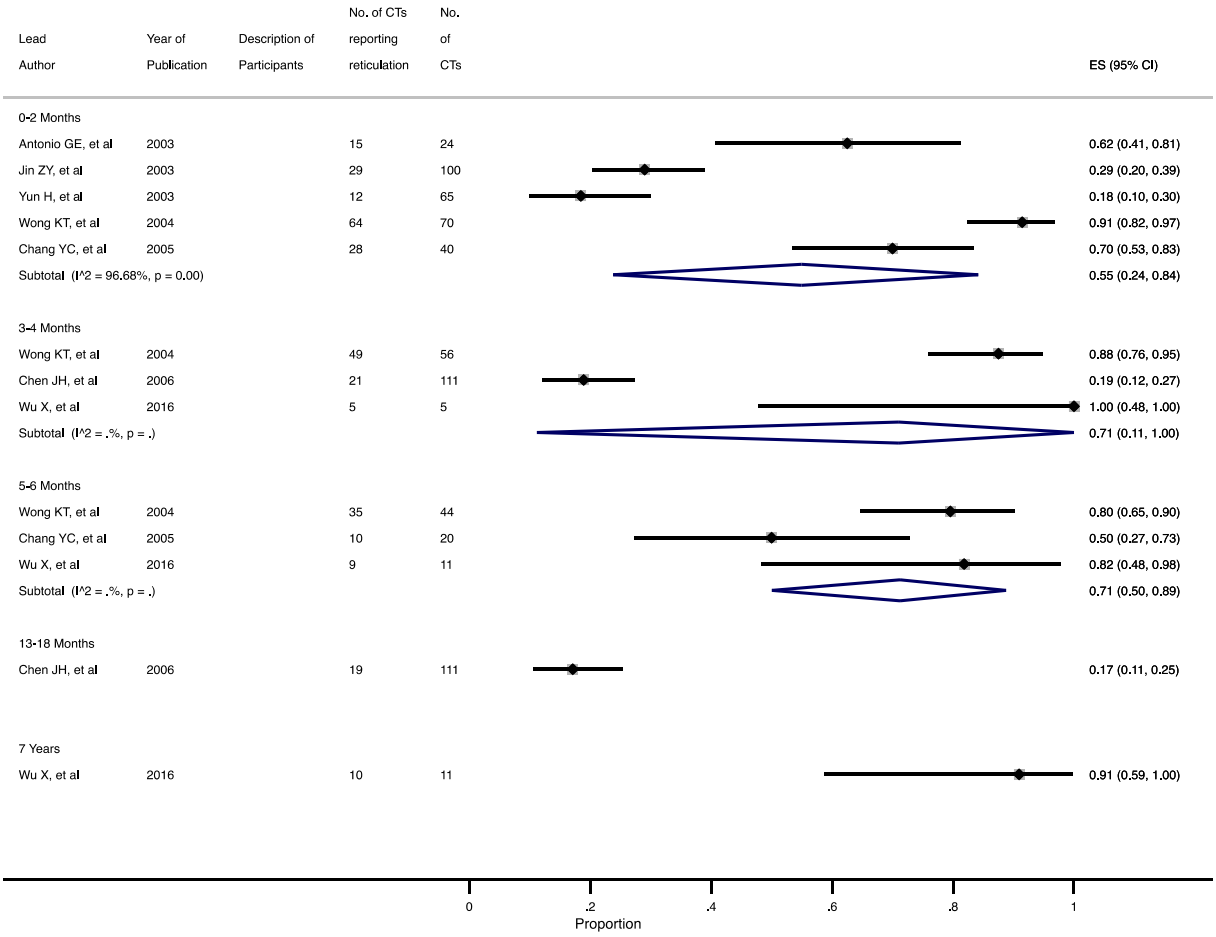


Figure S6f: Forest plot of estimated proportion of follow-up CTs showing traction bronchiectasis & bronchiolectasis by time after SARS infection

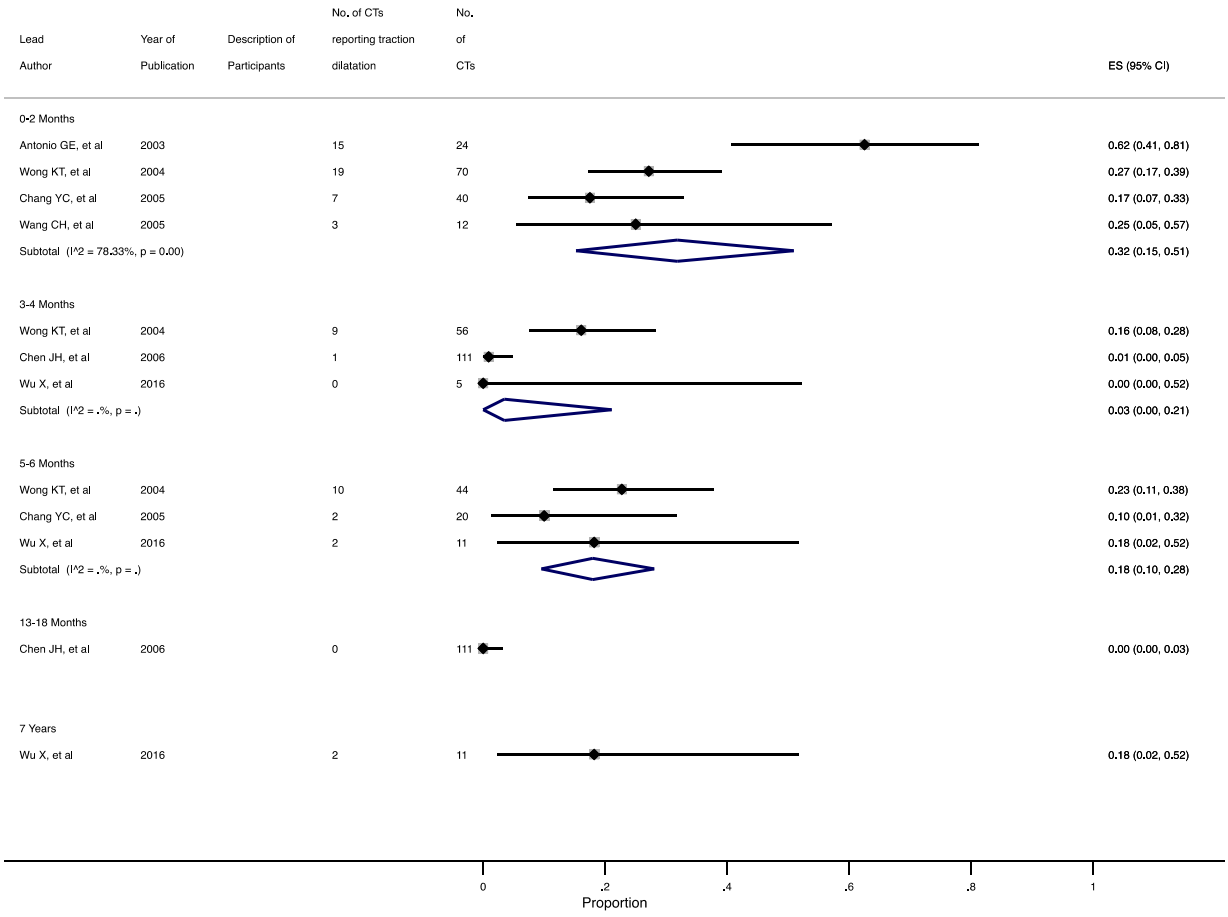


Figure S6g: Forest plot of estimated proportion of follow-up CTs showing non ground glass opacity nodules by time after SARS infection

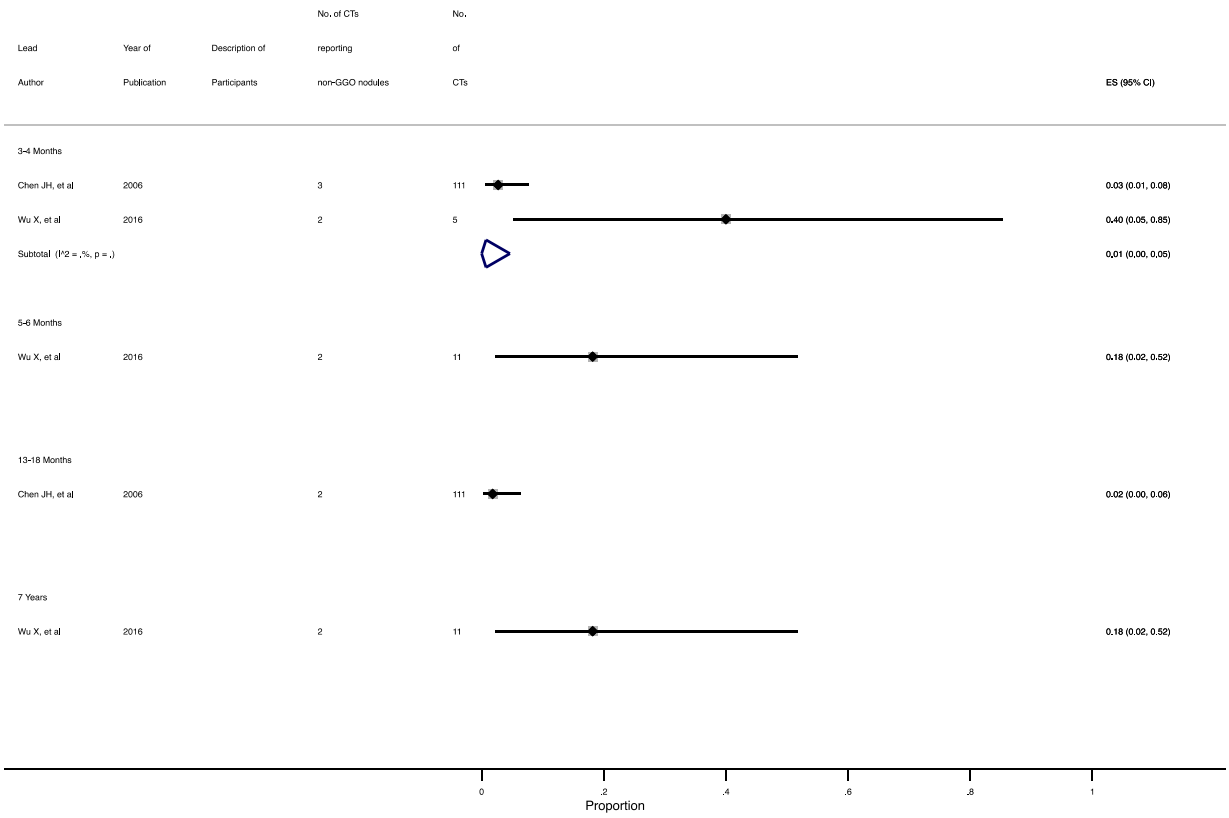


Figure S7a: Forest plot of estimated proportion of follow-up CTs showing consolidation by time after COVID-19 infection

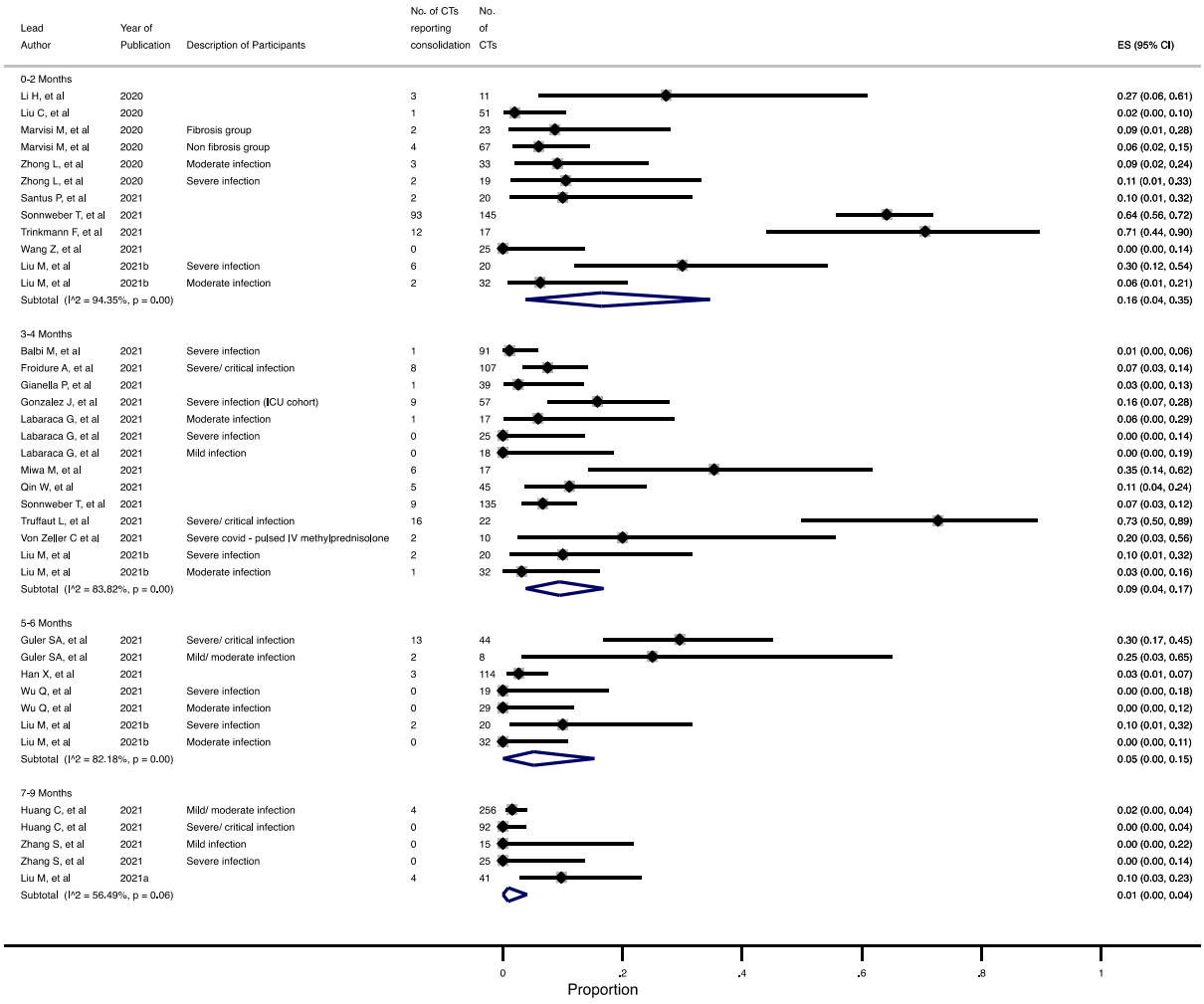


Figure S7b: Forest plot of estimated proportion of follow-up CTs showing ground-glass opacity by time after COVID-19 infection

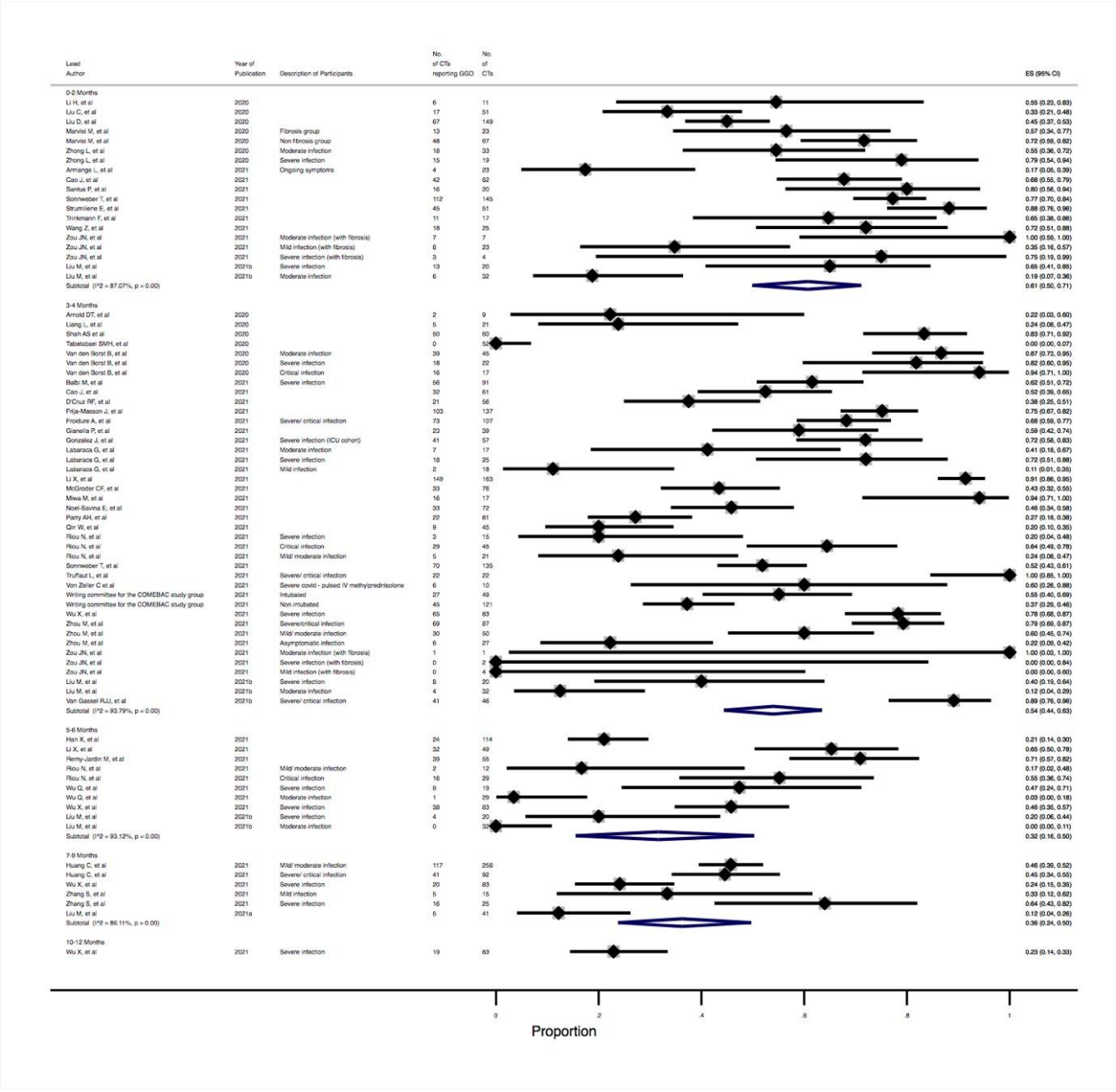


Figure S7c: Forest plot of estimated proportion of follow-up CTs showing honeycombing by time after COVID-19 infection

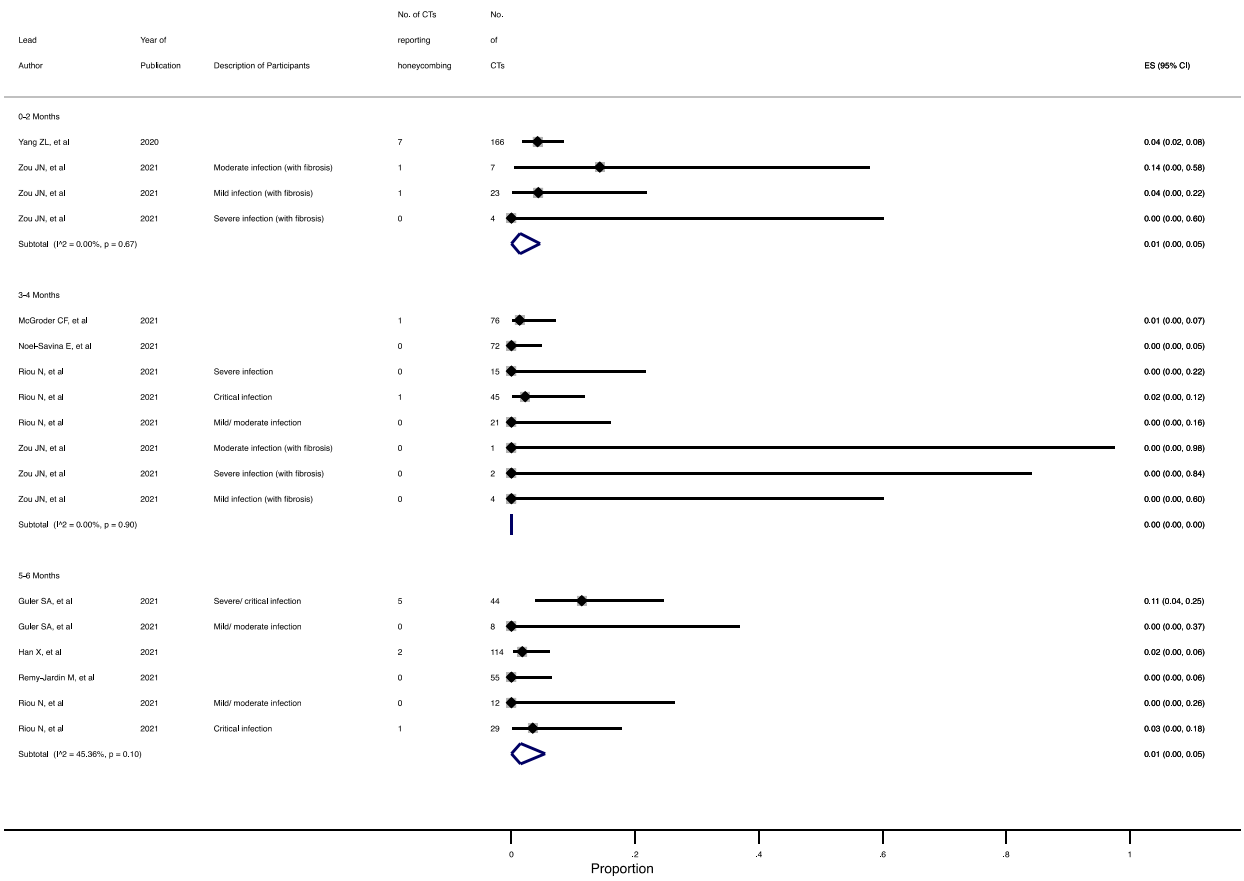


Figure S7d: Forest plot of estimated proportion of follow-up CTs showing linear opacities by time after COVID-19 infection

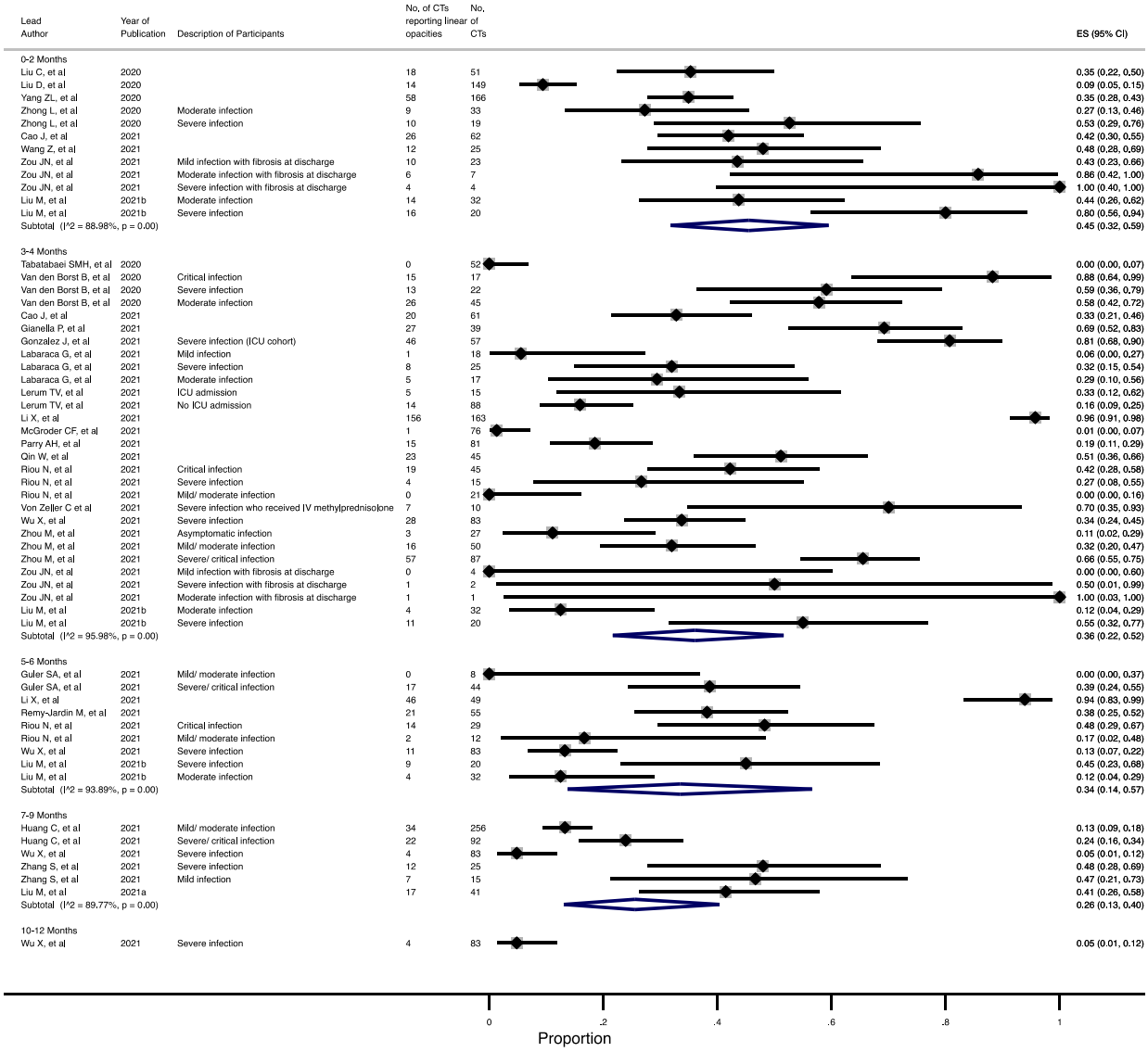


Figure S7e: Forest plot of estimated proportion of follow-up CTs showing reticulation by time after COVID-19 infection

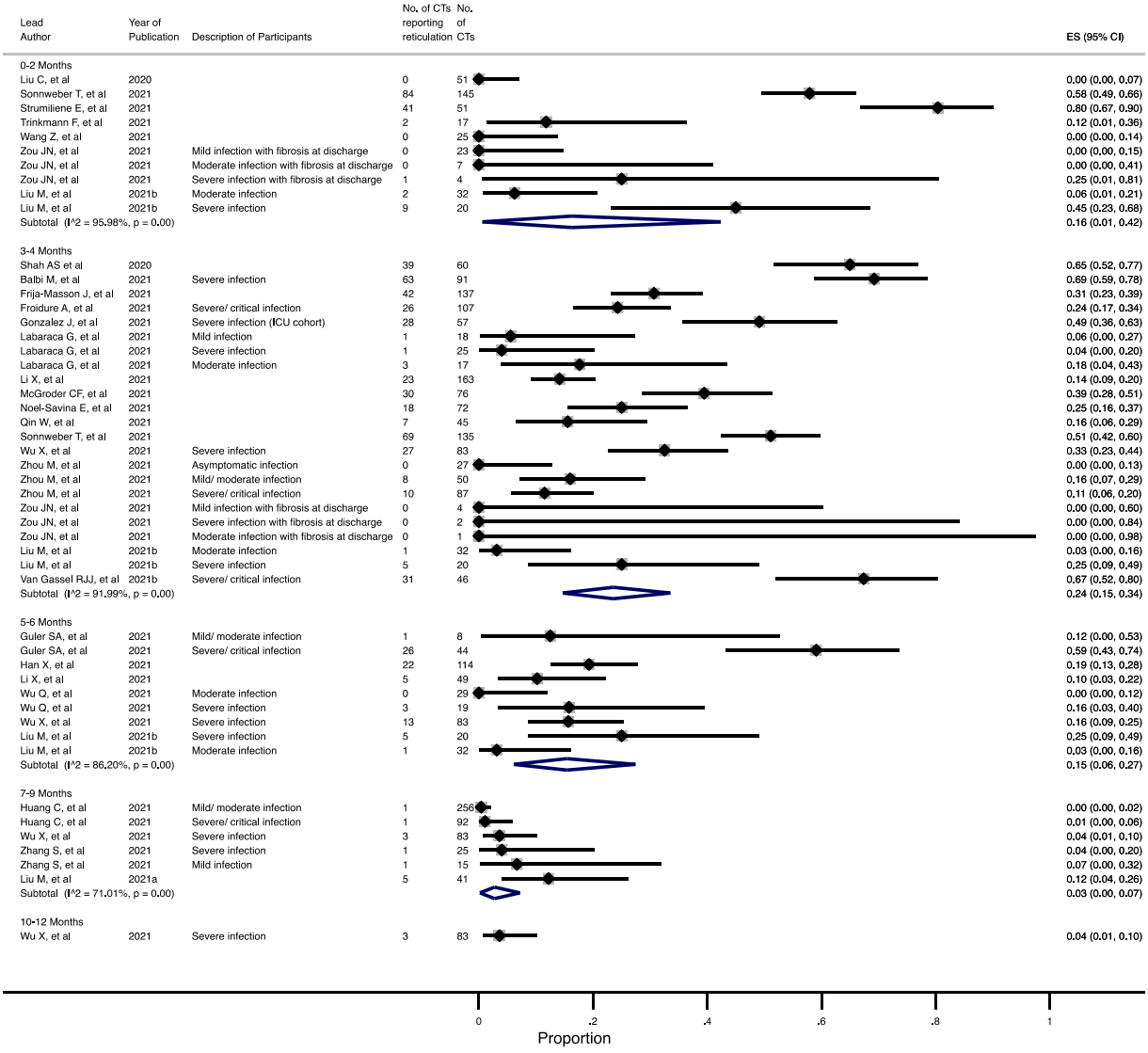


Figure S7f: Forest plot of estimated proportion of follow-up CTs showing architectural distortion by time after COVID-19 infection

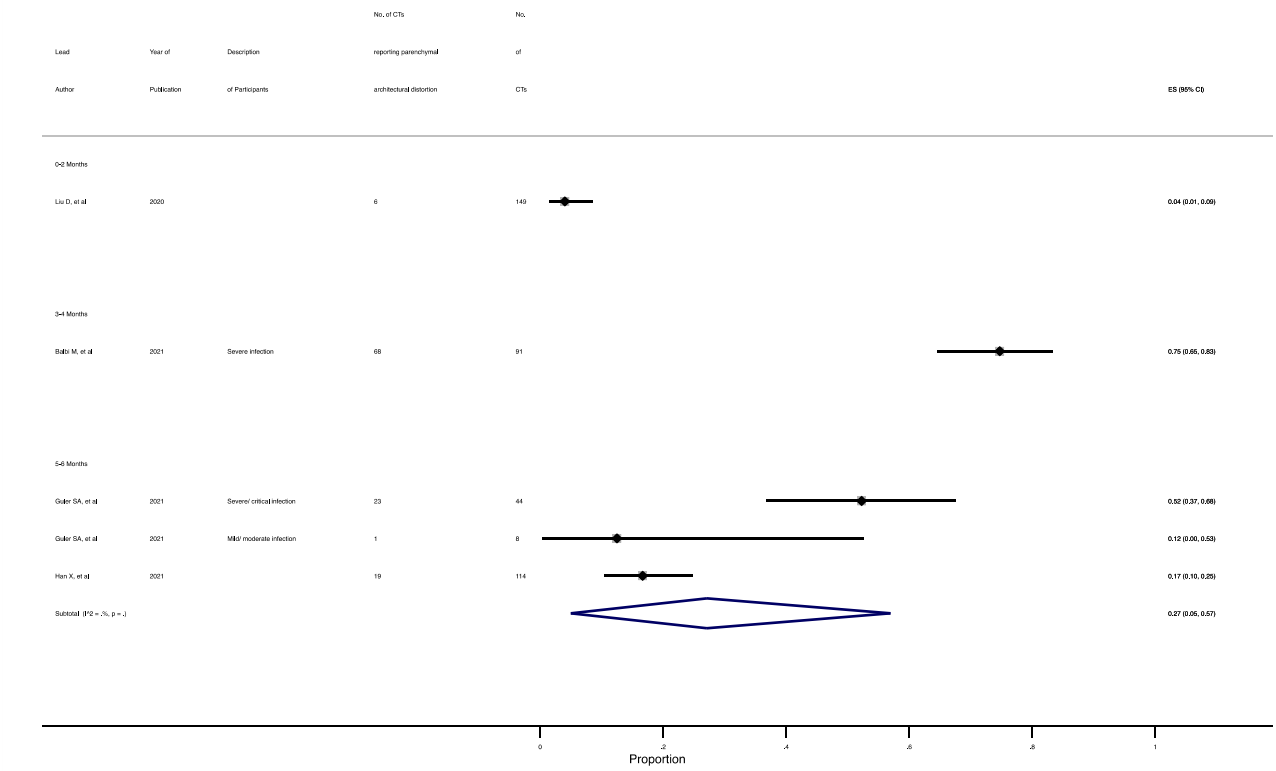


Figure S7g: Forest plot of estimated proportion of follow-up CTs showing pleural thickening by time after COVID-19 infection

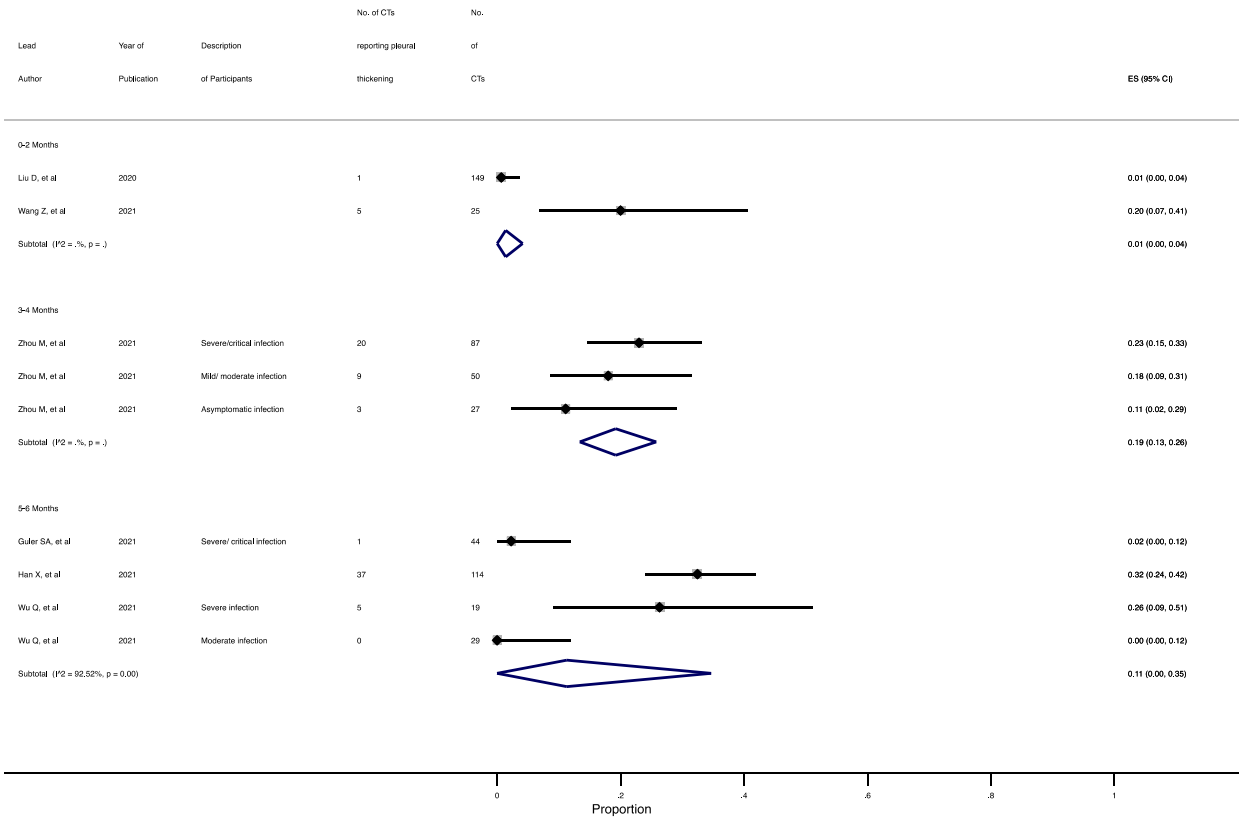


Figure S7h: Forest plot of estimated proportion of follow-up CTs showing traction bronchiectasis and bronchiolectasis by time after COVID-19 infection

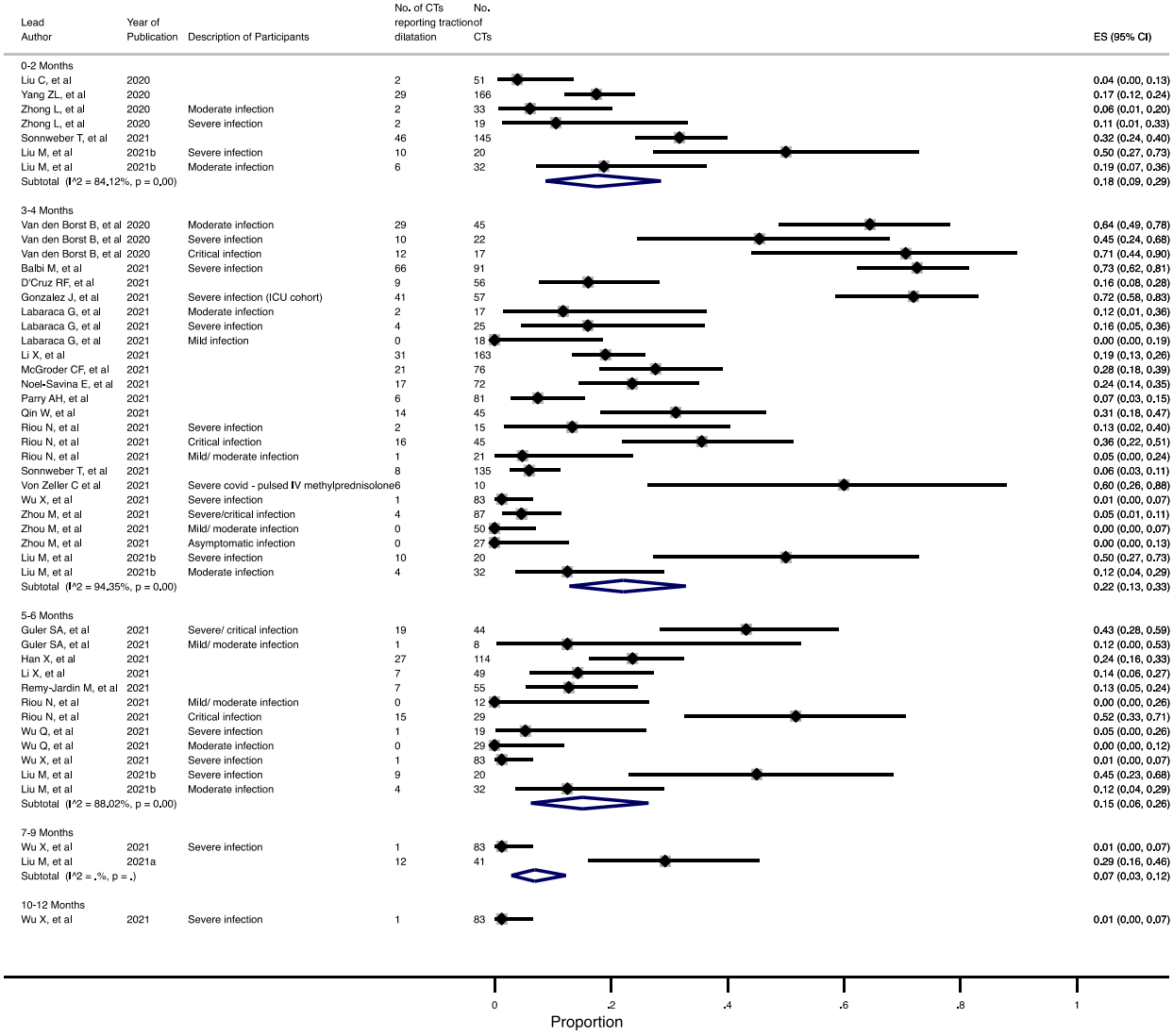


Figure S7i: Forest plot of estimated proportion of follow-up CTs showing non ground-glass opacity nodules by time after COVID-19 infection

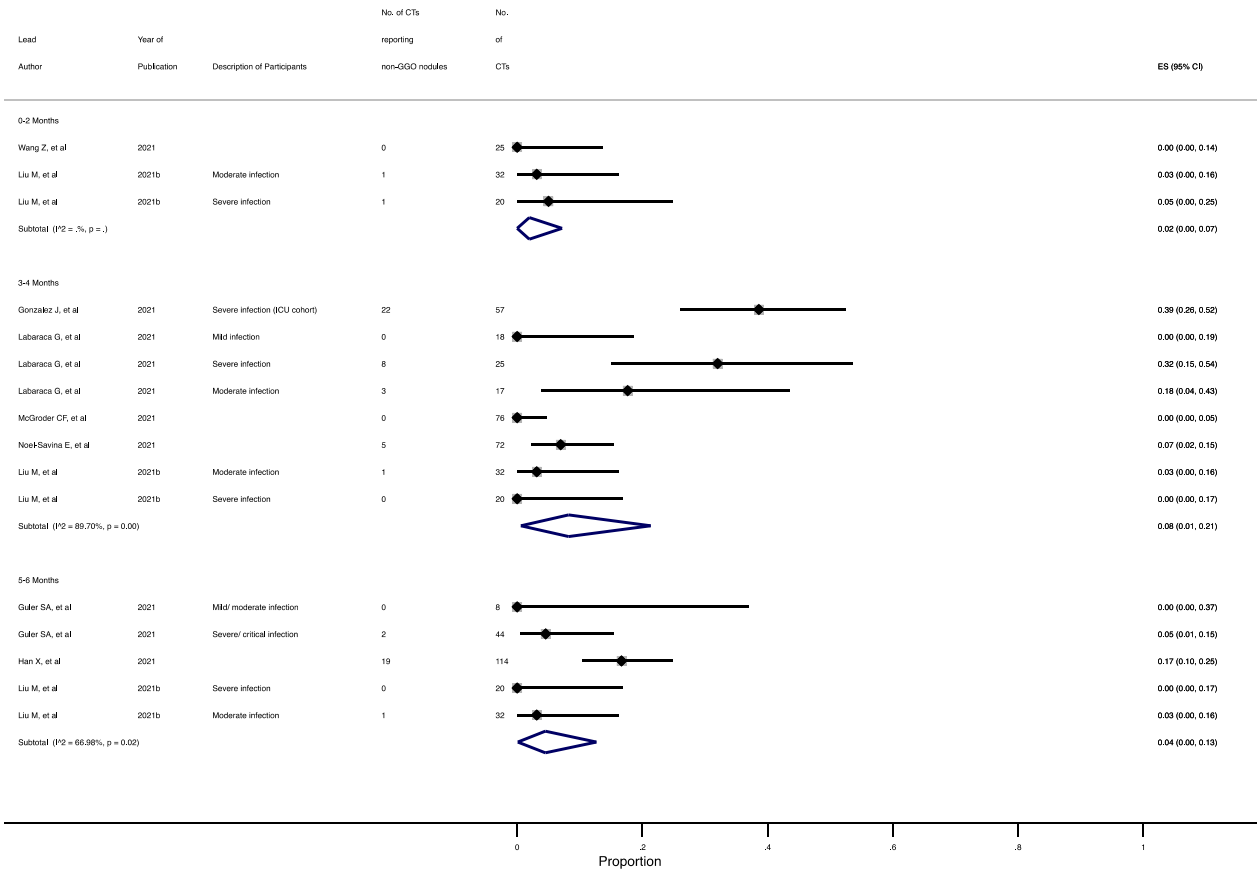


Figure S8a: Forest plot of estimated proportion of follow-up CTs showing ground-glass opacification by time after mild/ moderate COVID-19 infection

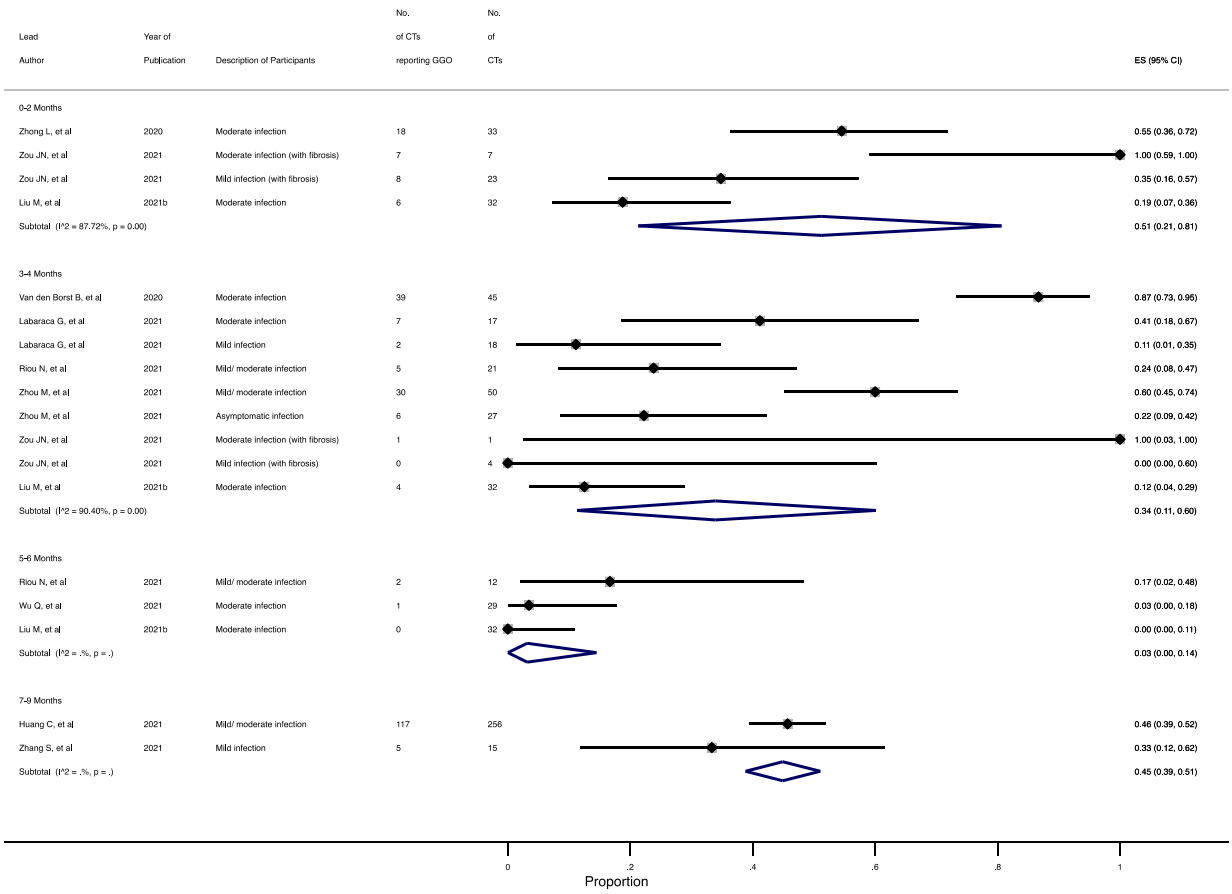


Figure S8b: Forest plot of estimated proportion of follow-up CTs showing linear opacities by time after mild/ moderate COVID-19 infection

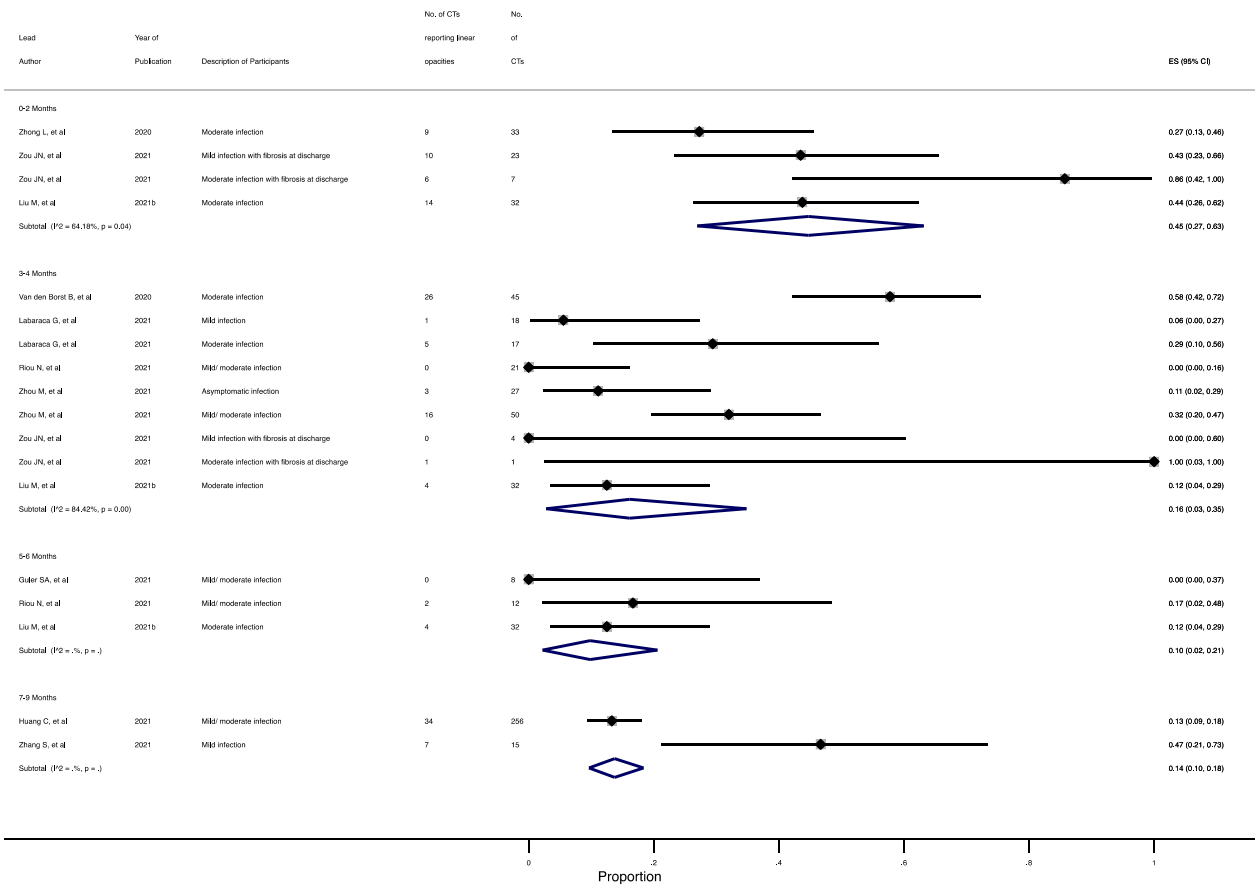


Figure S8c: Forest plot of estimated proportion of follow-up CTs showing reticulation by time after mild/ moderate COVID-19 infection

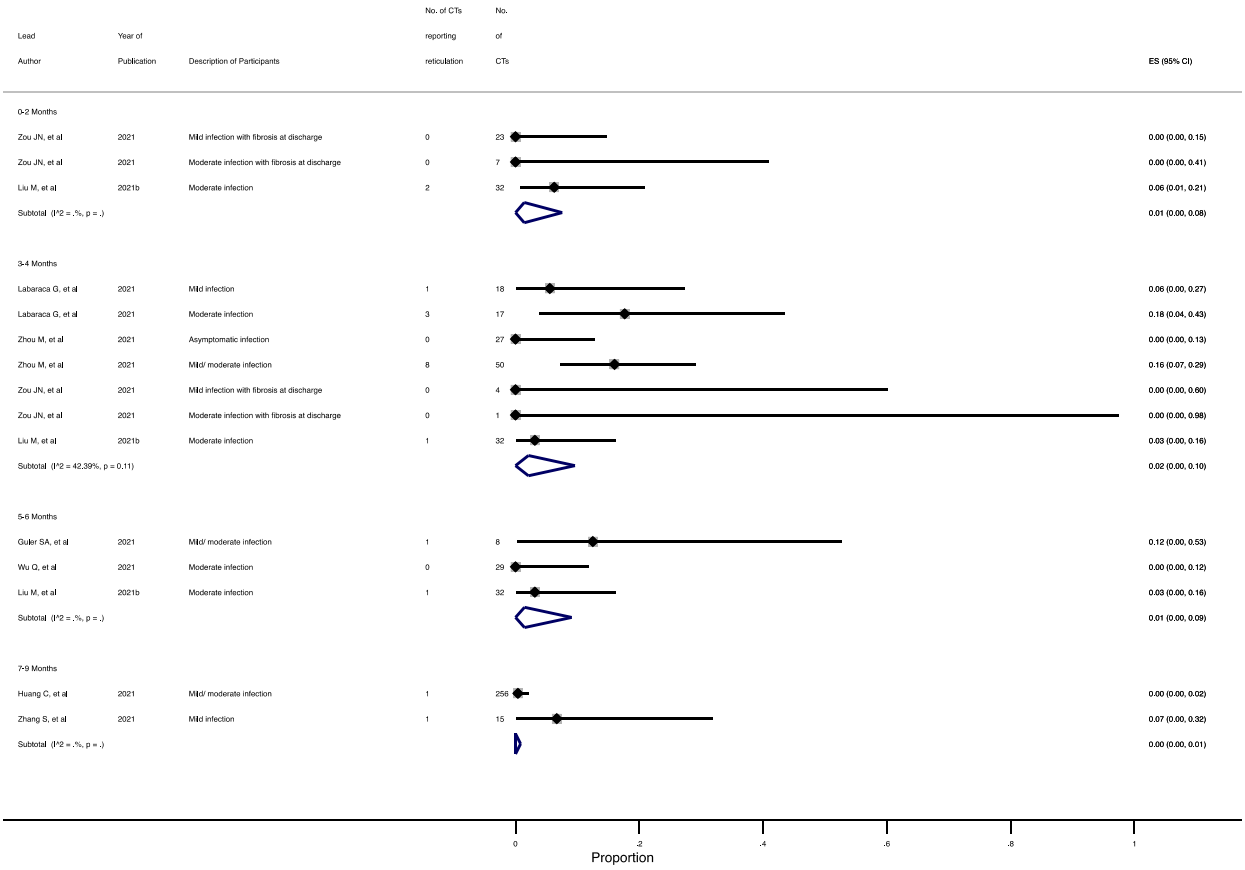


Figure S8d: Forest plot of estimated proportion of follow-up CTs showing traction bronchiectasis & bronchiolectasis by time after mild/ moderate COVID-19 infection

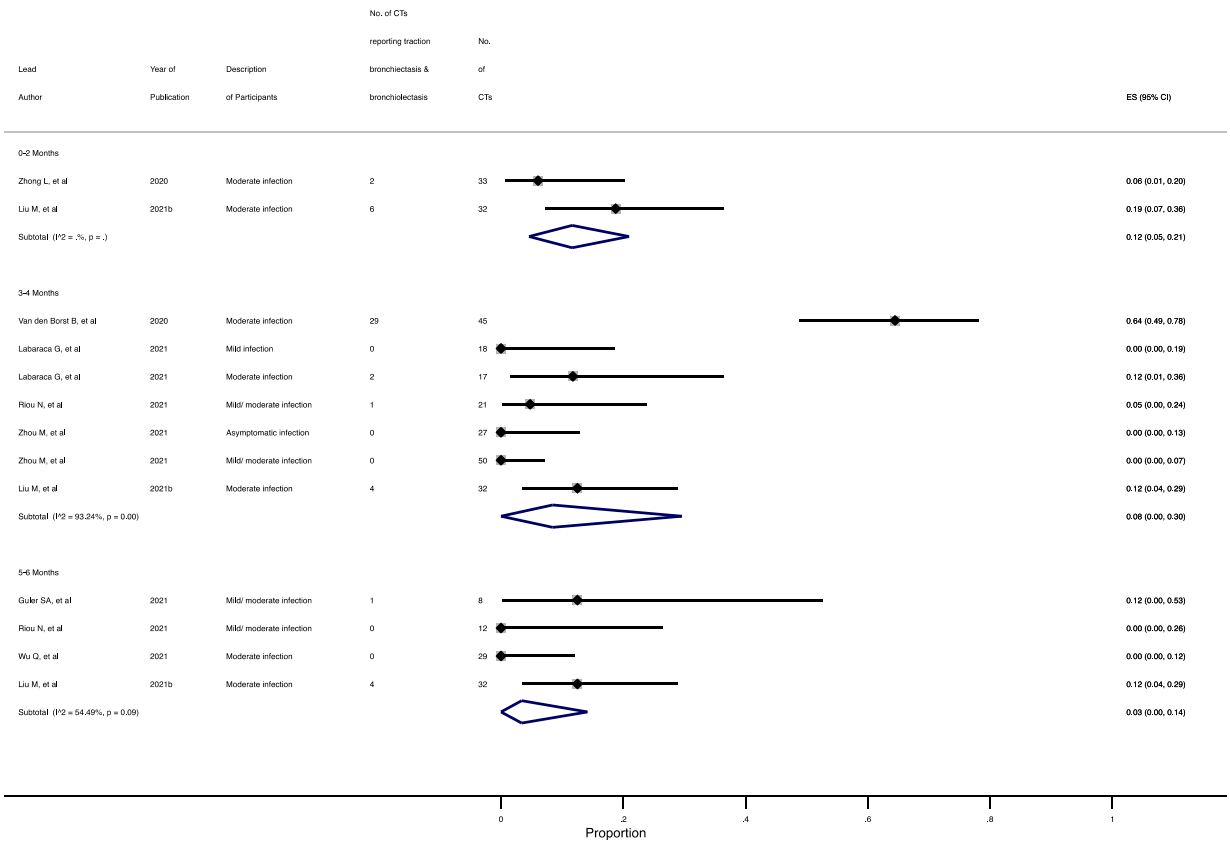


Figure S8e: Forest plot of estimated proportion of follow-up CTs showing honeycombing by time after mild/ moderate COVID-19 infection

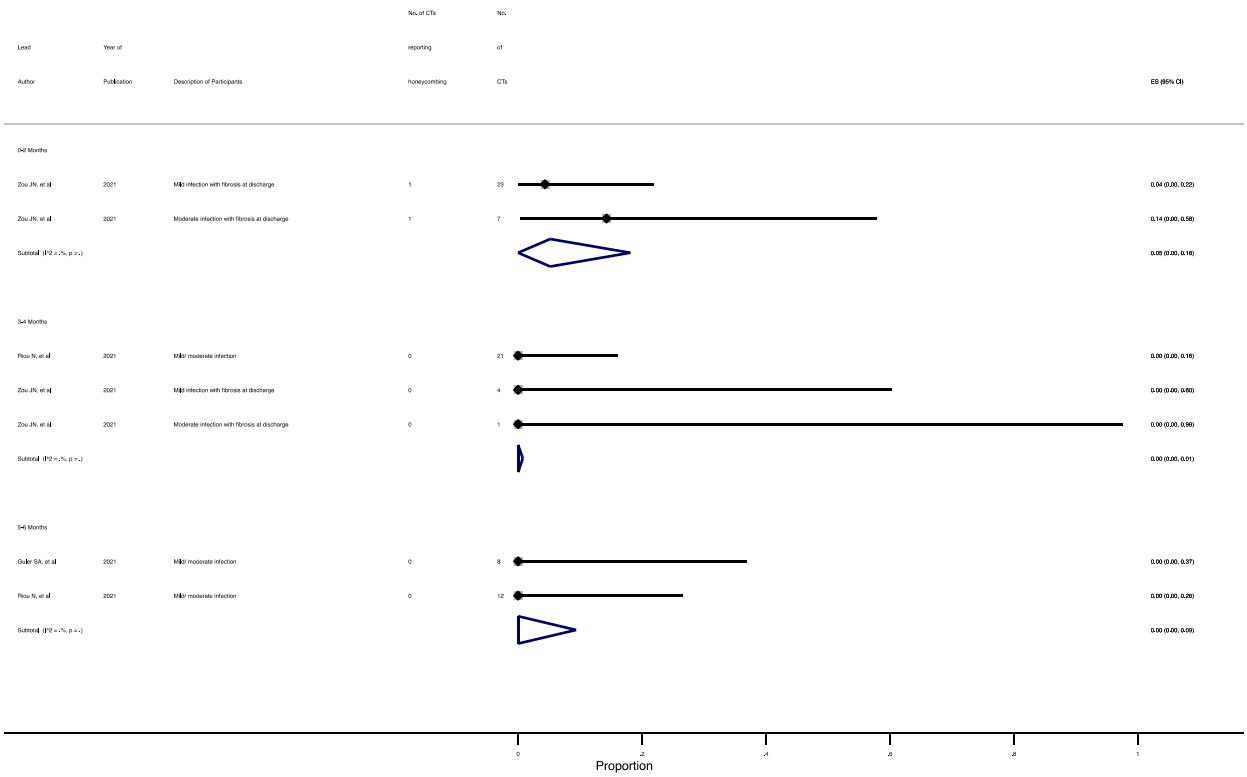


Figure S8f: Forest plot of estimated proportion of follow-up CTs showing consolidation by time after mild/ moderate COVID-19 infection

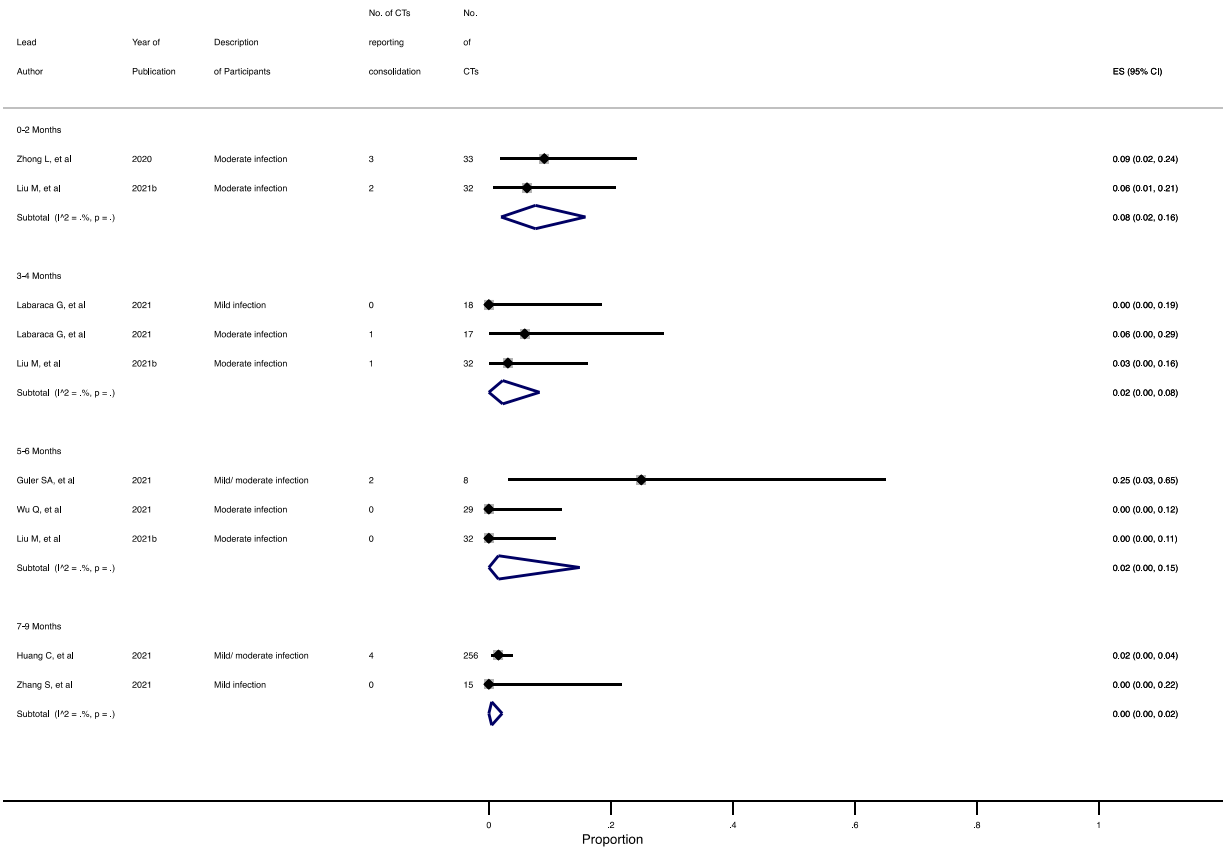


Figure S9a: Forest plot of estimated proportion of follow-up CTs showing ground-glass opacification by time after severe/ critical COVID-19 infection

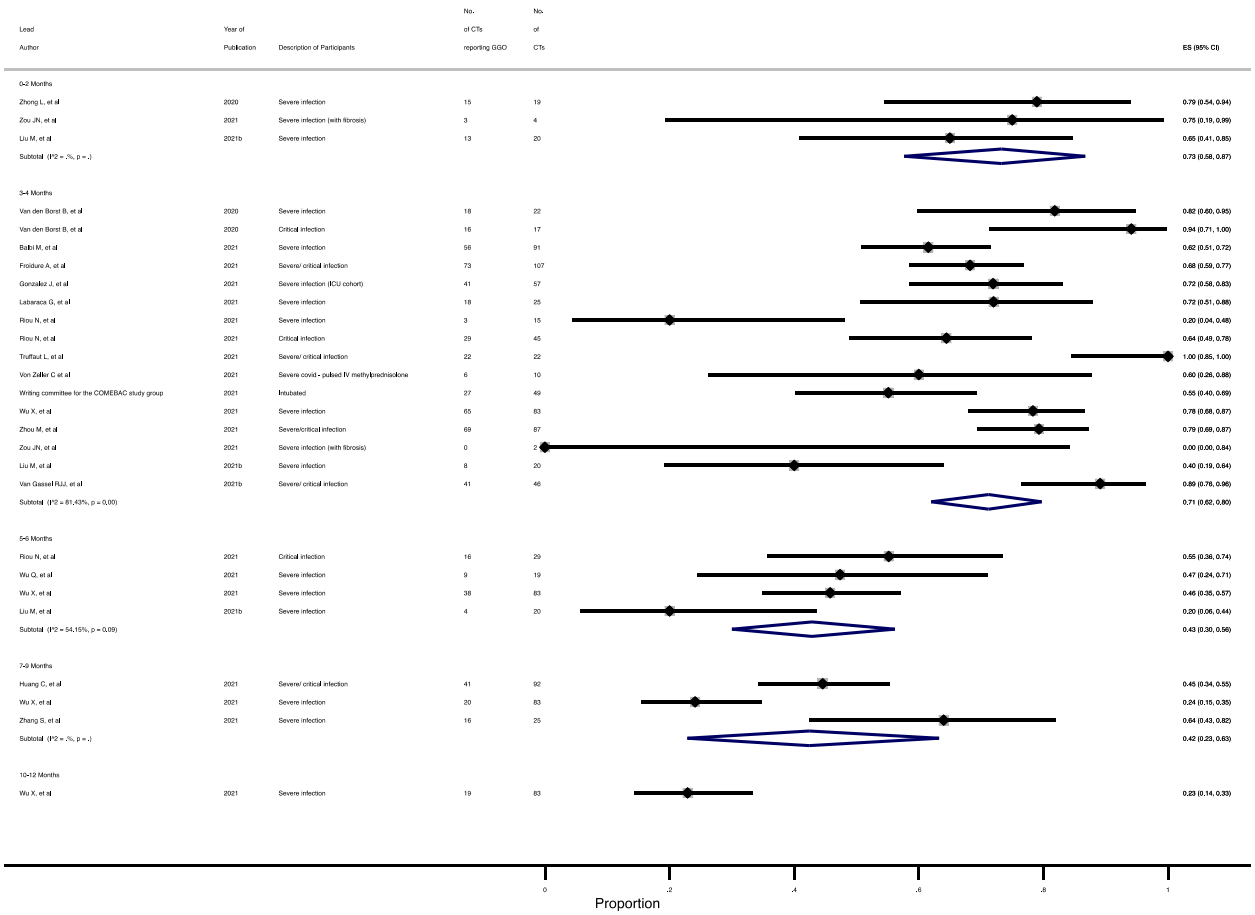


Figure S9b: Forest plot of estimated proportion of follow-up CTs showing linear opacities by time after severe/ critical COVID-19 infection

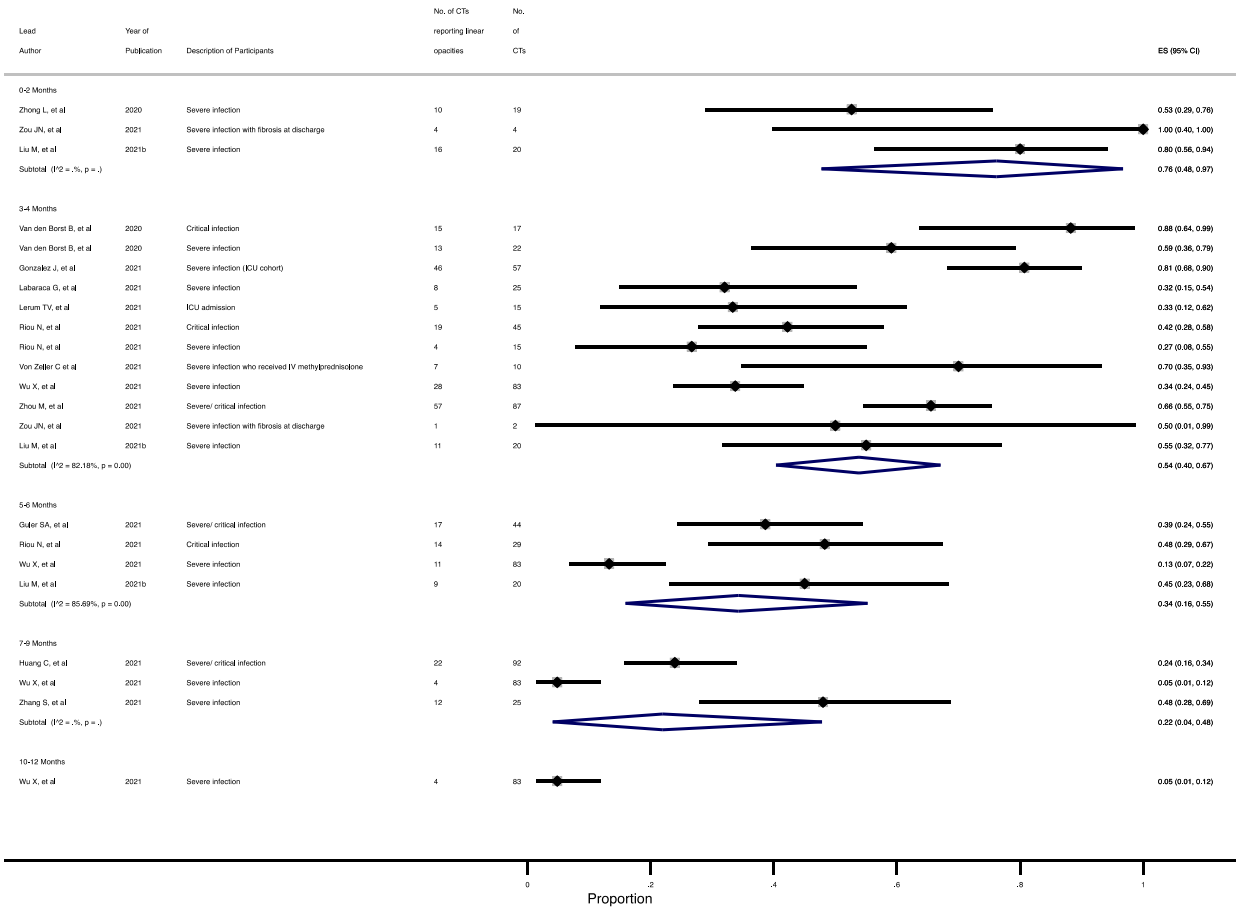


Figure S9c: Forest plot of estimated proportion of follow-up CTs showing reticulation by time after severe/ critical COVID-19 infection

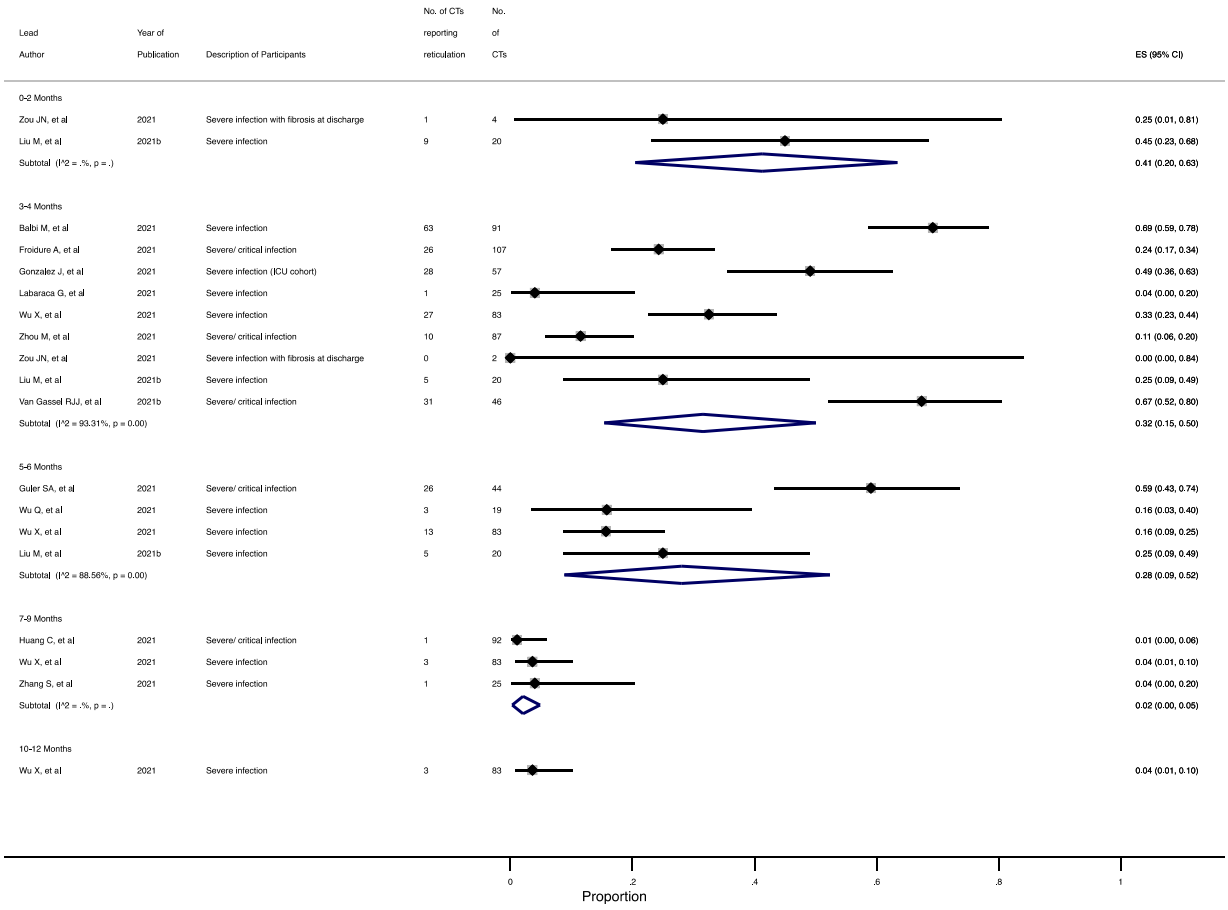


Figure S9d: Forest plot of estimated proportion of follow-up CTs showing traction bronchiectasis & bronchiolectasis by time after severe/ critical COVID-19 infection

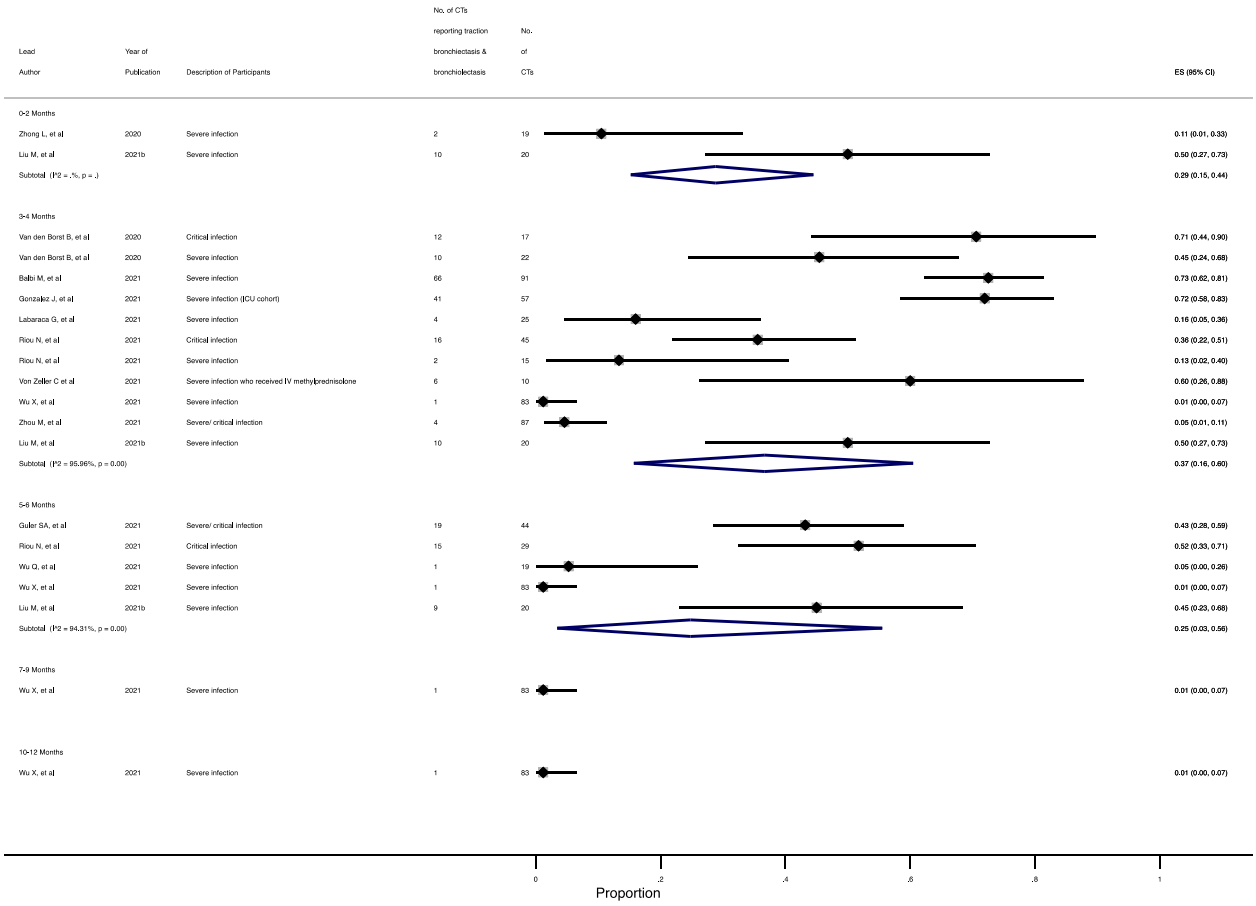


Figure S9e: Forest plot of estimated proportion of follow-up CTs showing honeycombing by time after severe/ critical COVID-19 infection

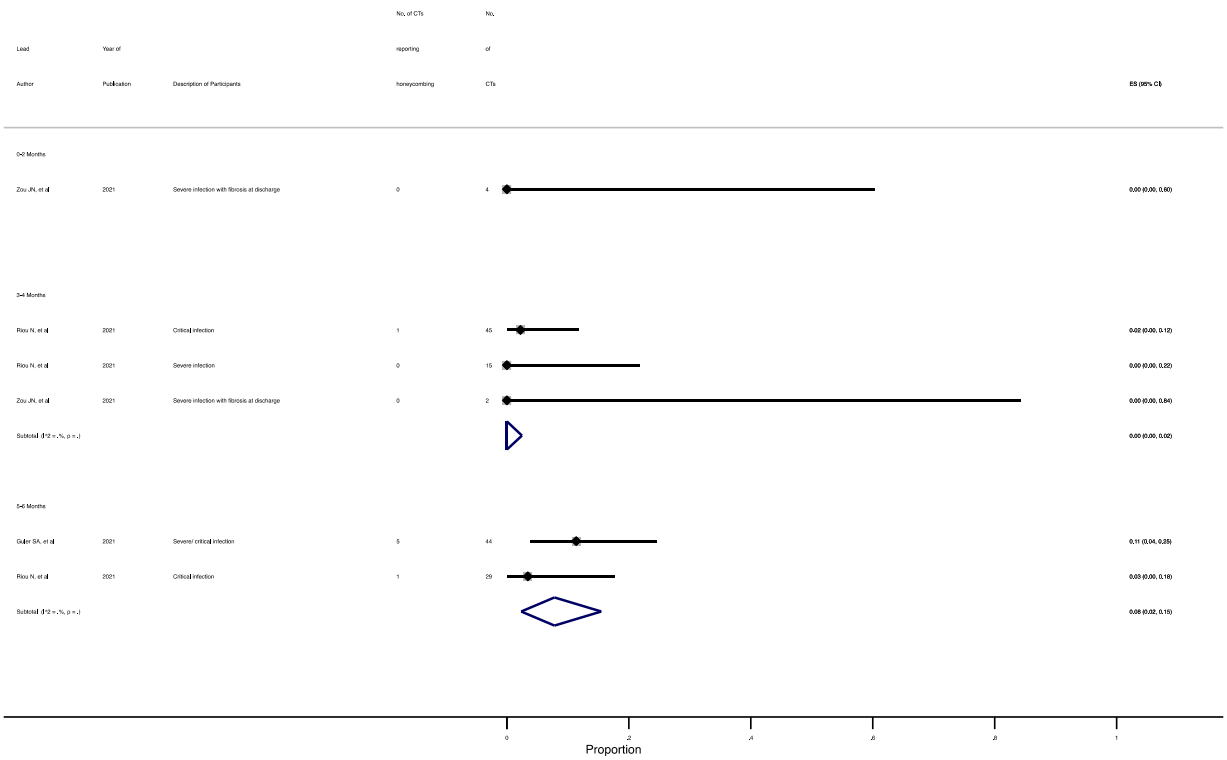


Figure S9f: Forest plot of estimated proportion of follow-up CTs showing consolidation by time after severe/ critical COVID-19 infection

